



Sovereign Ratings vs. Reality: India's Case for a Higher Rating

June 18, 2026

A cross-country comparison based on [latest IMF data](#) reveals important insights into India's fiscal position and sovereign rating relative to global peers. Advanced economies such as Japan, Italy, the United States, France and the United Kingdom continue to carry substantially higher public debt burdens—ranging from around 100% to over 200% of GDP—yet maintain significantly stronger sovereign ratings. In contrast, India's public debt ratio, is lower relative to these economies. This divergence suggests that sovereign ratings are influenced by a broader set of factors beyond debt levels alone, and that India's rating appears conservative when viewed purely through the lens of debt metrics.

India's macroeconomic fundamentals remain robust, characterized by sustained growth outperformance, anchored inflation, positive real interest rates, and a sound financial system. **Debt sustainability analysis** indicates that public debt is likely to follow a declining trajectory, supported by a favorable growth-interest rate differential.

India's sovereign credit profile reflects a combination of improving fiscal dynamics, strong growth prospects, and a resilient debt structure. While short-term fiscal deficits remain relatively high, medium-term consolidation, stable debt dynamics, and structural strengths suggest a stronger underlying credit position than implied by current ratings. **On balance, there appears to be a credible case for a reassessment of India's sovereign rating toward the upper end of the BBB range, particularly if fiscal consolidation continues broadly as envisaged.**

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Sovereign Ratings vs. Reality: India's Case for a Higher Rating

Introduction

The pandemic in 2020 had a severe impact on the deficit and debt dynamics of almost all countries in the world, though in varying degrees. India was also severely impacted, with the general government deficit and debt rising sharply in the immediate aftermath of the pandemic, which was unavoidable. Since then, thanks to the strong resolve of the authorities, significant improvement has been achieved on the fiscal front, culminating first in a [ratings outlook upgrade](#) to positive from stable by S&P Global Ratings on 29 May 2024 and then a subsequent [sovereign ratings upgrade to BBB from BBB-](#) on 14th August 2025 after a gap of 18 long years.

Fiscal consolidation has gathered significant momentum in the last few years, thanks to robust growth, fiscal marksmanship and particular focus on improving the quality of fiscal spending. Indeed, both the interim budget in Feb 2024 and the post-elections union budget in July 2024 [exceeded expectations](#) as regards to the fiscal deficit target, which is praiseworthy. The FY26 Union Budget also saw the central government announcing a further moderation in the fiscal deficit target to 4.4% of GDP, vs. a revised estimate outturn of 4.84% of GDP in FY25, which was finally met, despite concerns related to higher tariffs imposed by USA in 2025, a lower-than-expected nominal GDP growth profile and revenue uncertainty due to GST rates rationalisation.

Amidst ongoing concerns related to tariff uncertainty in the beginning of this year and in the backdrop of heightened geopolitical tensions, the government presented the Union Budget for FY27 on 1st February 2026. Striking a delicate balance between the dual objectives of supporting growth and employment, while remaining on the path of fiscal consolidation, the budget set a fiscal-deficit target of 4.3% of GDP for FY27, with the authorities reemphasizing the medium-term fiscal goal of reducing the centre's debt/GDP ratio to the pre-pandemic level of 50% by FY31, while setting the FY27 target at 55.6% of GDP.

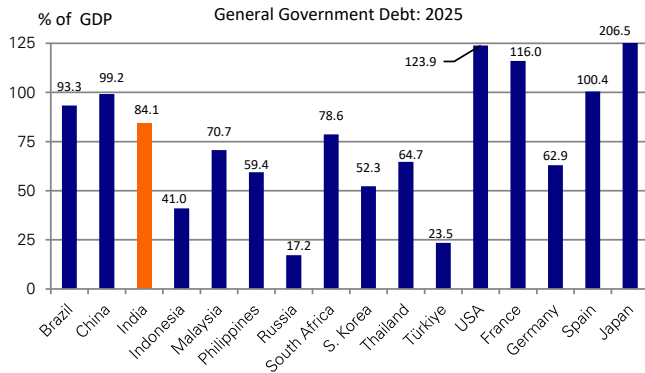
Indian sovereign ratings relative to rest of the world

In the following pages, we present a series of charts and tables using [latest IMF data](#) for major advanced and emerging economies (including India) covering six key dimensions:

1. General government debt as a % of GDP;
2. General government budget deficit as a % of GDP;
3. Expected change in debt and deficit between 2025 and 2031 based on IMF's forecasts;
4. Ranking of each of these economies by their size (2025 data; nominal GDP in USD trn terms);
5. Sovereign credit ratings assigned by S&P, Moody's and Fitch;
6. We also provide a table which show the IMF's interpretation of credit rating scale of major credit rating agencies.

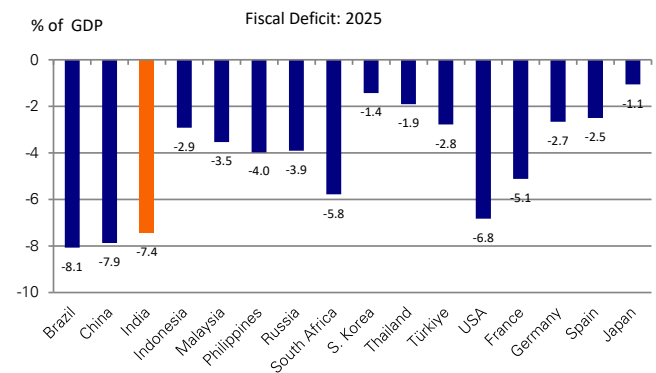


Figure 1: General Government Debt: 2025



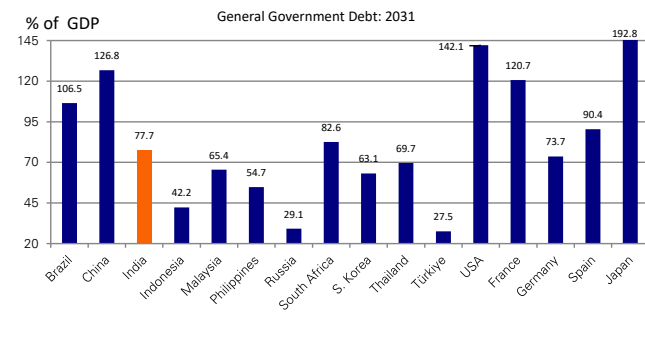
Source : IMF: World Economic Outlook Database, April 2026

Figure 2: General Government Deficit: 2025



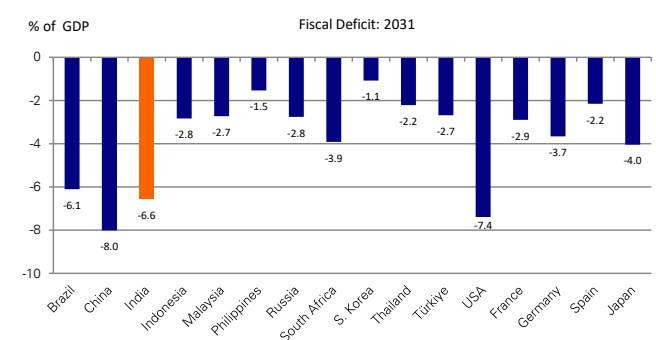
Source : IMF: World Economic Outlook Database, April 2026

Figure 3: General Government Debt: 2031 F



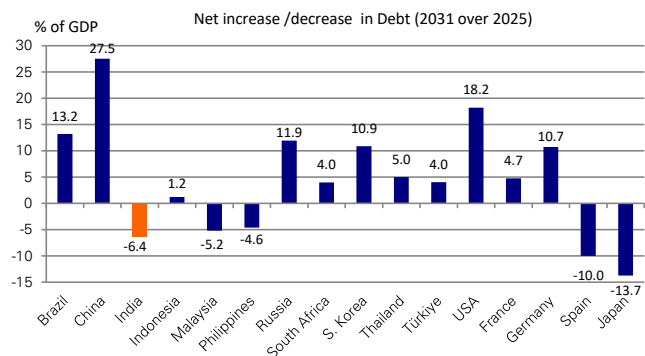
Source : IMF: World Economic Outlook Database, April 2026

Figure 4: General Government Deficit: 2031 F



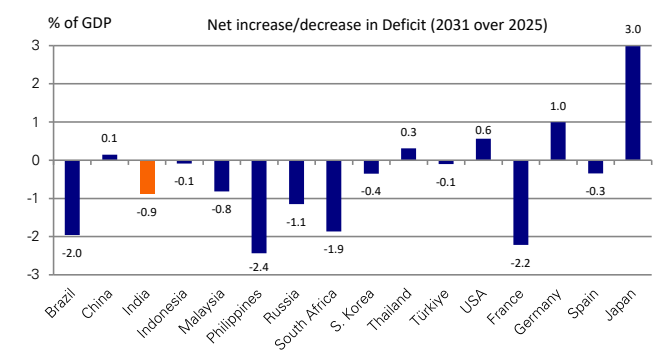
Source : IMF: World Economic Outlook Database, April 2026

Figure 5: Net change in Debt: 2031 over 2025



Source : IMF: World Economic Outlook Database, April 2026

Figure 6: Net change in Deficit: 2031 over 2025



Source : IMF: World Economic Outlook Database, April 2026



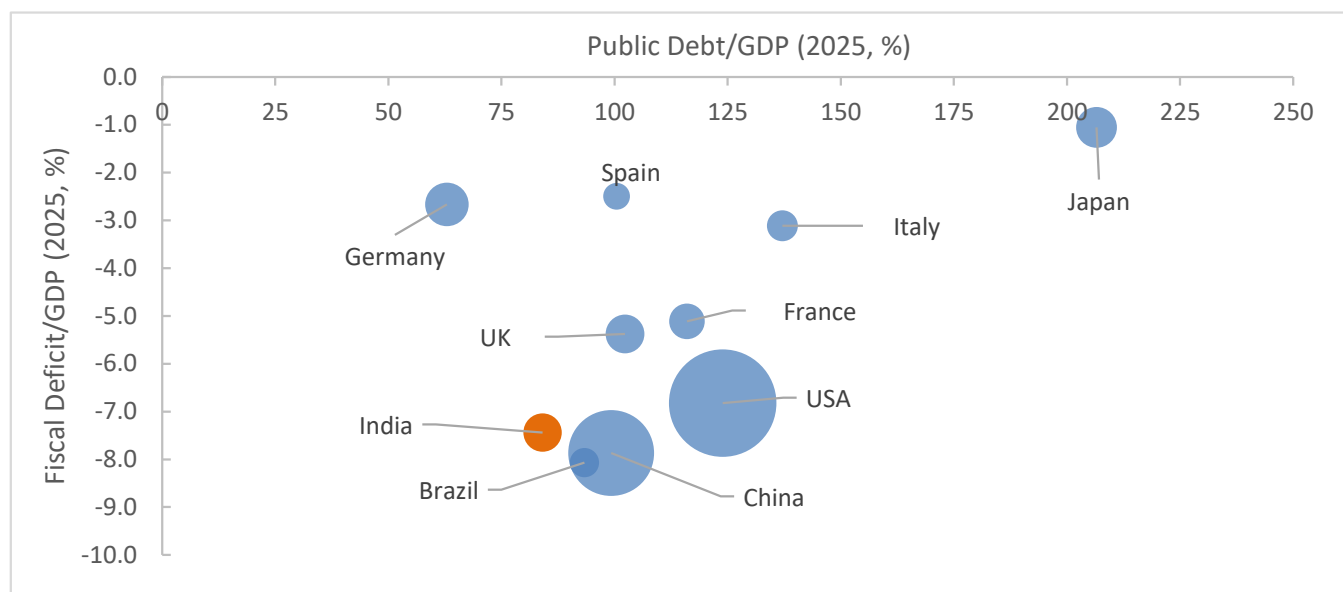
India's sovereign credit ratings, debt and deficit profile and economic size relative to other economies - a summary

Figure 7: Debt, deficit and creditworthiness - advanced and emerging economies

	Country	Nominal GDP USD trn, 2025	Rank By size	Public debt/GDP 2025	Fiscal deficit/GDP 2025	Credit Rating		
						S&P	Moody's	Fitch
1	USA	30.8	1	123.9	-6.8	AA+	Aa1	AA+
2	China	19.6	2	99.2	-7.9	A+	A1	A
3	Germany	5.0	3	62.9	-2.7	AAA	Aaa	AAA
4	Japan	4.4	4	206.5	-1.1	A+	A1	A
5	UK	4.0	5	102.3	-5.4	AA	Aa3	AA-
6	India	3.9	6	84.1	-7.4	BBB	Baa3	BBB-
7	France	3.4	7	116.0	-5.1	A+	Aa3	A+
8	Italy	2.6	8	137.1	-3.1	BBB+	Baa2	BBB+
9	Brazil	2.3	10	93.3	-8.1	BB	Ba1	BB
10	Spain	1.9	11	100.4	-2.5	A+	A3	A
11	S. Korea	1.9	12	52.3	-5.8	AA	Aa2	AA-
12	Turkiye	1.6	15	23.5	-2.8	BB-	B1	BB-
13	Indonesia	1.4	16	41.0	-2.9	BBB	Baa2	BBB
14	Thailand	0.6	29	64.7	-1.9	BBB+	Baa1	BBB+
15	Philippines	0.5	33	59.4	-4.0	BBB+	Baa2	BBB
16	Malaysia	0.5	34	70.7	-3.5	A-	A3	BBB+
17	South Africa	0.4	40	78.6	-5.8	BB	Ba2	BB-

Source : IMF: World Economic Outlook Database, April 2026, S&P, Moody's and Fitch

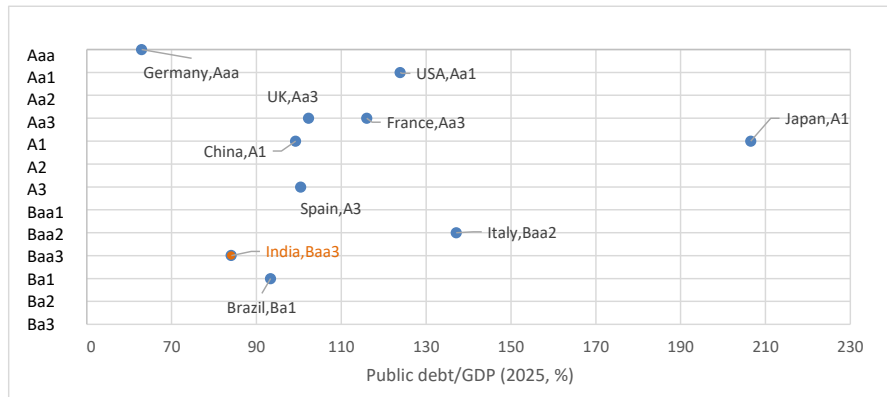
Figure 8: Debt, deficit and GDP size across major economies (2025)



Source : IMF, S&P, Moody's and Fitch. Note: Bubble size = Nominal GDP

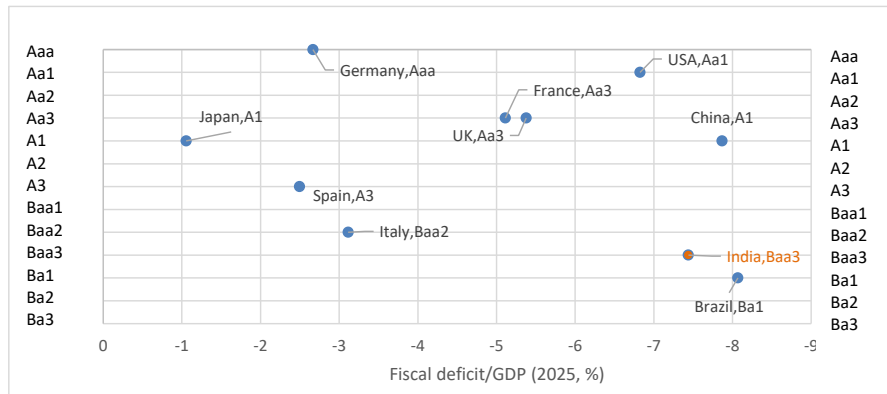


Figure 9: Debt/GDP vs. Moody's credit ratings across major economies (2025)



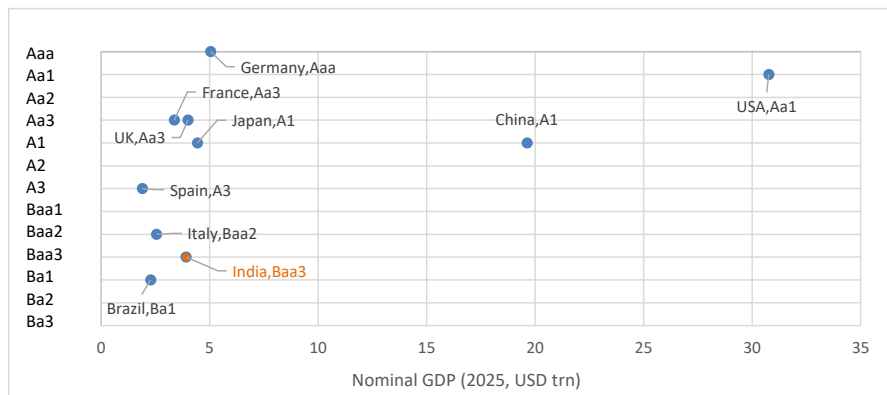
Source : IMF, Moody's

Figure 10: Deficit/GDP vs. Moody's credit ratings across major economies (2025)



Source : IMF, Moody's

Figure 11: Nominal GDP vs. Moody's credit ratings across major economies (2025)



Source : IMF, Moody's



Figure 12: IMF's interpretation of credit rating scale of some major credit rating agencies

Interpretation	Fitch and S&P	Moody's
Highest quality	AAA	Aaa
High quality	AA+	Aa1
	AA	Aa2
	AA–	Aa3
Strong payment capacity	A+	A1
	A	A2
	A–	A3
Adequate payment capacity	BBB+	Baa1
	BBB	Baa2
	BBB–	Baa3
Likely to fulfill obligations, on going uncertainty	BB+	Ba1
	BB	Ba2
	BB–	Ba3
High-risk obligations	B+	B1
	B	B2
	B–	B3
Vulnerable to default	CCC+	Caa1
	CCC	Caa2
	CCC–	Caa3
Near or in bankruptcy or default	CC	Ca
	C	C
	D	D

Source : Economic Survey 2020-21 Volume 1, [IMF: 2010: The Uses and Abuses of Sovereign Credit Ratings](#)



Observations

* **Do India's current sovereign ratings adequately reflect its economic fundamentals?** India appears fiscally stronger than several countries that enjoy significantly higher sovereign ratings, particularly when debt levels, economic growth prospects and future fiscal trends are considered together. While India still runs a relatively large fiscal deficit, its debt burden is moderate and is expected to improve over time. This raises a legitimate question as to whether India's current sovereign ratings fully reflect its economic fundamentals and growth potential.

* **Public debt/GDP and sovereign ratings:** The 2025 IMF data show a wide heterogeneity among countries. The highest debt/GDP ratios are observed in the case of Japan (206%), Italy (137%), USA (124%), France (116%), and UK (103%). These countries have accumulated substantial public debt over many years due to aging population dynamics, welfare commitments, financial crises and pandemic-related spending. Despite these elevated debt levels, they continue to enjoy substantially higher sovereign ratings compared to India. As per the IMF's data, India's public debt/GDP and general government budget deficit stood at 84% and 7.4% respectively. Only Germany and some emerging economies have materially lower debt levels compared to India.

* **General government deficit/GDP and sovereign ratings:** India's general government budget deficit remains elevated at around 7.4% of GDP, which is on the higher spectrum of the economies considered in our analysis. However, the IMF's projections till 2031 factor in a meaningful consolidation of India's deficit (from 7.4% of GDP in 2025 to 6.6% of GDP by 2031) and public debt (from 84.1% of GDP in 2025 to 77.7% of GDP by 2031) by the end of this decade. The improvement suggests i) continued economic growth (10.8% average nominal GDP growth between 2026-2031); ii) rising tax revenues; and iii) better fiscal and expenditure discipline. **Indeed, India's debt trajectory appears more favorable than that of several major economies, where debt ratios are projected to increase further in the coming years.**

* **Size of the economy and sovereign ratings:** As far as size of the economy is concerned, **India is currently the 6th-largest economy, comparable to the UK and larger than France, and slated to become the third largest before the end of this decade, as per our and IMF forecasts.** Yet, India's sovereign ratings currently remain at the lowest investment grade (Moody's and Fitch). Therefore, on the basis of the size of the economy, India appears underrated, in our view.

* **Funding of the public debt and sovereign ratings:** In this respect, it is instructive, in our view, to reflect on the IMF's interpretation of credit rating scale of major credit rating agencies. As per Fitch/S&P scale, A range (A+, A, A-) signifies "Strong payment capacity", while BBB range (BBB+, BBB, BBB-) reflects "Adequate payment capacity" and BB range and below highlight increasing uncertainty and risk, as per the IMF's interpretation. India currently sits at BBB as per S&P, while ranked at the lowest level of investment grade by Fitch (BBB-) and Moody's (Baa3). That places India at the lowest rung of "adequate payment capacity".

In our view, **there is little risk associated with the funding of the public debt** as bulk of India's debt is in domestic currency. Indeed, foreign currency denominated debt is small - less than 5% of GDP - and even within that, more than half of it is owed to multilateral institutions. A country, like India, with mostly domestic currency debt, does not face currency mismatch risk (unlike EM crisis cases), can rely on its



domestic financial system to absorb debt, has more monetary flexibility (central bank support) and is less exposed to external funding crises. **These are typically characteristics associated with higher creditworthiness (often A-category countries).**

India, in our view, shows traits consistent with “strong payment capacity”, being a large, diversified domestic economy (#6 globally; on its way to become #3 before the end of this decade); holding predominantly local-currency debt; deep and growing domestic financial market; limited external vulnerability compared to many EM peers. These characteristics align more closely with “A category” descriptors than the BBB range in our view. While S&P upgraded India's sovereign ratings one notch higher to BBB, from BBB- in August 2025, both Fitch (BBB-) and Moody's (Baa3) have kept India's sovereign ratings at the lowest level of investment grade.

Even if the rating agencies do not provide India with an A-category sovereign rating, reflecting "Strong payment capacity", due to lower-per-capita income relative to peers, **there could be merit to at least consider upgrading India to the upper rung of the "Adequate payment capacity" (BBB+ for S&P and Fitch) and Baa1 for Moody's, which will require one more rating upgrade by S&P and a double-notch ratings upgrade by Fitch and Moody's, in our view.**

Figure 13: India's sovereign credit rating (1998 -2024)

Date	S&P	Moody's	Fitch
June 1998		Ba2*	
October 1998	BB*		
March 2000			BB+*
November 2001			BB*
February 2003		Ba1*	
January 2004			BB+*
January 2004		Baa3	
February 2005	BB+*		
August 2006			BBB-
January 2007	BBB-		
November 2017		Baa2	
June 2020		Baa3	

*Speculative Grade; **Green highlights ratings upgrade**; **Red highlights ratings downgrade**, Black indicates first rating

Source : Economic Survey 2020-21 Volume 1, Compiled from S&P Global, Fitch and Moody's



A debt sustainability exercise for India

Next, we undertake a debt sustainability exercise for India, to ascertain whether the forecasted debt consolidation path has changed materially on account of the current oil-price shock. In our view, this is useful for providing a reference point of the progress that is likely under the baseline scenario. Then, using this reference point, we calculate the upside and downside risks, based on various scenarios. This assumes particular importance, as Government of [India's medium-term fiscal roadmap](#) is now aimed towards achieving a sustained debt consolidation path by the end of this decade, but without pre-committing explicitly to any fiscal deficit target. **As per the plan, the centre aims to reduce its debt/GDP ratio to the pre-pandemic level of 50% by FY31, from about 55.6% currently.**

Fiscal snapshot at the general government level

The two tables below summarize the fiscal performance at the general government level pre- and post-Covid. General government budget and primary deficits were on a path of consolidation until FY19, while interest payments remained broadly sticky at around 4.8% of GDP. The Covid-19 pandemic impacted the fiscal and debt position severely, with **general government deficit rising to 13.1% of GDP by FY21, from 5.8% of GDP in FY19.** This is despite the central government being extremely prudent with its targeted expenditure support for the absolutely vulnerable and middle-class section of the population. Interest payments as a % of GDP increased to 5.4% of GDP by FY21, from 4.8% of GDP in FY19, while **general government debt increased to 88% of GDP in FY21, from about 70% in FY19.**

Figure 14: General Government fiscal snapshot

% of GDP	FY20	FY21	FY22	FY23	FY24	FY25	FY26	FY27F
Revenue	20.7	19.8	21.6	22.2	22.5	22.5	22.6	22.7
Expenditure	27.9	32.9	32.0	31.2	30.9	30.0	29.9	30.0
Interest payment	4.8	5.4	5.3	5.2	5.3	5.2	5.3	5.2
General government deficit	7.2	13.1	10.3	9.0	8.4	7.5	7.3	7.3
Primary deficit	2.4	7.7	5.1	3.7	3.2	2.3	2.2	2.3

Source: Ministry of Finance, Deutsche Bank forecasts

Figure 15: Liability position of the centre and the states

% of GDP	FY15	FY16	FY17	FY18	FY19	FY20	FY21	FY22	FY23	FY24	FY25	FY26
I. Central govt debt	47.1	47.4	45.5	48.9	49.4	52.3	61.5	58.8	57.9	58.1	57.2	57.2
II. State govt debt	21.0	22.5	23.8	25.1	25.3	26.7	31.0	29.1	28.0	27.5	27.5	27.5
III. Central loans to state	1.2	1.1	1.0	3.7	3.2	2.8	3.5	3.6	3.4	3.4	3.3	3.3
IV. Investment in treasury bills by states	1.0	1.2	1.3	1.2	1.0	0.9	1.2	1.3	1.0	1.1	1.1	1.1
General govt debt = I + II - III - IV	65.9	67.6	67.1	69.0	70.4	75.2	87.8	83.0	81.3	81.1	80.3	80.3

Source: Ministry of Finance, Deutsche Bank forecasts

States' debt/GDP was also affected severely, rising from 21% of GDP in FY19 (excluding central loans to states and investment in treasury bills by states) to 26% of GDP by FY21. States' combined fiscal deficit also increased to 4.1% of GDP in FY21, from 2.5% of GDP in FY19. Meanwhile, the **centre's** fiscal deficit increased to 9.2% of GDP in FY21, from 3.4% of GDP in FY19 and debt/GDP worsened to 61.5% of GDP in FY21, from 49.4% of GDP in FY19.



Figure 16: Centre and state fiscal snapshot

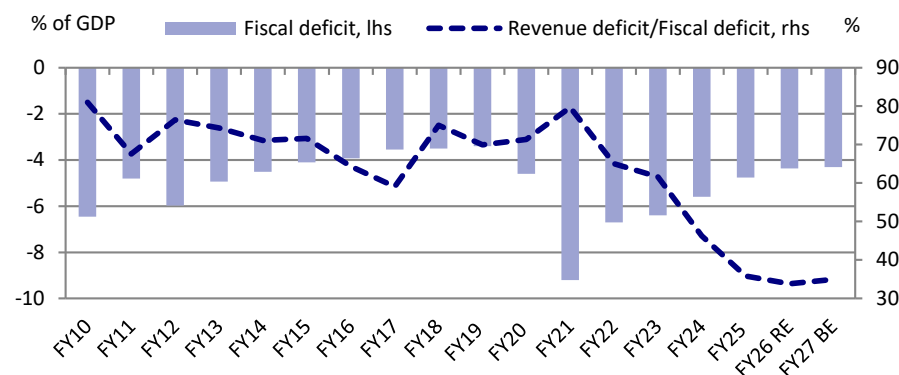
% of GDP	FY22	FY23	FY24	FY25	FY26	FY27F	FY28F
Central government balance	-6.7	-6.4	-5.7	-4.9	-4.4	-4.3	-4.2
Government revenue	9.4	9.0	9.6	9.7	9.9	9.4	9.8
Government expenditure	16.2	15.4	15.3	14.6	14.3	13.8	14.0
Central primary balance	-3.3	-3.0	-2.0	-1.4	-0.8	-0.6	-0.5
Consolidated deficit	-9.5	-9.1	-8.5	-7.7	-7.2	-7.1	-7.0
Memo							
State fiscal deficit	-2.8	-2.7	-2.8	-2.8	-2.8	-2.8	-2.8
Public sector debt/GDP	83.0	81.5	81.1	80.3	80.8	79.8	79.5

Source : Ministry of Finance, Deutsche Bank forecasts

Centre's quality of fiscal spending has improved significantly, but has peaked out

An indicator to track in this regard is the **revenue deficit/fiscal deficit ratio** (lower the ratio, the better). Since FY17 (which recorded a low of 59.1%), this ratio moved up to a peak of nearly 80% in FY21, but it has improved once again and fallen below 40%. Revenue deficit/fiscal deficit moderated to 33.8%, from 35.8% in FY25. Beyond this, we see little scope for this ratio to improve significantly, given that the centre's capex spending allocation is reaching a peak, while it may be difficult to reduce revenue expenditure further as most of these expenditures are by definition "committed expenditure". Indeed, the authorities are targeting a revenue deficit/fiscal deficit ratio of 34.9% in FY27, higher than 33.8% in FY26.

Figure 17: Centre's quality of fiscal consolidation has improved significantly



Source : Various budget documents, RBI, MOF and Deutsche Bank

Short-term fiscal projections

Since the peak deterioration of the fiscal and debt dynamic in FY21, owing to the adverse impact of the Covid-19 pandemic, significant progress has been made on the fiscal front, both at central and state government level.

- **Centre's fiscal deficit** has come down to 4.4% of GDP in FY26 and the government has set a target of 4.3% of GDP for FY27. **Despite the oil-price shock, we think the central government has enough buffers to be able to meet the FY27 fiscal deficit target of 4.3% of GDP.** The below table shows our latest projections for FY27 over the FY26 outturn.



Figure 18: Latest FY27 fiscal projections: Deutsche Bank vs. Government estimates

	Key items	2024-25 Actual	2025-26 RE	2026-27 DB Estimate	2026-27 Budget Estimate	2024-25 Actual	2025-26 RE	2026-27 DB Forecast	2026-27 Budget Estimate	FY26 RE over FY25 Actual	FY27 DB est over FY26 RE	FY27 BE over FY26 RE
S. no		INR bn	INR bn	INR bn	INR bn	% of GDP	% of GDP	% of GDP	% of GDP	% yoy	% yoy	% yoy
1	Revenue Receipts (2+3)	30366	33423	35280	35332	9.55	9.67	9.20	9.0	10.1	5.6	5.7
2	Tax Revenue (Net to Centre)	25000	26747	27840	28669	7.86	7.74	7.26	7.29	7.0	4.1	7.2
3	Non Tax Revenue	5366	6677	7439	6662	1.69	1.93	1.94	1.7	24.4	11.4	-0.2
4	Recovery of Loans	246	302	307	384	0.08	0.09	0.08	0.1	22.6	1.6	27.2
5	Other Receipts (disinvestments)	172	338	499	800	0.05	0.10	0.13	0.2	96.7	47.3	136.4
6	Total Receipts (1+4+5)	30784	34064	36085	36515	9.68	9.86	9.41	9.29	10.7	5.9	7.2
7	Total Expenditure (8+10)	46529	49648	52728	53473	14.63	14.37	13.75	13.61	6.7	6.2	7.7
8	On Revenue Account	36009	38691	41300	41255	11.32	11.20	10.77	10.5	7.4	6.7	6.6
9	of which: Interest Payments	11156	12743	14189	14040	3.51	3.69	3.70	3.6	14.2	11.3	10.2
10	On Capital Account	10520	10958	11428	12218	3.31	3.17	2.98	3.1	4.2	4.3	11.5
11	Revenue Deficit (8-1)	5643	5268	6021	5923	1.77	1.52	1.57	1.51			
12	Fiscal Deficit (7-6)	15744	15585	16643	16958	4.95	4.51	4.34	4.31			
13	Primary Deficit (12 - 9)	4589	2842	2454	2918	1.44	0.82	0.64	0.74			
	<i>Memo item</i>											
14	Nominal GDP, %yoy	318073	345472	383473	393004					8.6	11.0	13.8

Source : Various budget documents and Deutsche Bank

- **States' combined fiscal deficit** is likely to stay steady at 2.8% of GDP, as per our conservative baseline assumption. Prior to Covid, states' fiscal deficit was at 2.4% of GDP and 2.6% of GDP in FY19 and FY20, respectively.
- **General government deficit** is likely to moderate to 7.1% of GDP in FY27, from 8.5% of GDP in FY24.
- **General government debt** is likely to moderate to 79.8% of GDP in FY27, due to higher-than-expected nominal GDP growth, from 80.3% of GDP in FY26.
- **Central government deficit with and without EBR (Extra Budgetary Resources):** Prior to the Covid-19 pandemic, the central government accounted certain Extra Budgetary Resources as off-balance sheet items, which ideally should have been calculated as on-balance sheet expenditure, resulting in a higher-than-reported fiscal deficit. For example, in FY19, when the reported fiscal deficit was 3.4% of GDP, the EBR in that fiscal year was 0.9% of GDP; therefore, incorporating the EBR would have resulted in fiscal deficit rising to 4.3% of GDP. In FY17 and FY18, the reported fiscal deficit was 3.5% of GDP, but adding 0.5% of GDP of EBR for both those years, the augmented fiscal deficit of the central government stood at 4.0% of GDP. Post FY22, the central government has taken all expenditure on balance sheet and the practice of providing off-balance sheet EBR has stopped, in a bid to promote fiscal transparency. We assume that, in the coming years, the centre's fiscal deficit will not fall below 4.0% of GDP, which would be similar to the pre-pandemic low of 4.0% of GDP (after adding 0.50% of GDP worth EBR to the headline fiscal deficit of 3.5% of GDP). Assuming the states' fiscal deficit stabilizes at 2.8% of GDP, this will likely result in general government deficit stabilizing at the current 6.8% of GDP within the forecast horizon.

In the rest of this report, we use our own economic forecasts to present a debt sustainability analysis of the general government to ascertain the medium-term debt/GDP trajectory under various scenarios.

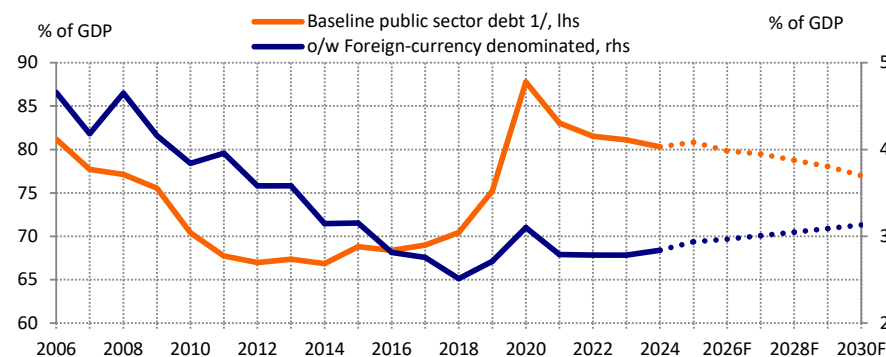


Baseline assumptions and projection

- Real GDP grows around 6.5% average in the rest of this decade (till FY31); nominal GDP growth averages about 10.5-11% during the forecasting period;
- With a flexible inflation targeting framework firmly in place, CPI inflation averages in the 4.0-4.5% range for the rest of this decade; we have assumed GDP deflator (mix of CPI and WPI) to average 4.5% in the years ahead;
- Real interest rate on public debt works out to 2.5% on an average (derived from 7.0% nominal rate and 4.5% GDP deflator-based inflation);
- We assume general government deficit (centre + state) to trough at 6.8% of GDP through the forecasting period, with the centre's fiscal deficit stabilizing around 4.0% of GDP, and 2.8% of GDP in case of states. There is scope for general government deficit to be lowered further beyond this, but we have assumed for now that, given the need for supporting growth through fiscal impulse and accommodating 8th Pay Commission recommended higher wages in the coming years, the combined deficit may not fall significantly below the 7.0% of GDP mark (pre-Covid 10-year average was 6.9% of GDP).
- The primary deficit eases in the period ahead, averaging close to 2.2% of GDP; primary spending rises by 5-6% in real terms, while revenues rise 10.5-11% on an average in nominal terms, in line with the projected nominal GDP growth for the rest of this decade.
- The rupee maintains a depreciation bias in the medium term; but given India's relatively small public external debt burden (2.8-3.0% of GDP), the exchange rate assumption matters little in the debt sustainability exercise.

Under this baseline scenario, public debt is projected to decline to about 75.1% of GDP by 2030 (FY31), from about 80.3% in FY26. This outcome is primarily driven by the assumption that the real growth-real interest rate differential stabilizes at 4-4.5% during the forecasting period. **India's debt burden does not look onerous under this scenario, but the interest cost of servicing the debt remains high during the forecasting period, accounting for 16-16.5% of total consolidated spending (or about a quarter of central government spending).** Given that India needs to ramp up total public spending on key sectors such as health, education and defense, the importance of reducing debt and interest costs is all too apparent.

Figure 19: Baseline Debt/GDP ratio - on a steady downward trajectory



Source : MOF, Deutsche Bank. Note: 1/ Combined Central and State level debt



Problems associated with long-term projections - a cautionary tale

A word of caution - medium-term forecasts are subject to various risks which are difficult to anticipate and therefore the actual outcome could be appreciably different from the projected path. The Covid-19 pandemic is a testimony to this fact. A few years prior to the Covid-19 pandemic in 2020, the central government had formed an expert committee in 2016 to recommend a desirable fiscal and debt consolidation path for India over the medium term.

The FRBM committee recommended the central government continues on the path of fiscal consolidation, with the medium-term goal of bringing the centre's fiscal deficit down to 2.5% of GDP by FY23 (from 3.5% of GDP in FY17), which would result in the centre's debt/GDP coming down to 40% of GDP (from about 49% in FY18). The committee also recommended a desirable fiscal path for the state governments, so that cumulatively the debt/GDP of states can remain constant at around 20% of GDP throughout the forecasting period. This would require state fiscal deficit to reduce from 3.0% of GDP in FY17 - by about 1% of GDP - to 2.0% by FY23. If this was achieved, then India's overall public debt/GDP ratio would have moderated to about 60% of GDP by FY23, a 10% drop from FY18 levels (general government deficit was close to 70% of GDP in FY18).

But then Covid-19 happened in 2020, and the seismic growth shock pushed general government debt/GDP close to 90% in FY21, a 20% increase from FY18 levels. By FY23, public debt moderated but it was still above 81.5% of GDP. India's public debt/GDP is expected to moderate to 75.1% by 2030 (FY30/31), but that will still be higher than the 70% debt/GDP that was recorded in FY18. Or, in other words, a seismic shock like the Covid-19 pandemic set India's debt/GDP trajectory back by a decade. This is true not just for India, but for most countries around the world.

Figure 20: FRBM committee recommended fiscal target

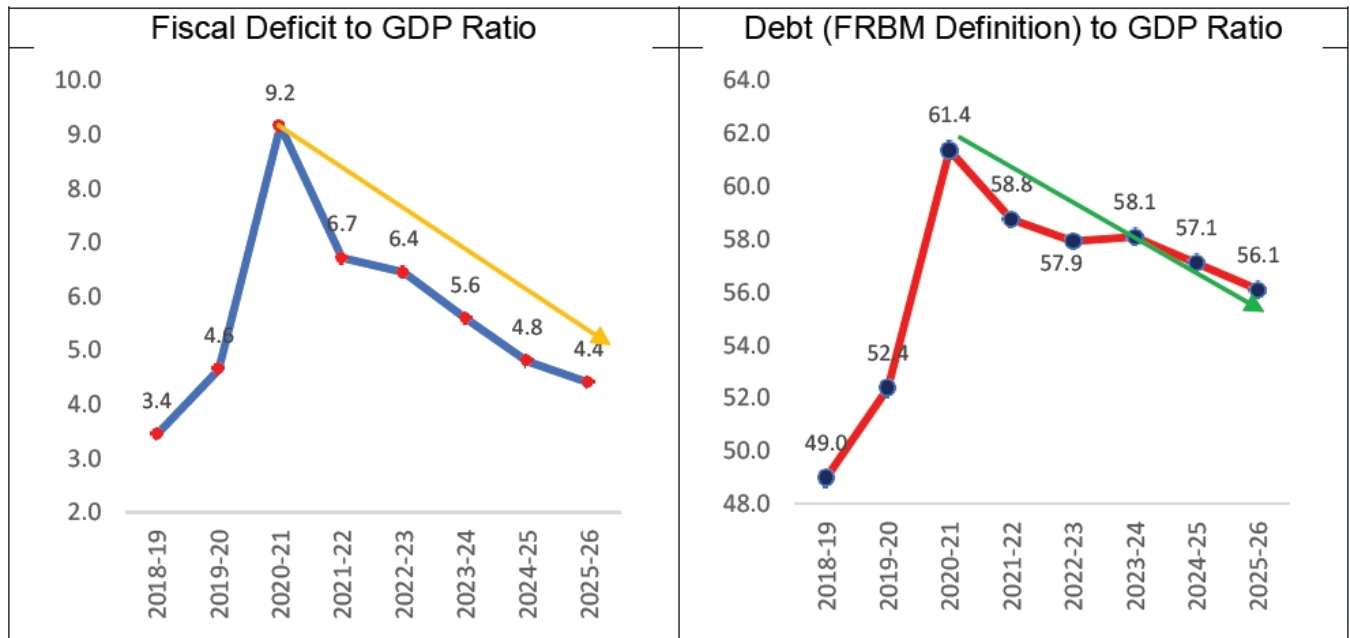
% of GDP	FRBM committee recommended target			DB estimate
	Central govt.	State govt. fiscal	General govt.	General govt.
FY17	3.5	3.0	6.5	6.5
FY18	3.0	2.8	5.8	6.2
FY19	3.0	2.7	5.7	6.0
FY20	3.0	2.5	5.5	6.0
FY21	2.8	2.3	5.1	6.0
FY22	2.6	2.2	4.8	6.0
FY23	2.5	2.0	4.5	6.0

Source: FRBM Review Committee Report, Deutsche Bank

Therefore, it is important to recognize that the baseline scenario of debt/GDP projections can easily be derailed, if there is a large growth shock within the forecasting horizon. **The baseline debt/GDP projection does not incorporate any domestic or external shock. But, as has been illustrated vividly through the last decade, India's fiscal regime is vulnerable to various shocks.** It has also been seen that, even after an extended high-growth period, fiscal consolidation proves to be transitory when growth shocks materialize, or an external shock (global tariff shock, for example) weakens the fiscal position indirectly by impacting growth and revenue adversely.

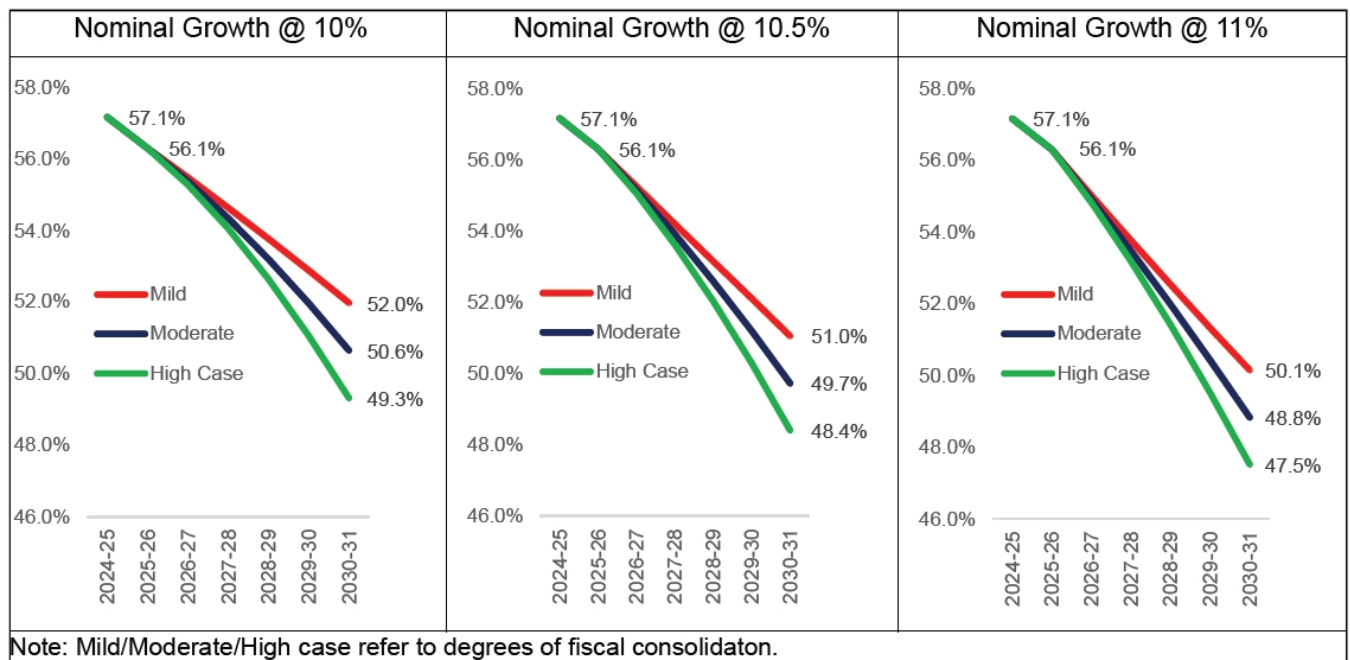


Figure 21: Central government fiscal deficit and debt/GDP



Source : Ministry of Finance - Government of India

Figure 22: Central govt. debt/GDP projection by Ministry of Finance, Government of India



Source : Ministry of Finance



Keeping this in mind, in the rest of this note, **we introduce some shocks to our baseline scenario and see how the projections change** if indeed some of the key macro variables associated with debt dynamic were to take a turn for the worse in the coming years.

Stress Tests

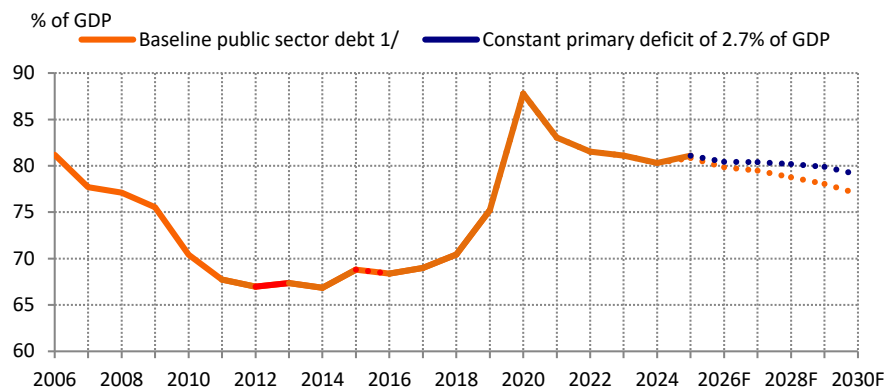
Shock 1: No change in fiscal effort - primary deficit stays around 2.7% of GDP

If the primary deficit stays at around 2.7% of GDP (higher than our baseline assumption of about 2.2% of GDP), and the growth/interest rate nexus discussed above remains in place, India's debt path would still be on downward path, but the **debt/GDP ratio would fall to around 77.3% of GDP by 2030 (FY30/31)**. A flatter path of debt adjustment should be a source of worry as fragility to shocks would rise.

State fiscal finances pose a big risk to the overall fiscal consolidation agenda.

While on an aggregate basis, states' combined fiscal deficit has moderated once again below 3% of GDP (after rising to 4.1% of GDP in FY21 due to the impact of Covid-19), there is a large degree of heterogeneity and vulnerability regarding the fiscal health of certain states, which warrant close monitoring. **The states that would likely remain the most vulnerable to debt sustainability risks are Punjab, West Bengal, Bihar, Rajasthan and Kerala, given their weak fiscal and debt metrics even prior to the Covid-19 crisis, as per our analysis.**

Figure 23: Debt dynamic favorable to achieve decline in debt path even with modest fiscal effort



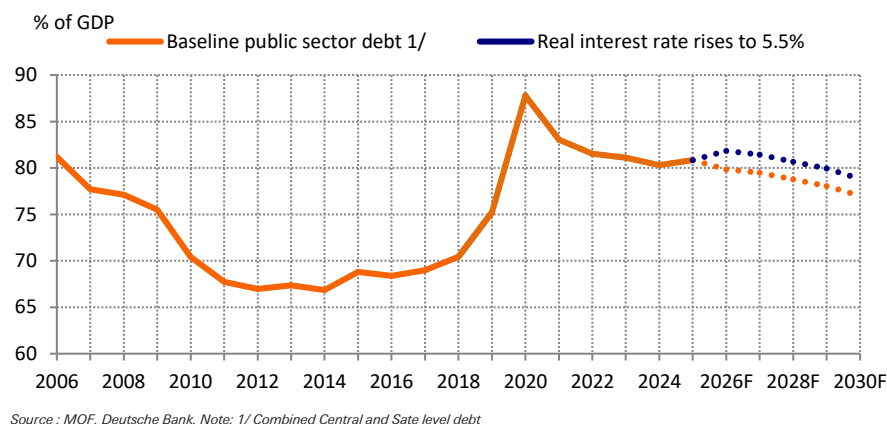
Source : MOF, Deutsche Bank. Note: 1/ Combined Central and State level debt

Shock 2: Interest rate shock - real interest rate rises to about 5.5% on an average

If CPI inflation increases above the 6% upper threshold inflation target range in the years ahead, RBI would be compelled to hike the policy rate, which may increase the real interest rate to mid single digit, when measured in terms of WPI and GDP deflator. Under the interest rate shock scenario, real interest rate paid on public debt is raised by one standard deviation of the past average; **under this scenario, the debt trajectory flattens and debt/GDP comes down only to 76.9% of GDP by FY31 (2030).**



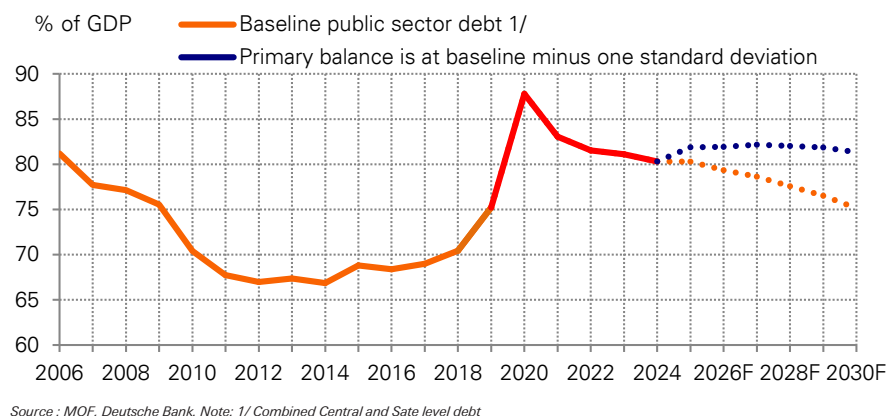
Figure 24: A rise in real interest rate could cause adverse debt dynamic



Shock 3: Fiscal shock - primary deficit rises to 3.5% of GDP within the forecasting period instead of coming down to 2.2% of GDP

If the primary deficit rises to around 3.5% of GDP (higher than our baseline assumption of about 2.2% of GDP), and the growth/interest rate nexus discussed above remains in place, India's debt path would flatten considerably, with the **debt/GDP ratio likely to remain unchanged at about 81.3% of GDP by 2030 (FY31)**.

Figure 25: A fiscal shock will affect the debt dynamic

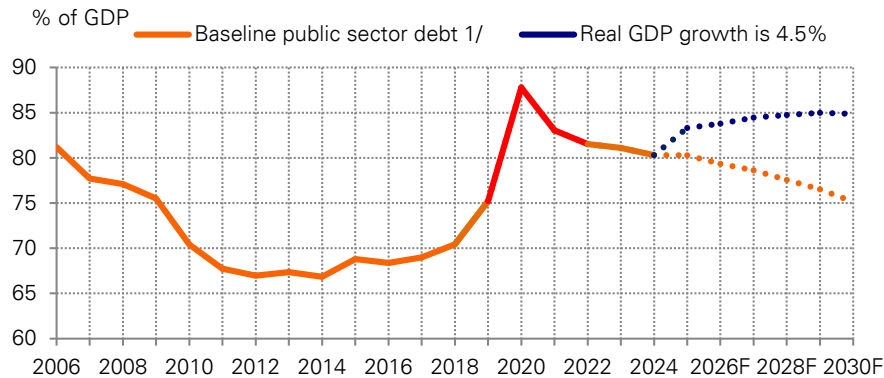


Shock 4: Growth shock - real GDP growth moderates to 4.5%

Here we lower real GDP growth by 200bps, which pegs economic growth to around 4.5%. The shock impacts the debt path severely, **pushing up the debt ratio to 84.9% by FY31 (2030)**, 10% higher than our baseline estimate of debt/GDP falling to 74.8% by the end of this decade. Indeed, the biggest risk to debt sustainability arises from a growth shock, as was witnessed during the time of Covid-19 pandemic. Of course, under such a scenario, interest rates would ease, real rates would fall, but the overall impact of lower growth and presumably higher fiscal deficit would push the general government debt/deficit significantly higher.



Figure 26: Debt sustainability hinges critically on sustained, strong growth



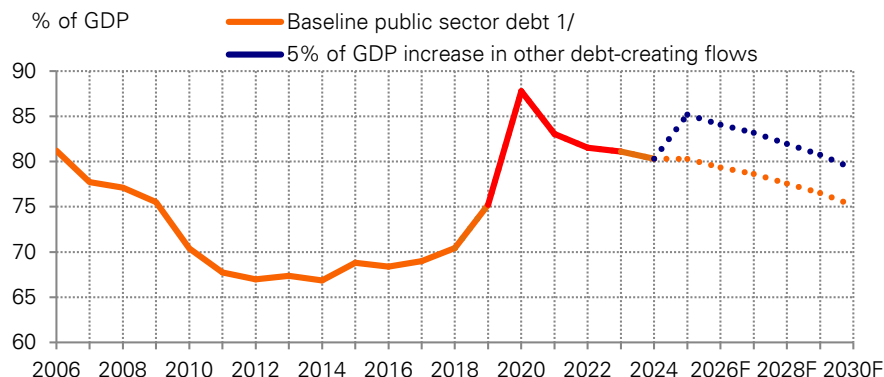
Source : MOF, Deutsche Bank. Note: 1/ Combined Central and State level debt

Shock 5: Contingent liability shock of 5% of GDP

In this scenario, we assume that the government's contingent liabilities would rise suddenly. To account for such a risk, we raise the government's liabilities by 5% of GDP in FY25/26. This first raises the debt/GDP to 85.2% in FY26, thereafter moderating to 79.2% by the end of FY30/31 (2030).

Earlier such risks arose mainly from banking sector NPAs which had deteriorated significantly in the last decade. However, the public sector bank balance sheets have been cleaned up and the NPAs have fallen below 3%, a record low. Now the risk of contingent liabilities arises mostly from state finances. As per RBI data, overall state guarantees amounted to about 3.7% of GDP by FY21. **The guarantees are likely to have moderated to 0.8% of GDP, as per FY23 budget estimate figures are concerned.** If these guarantees (contingent liabilities) are accounted as on balance sheet expenditure at any point of time, then both general-government deficit and debt would go up by a commensurate amount, ceteris paribus.

Figure 27: A one-off increase in liabilities would delay debt reduction



Source : MOF, Deutsche Bank. Note: 1/ Combined Central and State level debt

While the centre has stopped the practice of providing off-balance sheet resources (Extra Budgetary resources or EBR) and now everything is accounted as on-balance sheet expenditure, a similar accounting treatment by states at a later point of time could increase deficit and debt, a non-trivial risk, in our view. Recall that in past years, even before the Covid-19 crisis, a decision to accommodate the interest



burden of the UDAY scheme as an on-balance sheet expenditure item along with the costs of farm loan waivers given by certain states deteriorated the health of the states' fiscal balances materially, a risk which exists even today, given the guarantees that exist in the form of contingent liabilities.

Figure 28: 'Freebies' announced by the states in 2022-23

	% of GSDP	% of Revenue Receipts	% of Own Tax Revenue
Andhra Pradesh	2.1	14.1	30.3
Bihar	0.1	0.6	2.7
Haryana	0.1	0.6	0.9
Jharkhand	1.7	8.0	26.7
Kerala	0.0	0.0	0.1
Madhya Pradesh	1.6	10.8	28.8
Punjab*	2.7	17.8	45.4
Rajasthan	0.6	3.9	8.6
West Bengal	1.1	9.5	23.8

Source : RBI and Deutsche Bank. Note: * Dhasmana, I. (2022). "Not all states are so financially weak that they can't announce freebies". Business Standard. April 2022.

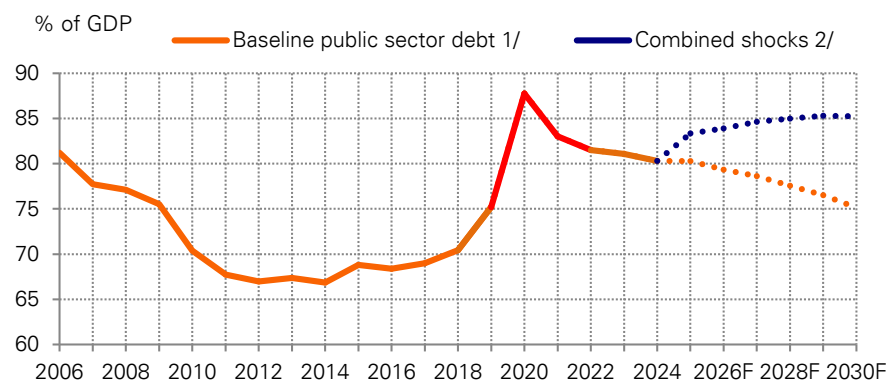
Figure 29: Guarantees issued by State Governments (% of GSDP)

	Bihar	Kerala	Punjab	Rajasthan	West Bengal	Andhra Pradesh	Uttar Pradesh	Haryana
2017-18	1.1	2.5	4.5	7.5	0.9	4.6	6.3	2.2
2018-19	1.0	3.4	0.9	7.6	0.6	6.2	6.9	2.6
2019-20	0.9	3.2	4.1	8.1	0.5	8.1	6.7	2.7
2020-21	2.6	4.6	5.3	8.2	0.6	11.0	7.9	2.3
2021-22 (RE)	3.6	-	5.7	-	1.1	9.8	-	-

Source : RBI and Deutsche Bank. Note: RE refers to revised estimates.

Shock 6: Combined shock - What if shocks were multiple in nature? There could be occasions when a growth shock eventually leads to a fiscal shock which then translates into an interest rate shock. What happens to the debt profile in case of such a combined shock scenario? With this in mind, we construct a scenario incorporating shocks half the size of the ones discussed above but in conjunction with one another. We find that the impact on debt is immediate and unambiguously adverse (the **debt/GDP ratio rises to 85.2% by FY31**). While the probability of such a manifestation is low, it is important to be cognizant of such eventualities.

Figure 30: Debt path can turn higher if hit by multiple shocks



Source : MOF, Deutsche Bank. Note: 1/ Combined Central and State level debt



India's debt servicing ability remains healthy

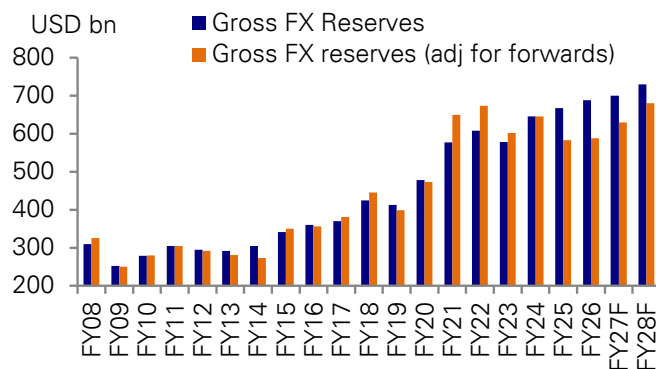
As our exercise shows, public debt/GDP is likely to be on a sustained downtrend due to positive growth-interest differential and there is little risk associated with the funding of the public debt as bulk of the debt is in domestic currency. Foreign currency denominated debt is small - less than 5% of GDP - and even within that more than half of it is owed to multilateral institutions.

India's macro outlook - both in the short and medium term - remains bright, with likelihood of sustained strong growth outperformance amidst anchored inflation, positive real rates, continued fiscal consolidation, healthy external and domestic financial sector and expected macro dividend from ongoing supply-side reforms. As our exercise shows, public debt/GDP seems likely to be on a sustained downtrend due to positive growth-interest differential and there is little risk associated with the funding of the public debt as bulk of the debt is in domestic currency.

Our analysis reveals that, despite GST rates rationalisation of 2025 and an oil price shock this year, **it is possible for India to lower its debt/GDP to 75.1% of GDP by FY31**, provided nominal GDP growth averages around 10.5-11% during the forecasting period. In our view, barring some unforeseen shocks, a 10.5-11% nominal GDP growth assumption is realistic based on an average 6.5% real GDP growth and 4-4.5% inflation (in terms of GDP deflator) over the forecasting period.

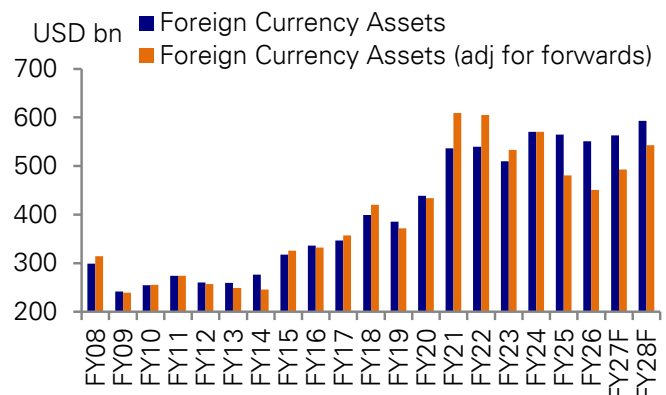
Our forecast is similar to IMF's estimate, which sees India's public sector debt/GDP moderating to 75.4% by 2031 (see, "[Staff Report for the 2025 Article IV Consultation for India](#)", Nov 2025), but is slightly higher compared to RBI's forecast of 73.4% over the same forecasting horizon (see, "[The Shape of Growth Compatible Fiscal Consolidation](#)", Feb 2024).

Figure 31: Gross Foreign Exchange Reserves



Source : CEIC, RBI and Deutsche Bank. Note: FY27,28 F refer to DB forecasts.

Figure 32: Foreign Currency Assets



Source : CEIC, RBI and Deutsche Bank. Note: FY27,28 F refer to DB forecasts.



Figure 33: India's key external debt (% unless indicated otherwise)

End-March	External Debt (US\$ billion)	Ratio of External Debt to GDP	Debt Service Ratio	Ratio of Foreign Exchange Reserves to Total Debt	Ratio of Concessional Debt to Total Debt	Ratio of Short-Term Debt to Foreign Exchange Reserves	Ratio of Short-Term Debt (original maturity) to Total Debt
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
1991	83.8	28.3	35.3	7.0	45.9	146.5	10.2
1996	93.7	26.6	26.2	23.1	44.7	23.2	5.4
2001	101.3	22.1	16.6	41.7	35.4	8.6	3.6
2006	139.1	17.1	10.1 [#]	109.0	28.4	12.9	14.0
2007	172.4	17.7	4.7	115.6	23.0	14.1	16.3
2008	224.4	18.3	4.8	138.0	19.7	14.8	20.4
2009	224.5	20.7	4.4	112.2	18.7	17.2	19.3
2010	260.9	18.5	5.8	106.9	16.8	18.8	20.1
2011	317.9	18.6	4.4	95.9	14.9	21.3	20.4
2012	360.8	21.1	6.0	81.6	13.3	26.6	21.7
2013	409.4	22.4	5.9	71.3	11.1	33.1	23.6
2014	446.2	23.9	5.9	68.2	10.4	30.1	20.5
2015	474.7	23.8	7.6	72.0	8.8	25.0	18.0
2016	484.8	23.4	8.8	74.3	9.0	23.2	17.2
2017	471.0	19.8	8.3	78.5	9.4	23.8	18.7
2018	529.3	20.1	7.5	80.2	9.1	24.1	19.3
2019	543.1	19.9	6.4	76.0	8.7	26.3	20.0
2020	558.3	20.9	6.5	85.6	8.8	22.4	19.1
2021	573.4	21.1	8.2	100.6	9.0	17.5	17.6
2022	618.8	19.9	5.2	98.1	8.3	20.0	19.7
2023	623.9	19.1	5.3	92.7	8.2	22.2	20.6
2024 PR	668.9	19.2	6.7	96.6	7.4	19.7	19.1
2025 PR	736.4	19.8	6.6	90.8	6.9	20.1	18.3
end-June 2025 PR	746.8	19.7	6.6	93.5	7.1	19.3	18.0
end-September 2025 PR	747.2	20.1	6.0	93.7	7.0	19.7	18.5
end-December 2025 P	765.5	20.4	5.8	89.8	6.7	21.9	19.7

Source : RBI. Note: PR: Partially Revised; P: Provisional. #: Works out to 6.3 per cent with the exclusion of India Millennium Deposits (IMDs) repayments of US\$ 7.1 billion and pre-payment of external debt of US\$ 23.5 million. Note 1: Figures are rounded to one decimal place. Note 2: Ratio of external debt to GDP from end March 2024 (P) onwards is based on the new GDP series (Base Year: 2022/23)



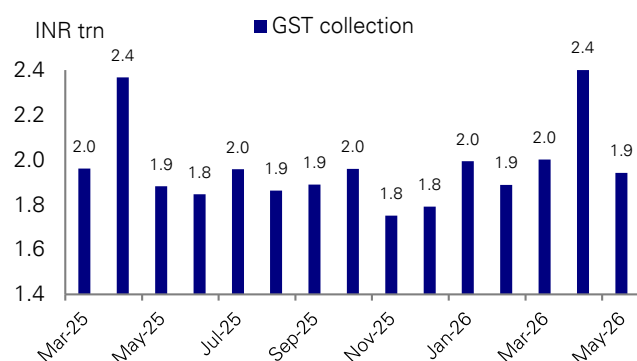
Appendix A: Tax collection trend

Figure 34: Monthly tax collection trend including Apr'26 (% yoy)

Revenue receipts, %yoy	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26	FY25	FY26
Gross Tax Revenue	8.4	13.8	-2.8	32.4	10.1	-19.2	2.1	-1.9	9.6	6.0
Corporate Tax	0.2	76.4	61.2	22.7	180.8	-96.2	7.7	17.4	8.3	11.4
Income Tax	33.7	26.9	5.8	-9.2	12.6	-45.2	-4.3	6.8	17.0	0.0
Central Goods and Service Tax	8.9	8.1	0.7	6.9	10.6	5.9	10.5	37.7	10.8	6.4
Union Territories Goods and Service Tax	56.5	61.6	253.5	72.1	31.3	90.1	-51.6	268.1	7.2	32.5
Goods and Service Tax Compensation Cess	-5.5	-39.7	-70.3	-55.1	-57.5	-63.5	-100.2	-101.5	6.5	-33.4
Customs	29.0	13.4	-36.3	229.7	19.7	50.9	-19.7	25.0	0.1	13.4
Union Excise Duties	10.3	6.9	17.5	14.6	10.9	23.4	26.5	-1246.2	-1.7	13.9
Service Tax	-145.2	12300.0	-322.2	-51.7	-110.3	-15850.0	1050.0	115.4	-103.5	11726.7
Other Taxes	-33.0	-53.2	-33.0	4.1	11.2	39.4	19.8	66.6	5.8	-14.5
Less: Transfer to state	14.1	14.0	14.1	14.0	-41.3	19.0	19.0	7.4	13.9	8.1
Net Tax Revenue	7.1	12.9	-13.6	36.5	156.1	-54.5	-1.3	-5.9	7.5	4.9
Memo item										
Direct tax	11.0	41.7	19.3	10.7	33.1	-53.8	2.1	8.6	12.9	5.2
Indirect tax	1.5	-5.0	-19.4	96.7	-3.1	6.9	2.2	-11.6	5.4	7.1

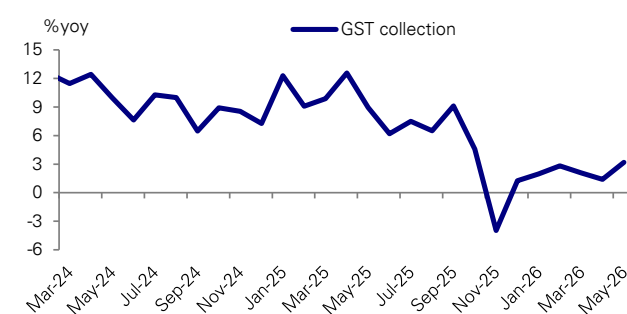
Source : Controller General of Accounts and Deutsche Bank

Figure 35: Monthly GST collection - INR trn



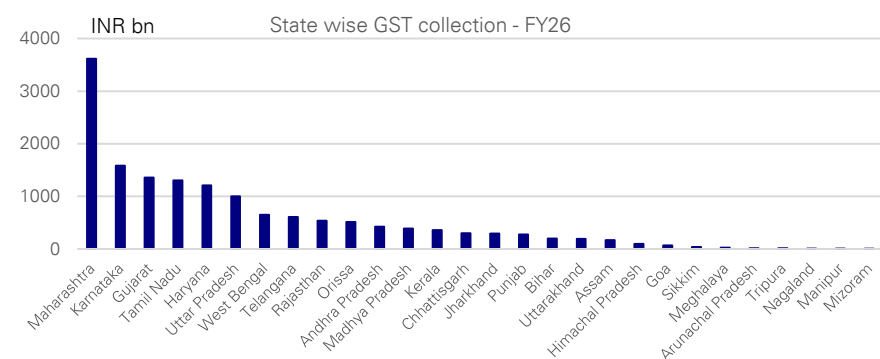
Source : CEIC and Deutsche Bank

Figure 36: Monthly GST collection - %yoy



Source : CEIC and Deutsche Bank

Figure 37: State-wise GST collection - FY26 (INR bn)



Source : CEIC and Deutsche Bank



Appendix B: A snapshot of state fiscal finances

Figure 38: Ranking of 17 key states based on FY26 BE

States	Ranking based on FY26 budget estimates	10 year SDL yield (%)						Fiscal deficit	Own tax revenue	Debt	Interest payment to revenue receipts
		2020-21	2021-22	2022-23	2023-24	2024-25	2025-26	% of GSDP	% of GSDP	% of GSDP	%
Maharashtra	1	7.0	6.8	7.9	-	-	-	2.8	7.8	19	11.5
Jharkhand	2	6.8	6.9	-	-	-	-	2.0	6.3	25	5.1
Telangana	3	7.0	-	-	-	-	-	3.0	8.1	28	8.4
Uttar Pradesh	4	6.9	7.0	-	-	-	7.5	3.0	9.6	29	9.7
Orissa	5	-	-	-	-	-	-	3.2	6.2	20	2.8
Chhattisgarh	6	-	-	-	-	-	-	3.8	8.5	30	6.7
Gujarat	7	6.7	7.0	7.8	-	-	7.4	2.0	5.3	18	12.4
Karnataka	8	6.6	7.0	-	-	-	7.4	2.9	6.8	27	15.6
Haryana	9	7.4	7.0	7.8	7.4	-	-	2.7	6.8	30	20.5
Tamil Nadu	10	6.7	7.0	7.8	-	7.4	7.4	3.0	6.2	29	20.8
Bihar	11	-	7.2	-	-	-	-	3.0	5.4	37	8.8
Madhya Pradesh	12	6.9	6.9	7.9	7.4	7.4	-	4.7	6.4	31	9.8
Rajasthan	13	6.7	7.0	7.8	7.4	-	-	4.3	7.2	36	13.6
Kerala	14	7.2	7.1	-	-	-	-	3.2	6.4	36	20.9
Punjab	15	6.8	7.0	7.9	-	-	-	3.8	7.1	46	22.4
Andhra Pradesh	16	6.5	-	7.7	7.4	-	-	4.4	6.0	35	16.1
West Bengal	17	6.7	7.1	7.8	-	-	-	3.6	5.5	49	18.3
Avg. of 17 states		6.8	7.0	7.8	7.4	7.4	7.4	3.4	6.8	32.0	13.8

Source : RBI, Various state budget documents and Deutsche Bank.

Figure 39: Fiscal ranking of states over time

States	Rank based on 2004-08 avg. fiscal data	Rank based on 2008-10 avg. fiscal data	Rank based on FY13 fiscal data	Rank based on FY14 fiscal data	Rank based on FY15 fiscal data	Rank based on FY16 fiscal data	Rank based on FY18 fiscal data	Rank based on FY19 fiscal data	Rank based on FY20 fiscal data	Rank based on FY21 fiscal data	Rank based on FY22 fiscal data	Rank based on FY23 fiscal data	Rank based on FY24 fiscal data	Rank based on fiscal data b/w 2004 to 2016	States
Maharashtra	6	6	1	2	4	4	1	1	1	2	2	-	1	1	Karnataka
Chhattisgarh	4	1	4	10	2	1	2	3	9	3	1	1	9	2	Tamil Nadu
Orissa	8	4	2	7	9	6	3	4	10	1	3	-	2	3	Chhattisgarh
Karnataka	1	3	5	1	1	2	4	5	2	7	7	2	4	4	Maharashtra
Uttar Pradesh	14	16	10	12	14	15	5	2	3	4	11	4	7	5	Orissa
Goa	11	13	12	13	10	12	6	8	-	-	-	-	-	6	Andhra Pradesh
Gujarat	9	10	7	9	6	8	7	6	5	6	5	3	5	7	Madhya Pradesh
Tamil Nadu	3	2	3	3	3	3	8	7	6	12	12	-	12	8	Gujarat
Madhya Pradesh	10	7	8	5	5	7	9	10	7	9	10	-	10	9	Haryana
Haryana	2	14	11	8	11	13	10	9	15	10	9	7	3	10	Jharkhand
Rajasthan	13	12	9	14	15	17	11	15	14	16	14	-	13	11	Punjab
Kerala	7	8	16	16	13	9	12	12	8	13	15	6	11	12	Kerala
Bihar	15	15	15	15	17	10	13	13	12	11	13	11	17	13	Goa
Punjab	12	11	14	11	8	11	14	17	16	17	17	10	15	14	Rajasthan
Andhra Pradesh	5	5	6	4	12	5	15	14	13	14	8	8	16	15	Uttar Pradesh
West Bengal	16	17	17	17	16	16	16	16	17	15	16	9	14	16	Bihar
Jharkhand	17	9	13	6	7	14	17	11	11	8	4	5	8	17	West Bengal

Source : Deutsche Bank



Figure 40: Fiscal ranking of states based on FY24 actual data & comparison with FY25 RE and FY26 BE (best to worst)

States	Ranking based on FY24 actual fiscal data	Ranking based on FY25 revised estimates	Ranking based on FY26 budget estimates
Maharashtra	1	1	1
Orissa	2	4	5
Jharkhand	3	2	2
Gujarat	4	5	7
Karnataka	5	8	8
Uttar Pradesh	6	7	4
Chhattisgarh	7	6	6
Telangana	8	3	3
Madhya Pradesh	9	11	12
Haryana	10	9	9
Kerala	11	13	14
Tamil Nadu	12	10	10
Bihar	13	14	11
Rajasthan	14	12	13
Andhra Pradesh	15	15	16
West Bengal	16	17	17
Punjab	17	16	15

Source : Various state budget documents and Deutsche Bank. Note: RE refers to revised estimates and BE refers to budget estimates.

Figure 41: Fiscal snapshot of states

States	Ranking based on FY24 actual fiscal data	2023-24 Actuals				2024-25 RE				2025-26 BE			
		Fiscal deficit (% of GSDP)	Own tax revenue (% of GSDP)	Debt (% of GSDP)	Interest payment to revenue receipts (%)	Fiscal deficit (% of GSDP)	Own tax revenue (% of GSDP)	Debt (% of GSDP)	Interest payment to revenue receipts (%)	Fiscal deficit (% of GSDP)	Own tax revenue (% of GSDP)	Debt (% of GSDP)	Interest payment to revenue receipts (%)
Maharashtra	1	2.2	7.5	18.5	10.6	2.9	8.1	18.6	10.2	2.8	7.8	19.0	11.5
Orissa	2	1.7	6.3	19.7	2.9	3.1	6.3	20.6	2.6	3.2	6.2	20.3	2.8
Jharkhand	3	1.4	6.1	27.2	7.8	2.3	6.7	25.5	6.5	2.0	6.3	25.4	5.1
Gujarat	4	1.0	5.5	18.3	12.2	1.9	5.5	17.8	12.4	2.0	5.3	17.9	12.4
Karnataka	5	2.6	6.4	25.3	13.2	2.9	6.3	25.4	14.2	2.9	6.8	26.5	15.6
Uttar Pradesh	6	3.2	7.5	30.9	10.1	3.4	7.6	30.0	9.8	3.0	9.6	28.7	9.7
Chhattisgarh	7	5.3	7.6	27.3	6.6	5.3	8.1	29.2	7.4	3.8	8.5	29.8	6.7
Telangana	8	3.4	7.6	27.0	14.4	2.9	8.0	27.3	8.8	3.0	8.1	27.5	8.4
Madhya Pradesh	9	3.3	6.7	30.1	9.9	4.2	6.4	30.8	10.3	4.7	6.4	31.3	9.8
Haryana	10	2.9	6.7	31.3	21.3	2.7	6.8	30.2	21.9	2.7	6.8	29.9	20.5
Kerala	11	3.0	6.5	36.3	21.7	3.5	6.4	36.2	22.4	3.2	6.4	35.5	20.9
Tamil Nadu	12	3.3	6.1	31.0	20.2	3.3	6.2	27.3	20.5	3.0	6.2	29.2	20.8
Bihar	13	4.2	5.7	38.7	9.1	9.2	5.6	38.0	8.4	3.0	5.4	36.8	8.8
Rajasthan	14	4.3	6.2	37.4	16.8	4.1	7.1	37.0	14.9	4.3	7.2	35.9	13.6
Andhra Pradesh	15	4.4	6.0	34.6	17.0	4.6	6.0	35.5	17.6	4.4	6.0	35.4	16.1
West Bengal	16	3.3	5.4	39.7	21.3	4.0	5.5	40.0	20.3	3.6	5.5	48.9	18.3
Punjab	17	4.4	6.3	46.5	25.3	4.5	7.2	46.9	23.1	3.8	7.1	46.4	22.4
Avg. of 17 states		3.2	6.5	30.6	14.1	3.8	6.7	30.4	13.6	3.3	6.8	30.8	13.1
All States and UTs		2.9	6.5	28.1	12.9	3.5	6.8	28.4	12.3	3.3	7.1	29.2	12.2

Source : Various state budget documents and Deutsche Bank. Note: RE refers to revised estimates and BE refers to budget estimates.



Annexure: Credit ratings methodology

Figure 42: Moody's credit ratings methodology

Sovereign Bond Ratings Sector Scorecard Overview

Factor	Sub-factor	Sub-factor Weighting	Metric/Sub-sub-factor	Metric / Sub-sub-Factor Weighting
Factor: Economic Strength	Growth Dynamics	35%	Average Real GDP Growth $t-4$ to $t+5$	25%
			Volatility in Real GDP Growth $t-9$ to t	10%
	Scale of the Economy	30%	Nominal GDP (US\$ bn) t	30%
	National Income	35%	GDP per Capita (PPP, Int. USD) t	35%
	Adjustment to Factor Score	0 - 9 notches	Other	
Factor: Institutions and Governance Strength	Quality of Institutions	40%	Quality of Legislative and Executive Institutions	20%
			Strength of Civil Society and the Judiciary	20%
	Policy Effectiveness	60%	Fiscal Policy Effectiveness	30%
			Monetary and Macroeconomic Policy Effectiveness	30%
	Adjustment to Factor Score	0 - 3 notches	Government Default History and Track Record of Arrears	
		0 - 3 notches	Other	
Factor: Fiscal Strength	Debt Burden	50% ¹	General Government Debt / GDP t	25%
			General Government Debt / Revenue t	25%
	Debt Affordability	50% ¹	General Government Interest Payments / Revenue t	25%
			General Government Interest Payments / GDP t	25%
	Adjustments to Factor Score	0 - 6 notches	Debt Trend $t-4$ to $t+1$	
			General Government Foreign Currency Debt / General Government Debt t	
			Other Non-Financial Public Sector Debt / GDP t	
			Public Sector Financial Assets and Sovereign Wealth Funds / General Government Debt t	
		0 - 3 notches	Other	
Factor: Susceptibility to Event Risk	Political Risk	Minimum Function ²	Domestic Political and Geopolitical Risk	
	Government Liquidity Risk	Minimum Function ²	Ease of Access to Funding	
			0 - 2 scoring categories	Adjustment to Sub-factor Score High Refinancing Risk
	Banking Sector Risk	Minimum Function ²	Risk of Banking Sector Credit Event (BSCE)	
			Total Domestic Bank Assets / GDP t	
	External Vulnerability Risk	Minimum Function ²	0 - 2 scoring categories	Adjustment to Sub-factor Score
			External Vulnerability Risk	
	Adjustment to Factor Score	0-2 scores	0 - 2 scoring categories	Adjustment to Sub-factor Score

Source : Economic Survey 2020-21, Volume 1



Figure 43: Fitch's credit ratings methodology

Analytical pillar	Structural features	Macroeconomic performance, policies & prospects	Public finances	External finances
SRM weights (%)	53.7	10.9	18.0	17.4
SRM variables	Measure	Impact	Weight (%)	Coefficient
Governance indicators	Latest	Positive	20.4	0.075
GDP per capita	Latest	Positive	12.3	0.040
Share in world GDP	Latest	Positive correlation with size	13.2	0.607
Years since default or restructuring event	Latest	Negative	6.4	-2.481
Broad money supply (% of GDP)	Latest	Positive	1.4	0.185
Overall weight in SRM			53.7	
SRM variables (%)	Measure	Impact	Weight (%)	Coefficient
Real GDP growth volatility	Latest	Negative	4.9	-0.767
Consumer price inflation	3-year centred average	Negative	3.1	-0.056
Real GDP growth	3-year centred average	Positive	2.9	0.093
Overall weight in SRM			10.9	
SRM variables	Measure	Impact	Weight (%)	Coefficient
Gross general govt debt/GDP	3-year centred avg	Negative	8.0	-0.021
General govt interest (% of revs)	3-year centred avg	Negative	4.7	-0.046
General govt fiscal bal./GDP	3-year centred avg	Directional	3.0	0.055
FC govt debt/gross govt debt (%)	3-year centred avg	Negative	2.4	-0.006
Overall weight in SRM			18.0	
SRM variables	Measure	Impact	Weight (%)	Coefficient
Reserve currency flexibility	Latest	Positive	7.8	0.551
Sovereign net foreign assets (% of GDP)	3-year centred avg.	Positive	6.7	0.011
Commodity dependence	Latest	Negative	0.8	-0.003
Foreign exchange reserves (months of CXP) ^a	Latest	Positive	1.3	0.027
External interest service (% of CXR)	3-year centred avg.	Negative	0.7	-0.012
Current account balance + foreign direct investment (% of GDP)	3-year centred avg.	Directional	0.2	0.002
Overall weight in SRM			17.4	

Source : Economic Survey 2020-21, Volume 1



Appendix 1

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AI's tightest bottleneck: Memory chips

June 18, 2026

Figure 1: Key Terminology

DRAM	Dynamic Random-Access Memory: One of two main memory technologies (50-60% of annual industry revenue). DRAM is fast, volatile working memory that serves as a temporary system memory. It offers low latency for frequent reads/writes, but is volatile and must be refreshed constantly, which adds to power use. It primarily sells into data centre, client PC, graphics, industrial and automotive end-markets, and is highly concentrated, dominated by three players: SK Hynix (36%), Samsung (33%), and Micron (24%).
NAND	NOT-AND: One of two main memory technologies (30-50% of annual industry revenue). NAND is non-volatile flash storage used to keep data when power is off. Because of this it is useful for applications in the data centre, client PC, consumer and automotive end markets.
HBM	High Bandwidth Memory: A specialized form of DRAM, built by stacking individual DRAM dies vertically to create the ultra-fast, high-bandwidth memory used directly in AI accelerators and high-performance computing. Key applications of HBM include AI/Machine Learning, Data Centres, Graphics Processing and High-Performance Computing.

Source: Deutsche Bank Research

In 2026, we can no longer ignore the memory market's parabolic rise. The largest memory chipmakers, SK Hynix, Micron and Samsung Electronics are each now valued above \$1trn in market cap – a feat considering they represent over 90% of the DRAM market. What is happening in the memory market to drive these geometric valuations? Behind the ascent, the answer is more worrying than current prices would seem to suggest.

The reason is that the global race for AI is driving a supply-displacement shock into the memory market. And as the industry shifts from standard large language model (LLM) response to autonomous agents that observe, reason, plan and act (Agentic AI), this problem will only get more acute.

While memory has historically suffered a boom-and-bust cycle, the exponential demand from AI suggests the current cycle is more structural than cyclical. In this case, chipmakers' ability to deliver cannot keep pace with the surge in hyperscale demand for memory used in AI. And given it takes years for chipmakers to build fabs for memory chip production, we should expect a multi-year supply shock that constrains the rate of progress in AI. More worryingly for corporations and consumers, the vast amount of memory being swallowed to produce HBMs means less is being allotted to produce other types such as standard DRAM, which is used in the everyday applications the global economy depends on.

As a result, the shortage facing the memory industry has massive repercussions for the individual consumer. Just this month, nine US trade associations representing the automotive, consumer electronics, medical devices, telecoms and retail industries sent a letter to US Treasury Secretary Scott Bessent and Commerce Secretary Howard Lutnick warning of the consequences the AI-fuelled rush for memory will have on the US economy.¹

Lastly, the memory shortage creates a new dimension in US-China tech competition. For China, the repercussions are double-edged. On the one hand, the shortage constrains

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¹ <https://cryptobriefing.com/us-memory-chip-shortage-ai-boom/>



Chinese chipmakers further in the broader logic-chip and AI build-out. China's AI self-sufficiency is constrained by its access to both high-end compute and memory. Without them, its performance in AI processing and inference remains limited. On the other hand, the vacuum of memory chips needed to power traditional devices opens a space for lagging-edge (older, less advanced process node) Chinese memory makers like ChangXin Memory Technologies and Yangtze Memory Technologies Corporation. As Western non-AI corporations are squeezed out of the memory market, the possibility they could turn to Chinese memory manufacturers instead is growing.

1. The squeeze: What's actually happening in memory?

(i) The demand side: AI's bottomless appetite

The problem facing the memory industry is that there is not enough supply to meet the exponential demand for AI inference.

The crisis partially lies in high-bandwidth memory (HBM), a "memory stack" that layers individual DRAM dies into the ultra-fast, 3D memory needed to process AI models. For example, an AI chip, such as an Nvidia GPU, can only compute on data that is loaded onto it; memory does the holding and feeding of that data into the models. Memory has two dimensions: capacity (how much data fits) and bandwidth (how fast it moves). Without memory, chips cannot compute or train AI models – and without it, companies cannot run a trained model to perform an agentic task (inference).

In addition, the fast-evolving AI landscape is beginning to require more expansive memory consumption across other areas of the memory stack beyond. In particular, the shift from human-driven generative AI to inference and agent-driven AI is driving another leg up for DRAM bit demand. Agentic AI systems, unlike traditional AI models, can store and recall past experiences, learn from interactions, and maintain context, leading to more coherent outputs. This requires various types of memory to function effectively, spanning across other memory devices (DDR5, LPDDR, NAND).

Memory also dictates the pace of AI advancement, because companies cannot scale on compute alone. This is the "memory wall" problem: beyond a point, peak computational performance cannot be reached without expanding memory bandwidth. In other words, increasing computing power without increasing memory bandwidth eventually limits the benefits of the compute itself.

Thus, the accelerating need for memory is – alongside compute – a defining bottleneck for AI progress². But memory has arguably become a tighter near-term constraint, given its manufacturing and supply limits: HBM bandwidth for inference-heavy workloads, HBM/packaging capacity, and capacity for traditional DRAM for long-context queries as the determining factor for how fast compute can be deployed. Micron's CEO has called memory the backbone of AI, noting: "as AI models grow larger and inference requirements continue to rise – AI moving from training to inference, and from data centers to the edge – the demand for storage will only increase. It requires greater capacity, higher performance, and lower power consumption."³ As a result, hyperscalers (the largest cloud operators) like Meta, Amazon and Microsoft need vast quantities of HBM both to hold the model and to keep the compute from sitting "idle." In recent years, they have driven the memory market and last year total revenues increased by 35% y/y to a record \$223bn⁴.

² In the AI tech stack, compute is made of GPUs, TPUs, and other AI accelerators, while memory refers to where the data sits. That can include the model's weights, the intermediate activations produced during a forward pass, the KV cache used during inference, etc. Memory has two relevant dimensions: capacity (how much data it can hold) and bandwidth (how fast data can move between memory and the compute units) For those interested in reading more of our work on the compute bottleneck, please reach out.

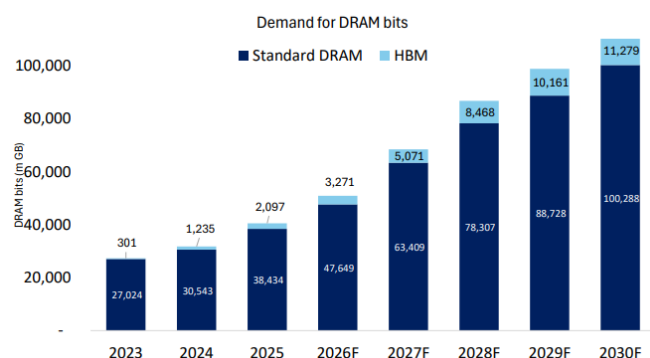
³³ https://news.futunn.com/en/post/74269829/micron-ceo-memory-is-the-overlooked-bottleneck-in-ai-supply?level=1&data_ticket=1781437411630811

⁴ [Global Automotive: DRAMa ahead? - Deutsche Bank Research](#)

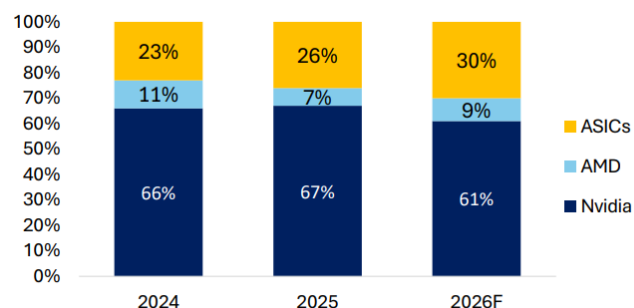


Our equity analysts forecast HBM demand to grow at a ~40% CAGR through 2030, versus a ~21% CAGR in demand for standard DRAM (the chips used for everyday consumer computing and standard cloud databases).⁵ This is because hyperscalers will pay premiums over automotive, consumer-electronics, and industrial buyers, and will sign multi-year lock-ins to secure supply.

Figure 2: Standard DRAM demand grows at +21% CAGR through 2030, vs HBM at +40%



Source: Deutsche Bank equity analyst research.



Source: Deutsche Bank Equity Team. For more, see [here](#) and [here](#).

(ii) The supply side: Why fabs can't follow

The inability of supply to meet outsized demand is compounded by the lead-time needed to construct more fabs. Memory fabs (fabrication plants) are among the most expensive and complex facilities to build, with a 2-3 years construction timeline. Most announced projects will not contribute meaningfully to HBM capacity until 2027 at the earliest.

The complications of HBM production exacerbate the memory shortage crisis. HBM is made from DRAM, but producing one additional bit of HBM requires around 3x more silicon.⁶ That means every wafer directed to HBM consumes multiple wafers' worth of standard DRAM/NAND capacity needed for end-markets like automotives and PCs – creating shortage risk across not just the AI segment but all memory types. As this ratio moves from 3x to 4x silicon needed with HBM4/HBM4e, the crowding-out effect will worsen. For example, facilities and tools that would have served conventional memory are being redirected to HBM; “clean room space” – the ultra-sterile environments where wafers are processed – is at its limits, forcing manufacturers to choose what to build on finite factory floors. Anecdotally, Micron has said it can fulfil only 50% to 2/3 of demand from several key customers, describing the demand-supply gap as the largest it has ever seen.⁷ Qualcomm has also said that the size of the 2026 handset market will be defined by DRAM availability rather than consumer demand.

Our Deutsche Bank equity analysts' DRAM demand-supply model projects continually tight supply demand conditions in the coming years (beyond 2028). Growth in WSPM is expected to still lag accelerating demand for DRAM wafers over the coming years, even despite new fab announcements and project expansions by top DRAM suppliers.⁸

DRAM now represents ~70% of the memory market – up from its historical 50-60% range, as the HBM boom lifts DRAM's share of revenue.

⁵ [Global Memory: Updating our DRAM D-S model; Industry imbalance set to persist](#)

⁶ Otherwise known as the HBM trade ratio.

⁷ <https://www.techradar.com/pro/we-are-only-able-to-supply-for-our-key-customers-in-the-midterm-about-50-percent-to-two-thirds-of-their-requirements-micron-ceo-forecasts-production-spend-increase-to-meet-the-insane-demand-for-memory-but-the-ram-crisis-will-only-get-worse>

⁸ WSPM stands for Wafer Starts Per Month, which measures the number of raw silicon wafers that enter a fab's manufacturing pipeline every month. That is the university industry standard metric to measure and express the max production capacity of a fab.



2. The spillover: Why it matters beyond AI

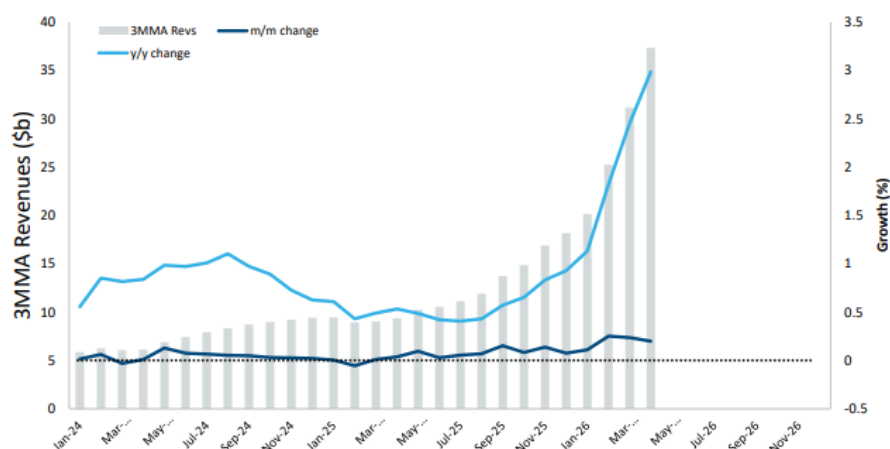
(i) Prices, mix shift, and the HBM-vs.-DRAM squeeze

The overwhelming demand for AI compute and inference means US hyperscalers will pay premiums to lock in future supply. Examples include SK Hynix's supply agreement with OpenAI for the Stargate Project, Samsung reported long-term deals with Google and Microsoft, and Micron's first five-year strategic agreement.

To meet surging AI demand, chip suppliers are reallocating their capacity towards more HBM production and pulling out of older segments. The result is an industry cycle shift, in which standard DRAM and NAND are also pulled into data centres and tighten supply, hitting non-HBM buyers in energy, consumer electronics and appliances. TrendForce predicts standard DRAM contract prices will rise 58-63% q/q in Q2 2026, with NAND flash contract prices up 70-75% q/q.⁹

We are seeing this today not just in leading-edge pricing: in Q1 2026, there was an uptick in lagging-edge pricing for the first time in two years, as increases filter into mature chip categories.¹⁰ Price rises are set to spread to server CPU, storage, substrates and passives, while several of the largest mature foundries – Vanguard, Powerchip and Nexchip – reported they would raise prices by up to 10% or more, driven by a 5% increase in costs.¹¹

Figure 4: 3MMA DRAM revenues continued to surge in April, growing to \$37.3bn (+299% y/y)



Source: Deutsche Bank equity analyst research, SIA. Notes: 3MMA = 3 month moving average.

(ii) Who can cope, and who gets rationed?

The shortage is affecting all direct users of memory. But hyperscalers and AI cloud providers remain the best positioned overall to weather it, because they can price AI compute services to reflect the underlying input costs.

With AI absorbing all the silicon, broader corporates and consumers are being squeezed and rationed. Lenovo, Dell Technologies, Apple and HP have sounded alarms on rising memory costs. The effect is global: Taiwanese manufacturers Acer and ASUS, and Chinese smartphone company Xiaomi have also issued warnings.

(iii) The most exposed sectors, bypass-through power

Of the key memory end-markets – data processing, communications, consumer and industrial – those most exposed to the AI squeeze are PCs (within data processing), consumer electronics (within consumer), and automotives (within industrials).

⁹ <https://www.trendforce.com/presscenter/news/20260331-12995.html>

¹⁰ Semis skimmed: Pricing and lead time tracker: Mar 2026. Deutsche Bank Equity Research.

¹¹ Semis skimmed: Pricing and lead time tracker: May 2026 - Deutsche Bank Equity Research



Starting with automotives: suppliers that sell complete technology systems to manufacturers such as Ford and Toyota have among the highest exposure to memory chips. These suppliers embed DRAM across the vehicles to help drivers use cockpit infotainment and navigation, advanced driver-assistance systems, and more. In 2026, companies such as Aptiv, Aumovio and Ford have signalled low DRAM supply despite efforts to secure inventory and contracts. With inventories set to drain through 2026, our equity analysts see potential impact on production volumes from 2027.¹²

Still, these suppliers may be best positioned among non-AI buyers, given their recovery mechanisms established after the 2021 chip shortages and these companies' ability to pass costs to the equipment manufacturers like Ford and Toyota, who have more limited consumer pricing power. Our DB models estimate the inflated cost of DRAM raises the price of an average vehicle by \$150-300 for luxury high-end models, and \$400-600 for vehicles with higher-level autonomy.¹³ These carmakers therefore can either absorb the cost and compress margins, pass it to consumers, or strip out some DRAM-intensive features such as L2+ autopilot capabilities or GenAI chatbots.

Consumer-electronics manufacturers and PCs are also highly exposed to the memory crisis, via margin compression and softer unit volume. Compared with automotives, this sector faces a greater challenge because it has the weakest pass-through capacity among the sectors. Price sensitive categories – smartphones, traditional phones, PCs, smart TVs, gaming consoles and tablets – all face rising memory costs per unit. Total consumer revenues also fell -10% q/q in Q1 this year, with most weakness from AMD's and Nvidia's consumer businesses. Our equity team expects total consumer end-market revenues down -15% y/y in 2026, recovering +9% y/y in 2027.¹⁴

The consequences of memory's supply disruption amongst consumer electronics are being echoed publicly across large corporations. In Apple's Q1 earnings call, its CEO warned that higher memory costs are just the beginning and that the company was facing supply constraints.¹⁵ While Apple has not detailed its options, it has reportedly quietly reduced the maximum memory capacity offered on several of its computers.¹⁶ It is not just Apple: Dell and Microsoft are also paring back their configurations to manage input costs, rather than pass them onto the individual buyer. or Microsoft, even as the company advances in AI with its Copilot+ – quietly cut its new surface for Business laptops to as little as 8GB of RAM (down from 16GB previously) on entry-level configurations to mitigate the DDR5 costs.¹⁷

Figure 5: Who is exposed?

Data Processing	Communications	Consumer	Industrials
PCs Servers	Wireless Wireline	Consumer electronics	Manufacturing Automotive Aerospace & Defence MedTech

Source: Deutsche Bank Research

¹² [Global Automotive: DRAMa ahead? - Deutsche Bank Research](#)

¹³ [Global Automotive: DRAMa ahead? - Deutsche Bank Research](#)

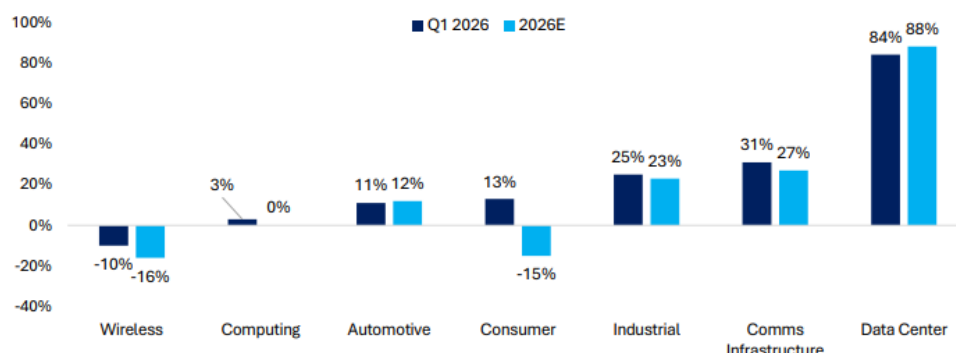
¹⁴ [US Semiconductors: Quarterly End-Market Monitor - Deutsche Bank Research](#)

¹⁵ [Apple CEO warns of memory crunch. 'We'll look at a range of options'](#)

¹⁶ [Apple quietly axes 128GB Mac Studio amid supply constraints and local AI frenzy — highest memory capacity reduced to 96GB, two months after discontinuation of 512GB model | Tom's Hardware](#)

¹⁷ <https://www.tomshardware.com/laptops/8gb-of-ram-is-back-on-laptops-companies-are-lowering-memory-offerings-to-make-affordable-notebooks-during-component-crisis>

Figure 6: End-market growth, Q1 vs. 2026 estimates



Source: Deutsche Bank equity analyst research

(iv) The macro spillover: the chipflation tax

With companies in consumer electronics and appliances for example forced to bear rising costs, the risk of a “memory inflation tax” on consumers is growing.

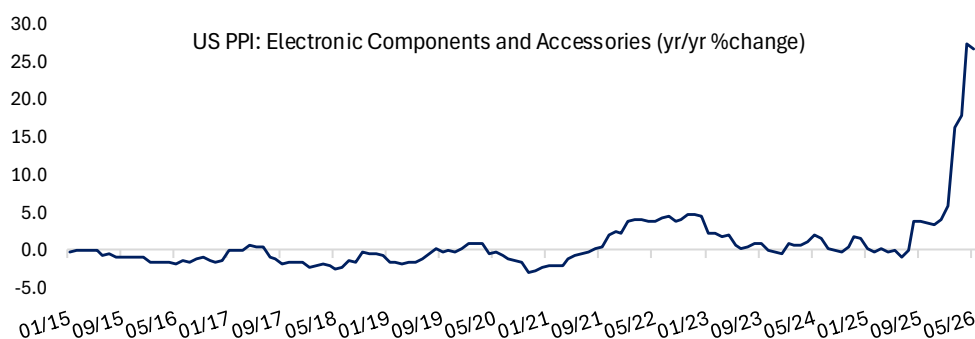
This is happening because end-product manufacturers – across PCs and smartphones for example – face severe Bill of Materials (BOM) cost pressure from the reallocation of silicon towards AI infrastructure. When memory prices surge, products with memory-heavy BOMs meet sharp production-cost increases, forcing companies to redesign or pass the “tax” down to consumers. We detailed the automotive and smartphone trade-offs above; if firms pass costs through, the average person takes a direct price impact. PC vendors Lenovo, Dell and ASUS have already warned of potential 15-20% price hikes and contract resets, with Lenovo and Dell starting increases this July.¹⁸

Consumers may also face financing and long-term stress: higher vehicle prices, for example, could push buyers into longer loan terms and higher total interest payments over the life of the asset.

In short, the memory-supply crisis has the potential to inflict acute second-order macroeconomic effects. The AI squeeze is crowding out the economy’s ability to serve the day-to-day needs of ordinary people. Qualcomm and AMD have both flagged memory constraints actively hitting consumer-device demand.

We are already seeing it in US purchasing price inflation (PPI): the electronic components and accessories – which captures semiconductor and memory-devices prices – rose +26.9% y/y in May, up from +5.9% y/y rise in January. That, in turn, is pushing up upstream input costs between the businesses that trade these materials.

Figure 7: The sharp rise in PPI for electronics is driven by surging memory-driven chip costs



Source: Deutsche Bank Research, Haver.

¹⁸ <https://www.trendforce.com/news/2024/06/10/news-lenovo-reportedly-set-for-july-price-hikes-across-product-portfolio-as-memory-costs-pressure-pc-market/>



3. Fault lines: The geopolitics of memory

(i) Winners and losers, country by country

South Korea is heavily exposed to the energy-intensive global AI capex boom, as its two tech giants (SK Hynix and Samsung) benefit from specializing in memory for AI data centres. The two companies account for around 69% of global DRAM production. However, that success heavily dictates Korea's economy, and when memory falls it drags Korean markets down with it. For example, after semiconductors melted down on June 8, the tech-heavy Kospi fell -8.29%, its ninth worst day in more than 12,000 trading sessions since Bloomberg data began in 1980. And earlier, in late March, Google's publication of a research algorithm called TurboQuant – which reduces the memory needed to run AI inference – also sent SK Hynix and Samsung shares lower. In short, the winners of the AI trade can quickly become the biggest losers if the gamble for global AI buildout fails.

China's semiconductor exports have also risen, as firms like CXMT (which specializes in DRAM), and Yangtze Memory Technologies (which specializes in NAND) capitalize on industry's shift towards HBMs, which has left supply for standard DRAMs tighter. Exports surged by 72.6% y/y (in dollar terms) between Jan-Feb 2026, with memory the largest and fastest-growing category: customs data show memory exports soared 81.6% y/y (in dollar value) in Dec 2025.¹⁹

At the same time, Chinese companies are also hurt by the memory squeeze, as waiting lists for advanced memory and compute lengthen. Vendors (CXMT, JHICC) are expanding aggressively but are constrained to lagging-edge DRAM by US export controls. For example, CXMT's Shanghai fab expansion will add up to 200k wafers per month by 2030 but only begins high-volume manufacturing in 2027. Chinese producers will relieve some pressure in low-end commodity segments but cannot address the HBM or automotive-grade DRAM shortage. US export controls have severely limited domestic fabrication capacity, so Chinese supply cannot substitute for the premium segments most affected. Ultimately, Chinese memory companies will be confined to lower-performance end-uses and will remain well behind in HBM.

The United States is home to Micron (24% of global market share) and is the largest region by memory revenue. Outside AI buyers, there is real potential for US corporates and government to lean on China's lagging-edge advantage as a partial answer to the AI memory squeeze. For instance, in February, the Pentagon quietly removed China's CXMT and YMTC from its restricted-companies list, which makes US sourcing Chinese memory and NAND more plausible – though it is too early to judge whether adoption follows.²⁰ Several tech companies, including Apple and major PC OEMs like HP and Dell, are reportedly exploring or qualifying partnerships with Chinese memory makers to diversify supply chains and combat rising memory costs.

The geopolitical stakes are serious. Every dollar into CXMT and YMTC also benefits companies widely viewed as national-security risks – a hard case to make when the current US administration's priority has been to decouple from China (as evident from the MATCH Act). Whether US tech companies actually embark on these partnerships is uncertain. As noted earlier, China is confined to the lagging edge, so it would be difficult to justify a company like Apple shipping more phones built on a less advanced node than prior versions.

Congress has introduced the US MATCH Act, which aims to restrict countries from buying semiconductor manufacturing equipment technology from the US and its trade partners that they cannot build themselves. Importantly, MATCH designates restrictions on CXMT and YMTC across exports, servicing, and technical support. It has received broad bipartisan support but blowback from US chipmakers, who warn of further memory price

¹⁹ Gavekal, China Customs Administration

²⁰ https://wccftech.com/cxmt-ymtc-removed-from-pentagon-list-opening-door-for-chinese-dram-adoption/?utm_source=substack&utm_medium=email



rises, as the Act could cap global production capacity and shift pricing power back to a dominant Western-aligned oligopoly. Micron alone has publicly backed MATCH – unsurprisingly given it is in the company’s interest to maintain price advantage.²¹ If MATCH were to be implemented, it would have severe consequences for China’s ability to industrialize.²²

4. Breaking the bottleneck: What could actually fix it

(i) Building more fabs (and why it’s slow)

To meet ongoing demand, the big 3 players are raising capex to help break the memory bottleneck. Our latest DB estimates place total DRAM wafers per month estimate to increase by +1,475k WSPM over the next five years, due to a variety of new fab announcements/expansions/accelerations.²³ Micron remains at the heart of a transformative moment in the memory industry, with a \$200bn commitment to expand its memory capacity in the US, and will benefit from sustained tightness in demand. As we expressed earlier, however, fab construction is a multi-year process and will come online only after the peak of the shortage in the industry occurs.

Samsung, SK Hynix and Micron are trying to pull the timeline forward by buying partially finished or existing facilities. One example is Micron’s \$1.8bn acquisition of a used fab from Powerchip Semiconductor Manufacturing Corporation (PSMC) in Taiwan earlier this year, which lets Micron skip the roughly 2-year timeline needed to build a facility from scratch.²⁴ We expect other memory companies to follow in Micron’s lead. Samsung is also reportedly negotiating long-term deals with Google and Microsoft, while Micron has disclosed its first five-year strategic customer agreement.

Whether China can be a credible player is still hard to gauge, as many new Chinese fabs remain under construction. But as we said before, these would primarily address lagging-edge memory demand, and the US export controls have severely limited China’s ability to fabricate at more advanced nodes.

(ii) Beyond fabs: Algorithms, SRAM and yield

Google’s unveiling in March of an algorithm that requires less memory to serve LLMs (“TurboQuant”) caused a sharp decline in the share price of Samsung (-6%), Micron (-7%) and SK Hynix (-7%) on the day.

However, the subsequent recovery in the memory market underscores that experiments like TurboQuant have yet to convince investors or AI accelerators that they a breakthrough is at hand that would turn the memory boom to bust. The key reason is scope: TurboQuant compress the inference-time KV cache – not model weights or training memory. While both focus on reducing memory bottlenecks, KV caching speeds up generative inference, whereas training memory optimization allows models to process massive datasets during the learning phase. Therefore, TurboQuant’s potential impact was only limited to a part of total AI memory demand. More fundamentally, because cheaper inference historically expands total usage – the Jevons Paradox – such efficiency gains may ultimately lift, not cut, aggregate memory demand.

Still, TurboQuant is likely the first of several attempts to reduce reliance on HBM. Other strategic efforts include improved KV-cache eviction (such as Nvidia’s inference context memory storage system) and greater reliance on SRAM.

There will likely be more attempts to upgrade node conversion (upgrading the manufacturing process to print smaller circuits) and yield improvements (the share of chips on a wafer that actually work). Both aim to lift the capacity of existing memory

²¹ <https://finance.yahoo.com/economy/policy/articles/micron-backs-match-act-policy-180733909.html>

²² https://www.kim.senate.gov/press_release/senators-kim-and-ricketts-introduce-match-act-to-level-the-global-playing-field-for-u-s-tech/

²³ [Global Memory: Updating our DRAM D-S model; Industry imbalance set to persist - Deutsche Bank Research](#)

²⁴ <https://investors.micron.com/news-releases/news-release-details/micron-signs-letter-intent-purchase-tongluo-site-begin-0>



production rather than building new fabs. The memory industry will nonetheless continue to face headwinds in bringing incremental capacity online.

Conclusion: From commodity to macro variable

The production of memory chips is becoming a zero-sum game. For every wafer devoted to HBM stacks for AI servers, others are unavailable for smartphones, PCs, or vehicles. And because the shortage will reverberate across key economic systems well beyond AI – hurting the average consumer via memory-cost inflation – memory chips have transitioned from a pure commodity to a distinctly macroeconomic variable.

The companies best positioned to survive the structural dynamics of the memory industry are, of course, the hyperscalers and AI cloud providers. And Micron stand out as a clear winner of the current shortage. For traditional PCs and smartphone manufacturers, memory is critical to device function, but unlike in AI, adds little value at higher performance or capacity; their objective is to minimize cost per bit while keeping systems efficient. As a result, the insatiable needs of the AI buildout are consuming capacity that would have gone to consumer-end markets. Consumer electronics, automobiles and automotive-system suppliers will face headwinds large enough that sustained DRAM inflation will have a measurable inflationary impact in 2027, hurting average customers.

Looking ahead, the question worth considering is what the macroeconomic consequences would be if the AI gamble fails. Where would all the HBM and silicon capacity go? The sharp swings triggered by events like TurboQuant show that if AI demand moderates even modestly, the market could swing from severe shortage to meaningful oversupply relatively quickly.

References:

[Thematic Research: Chips: The Biggest AI Bottleneck](#)

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[Monthly SIA Data Digest: April 2026: Slight m/m dip, but better than expected vs. seasonal norms](#)

[Micron: A trip down memory lane; Initiate with Buy rating](#)

[US Semiconductors: Quarterly End-Market Monitor](#)

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Is the ceasefire a reset? Iran, the US and the Middle East they can't restore

June 15, 2026

The US and Iran have announced a framework agreement to end hostilities, setting the stage for a formal signing in Switzerland and sending markets rallying with oil tumbling. This landmark deal, a potential "win" for the Trump administration, Iran's resilience, and Israel's strategic gains, redefines the Middle East landscape. However, the path ahead is complex: watch for implementation disagreements, Gulf states' security demands, and Iran's continued leverage over the Strait of Hormuz. We're also seeing accelerated UAE-Saudi tensions and a multipolarizing Gulf as China and Russia position themselves, marking a critical juncture for regional security and global energy markets.

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Context

On 14 June, the US and Iran announced a framework agreement to end hostilities, with a formal signing ceremony to take place on 19 June in Switzerland. Under the tentative agreement, the US would gradually lift its naval blockade on Iranian ports and relax sanctions/release frozen Iranian assets, in exchange for progress towards a technical agreement on Iran's nuclear programme. The agreement is meant to reopen the Strait of Hormuz and address the stickier questions of freedom of navigation through the Strait, Iran's nuclear programme, reconstruction, and sanctions relief among other issues.

Everyone "Wins"?

We have suggested since early March that President Trump would seek an off-ramp (see Thematic Research: The Persian Chessboard: Decoding Iran's Strategy Beyond a US "Victory") given the greater incremental gains for the US versus diminishing returns.

Despite only a framework agreement and the tremendous costs of military actions and the Iranian responses, the Trump administration can likely present a "mission accomplished" based on the extensive damage to Iran's military infrastructure, nuclear sites and air defences, a route to reopening the Strait of Hormuz, and a pathway to some nuclear agreement.

For the Iranians, Tehran's strategy of weaponizing regional and economic instability to make the conflict prohibitively expensive for its adversaries was largely successful (though much will depend on concessions to be made during the forthcoming negotiations). Nevertheless, the regime has demonstrated its resilience. For Israel, it was able to significantly harm Iran's military capabilities, address Iran's nuclear programme, and degrade Hezbollah's capabilities in Lebanon. This may provide President Netanyahu with a political boost in advance of competitive Israeli elections later this year.

What should we be watching next?

While markets are celebrating with significant rallies and oil tumbling to a three-month low, the reversion to the old Middle East status quo may take time to restore, if indeed that is possible.

What could derail the deal? There are many moving parts in negotiations and we should watch closely for disagreements over implementation (timelines, verification, etc.), Gulf states' requirements for security guarantees, and domestic political opposition from key parties. However, we assess that both the US and Iran are keen to press ahead with the framework agreement even if this doesn't lead to a full-scale peace agreement. A critical factor to watch will be the potential for Israeli aspirations in Lebanon to upend the deal. We suspect that there is significant US pressure on President Netanyahu to cease operations and accept the accrued gains to prevent a derailing of the deal.

Strait of Hormuz: Shipping isn't likely to be restored immediately given the challenges of de-mining, insurance premia remaining high, lack of trust until we see sustained stability over time, infrastructure repair to ports, terminals and regional energy infrastructure, and security guarantees. While Iran will probably be keen to



show goodwill, the Strait of Hormuz is its most important bargaining chip in negotiations – one that it will likely seek to protect. Whether any agreement on the Strait of Hormuz entails a “toll”, Iran's strategic leverage over the Strait could translate into long-term, persistent, albeit possibly intermittent, threats to navigation.

Why Iran may have won more than it lost: Despite significant military losses, Iran has demonstrated resilience through its military actions and a willingness to close the Strait of Hormuz. Crucially, the mere fact that the US came to the table validates Iran's long-standing insistence that it cannot simply be bombed into submission. Iran's ballistic missile programme remains and its proxy relationships with groups across the region, while weakened, have not been dismantled.

The regime that enters these negotiations is a harder one – the strikes killed many moderate figures – leaving a leadership cadre that appears more hawkish and defiant. In addition, surviving a confrontation with the US and Israel without surrendering also strengthens the regime's hand domestically. The deal's promises of potential sanctions relief and access to \$24 billion in frozen assets provides a much-needed economic lifeline to an Iranian government under severe financial pressure. For the broader Middle East, this sets an uncomfortable precedent for engagement with Iran in the longer term.

The New Middle East?

UAE-Saudi tensions: The conflict has exposed and accelerated fault lines between the UAE and Saudi Arabia that have simmered for years. The UAE's decision to withdraw from OPEC in May 2026, reportedly taken without prior consultation with Riyadh, is the most visible sign. This deepening rupture is driven not just by energy policy but also by broader strategic rivalries and competing visions for the region's future. We will be watching what that rivalry means for oil markets, regional security arrangements and Washington's ability to manage its Gulf alliances as a coherent bloc once the Iran crisis settles.

Gulf-US Relations: The Gulf states have responded cautiously but positively to the framework agreement, welcoming any steps that could reduce regional tension and improve stability. That said, we judge that they are keen to ensure that any lasting agreement entails enforceable security assurances and regional participation in future arrangements. In the aftermath of this crisis, Gulf ties with the US may be more complex, and we could see efforts toward greater economic and security diversification. Moreover, any outcome that leaves the Gulf states to deal with the consequences of a conflict that has already caused significant regional disruption would risk deepening existing fissures between the Gulf and the US and undermine confidence in long-term US commitments to regional security.

Multipolarizing the Gulf? Both China and Russia have welcomed the framework agreement. While Beijing has focussed on restoring stability in the Gulf and safeguarding international trade routes, Moscow has underscored respect for Iran's sovereignty and the importance of a negotiated political settlement. Both seem keen to ensure that any final settlement does not reinforce long-term US dominance in the Gulf. Indeed, both are benefiting from positioning themselves as defenders of international law and the post-WWII order, and may be able to make inroads with regional Gulf allies in the aftermath of the crisis, potentially at the US' expense.



Appendix 1

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AI twist in the tale for Anthropic's Fable and Mythos

June 16, 2026

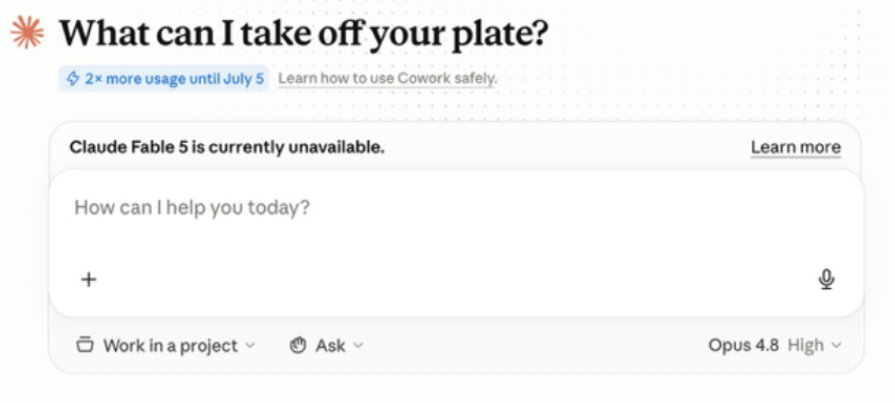
The story so far...

If you want to understand the implications of the abrupt switching off of Anthropic's Claude Fable 5 and Mythos 5 AI models globally due to US national security orders, just consider the likely reaction in the corridors of OpenAI, Washington and Brussels.

The US Commerce Department issued an export control directive late on Friday to suspend all access by foreign nationals to Anthropic's most advanced models, prompting Anthropic to disable access to all customers to ensure compliance.

The US said it had become aware of a method of bypassing, or "jailbreaking", Fable 5, the public version of Mythos 5, according to Anthropic. Mythos has only been rolled out to a few partners because of fears that it could be misused to exploit cybersecurity vulnerabilities. The startup disputed the concerns, which were reportedly raised by its partner Amazon with administration officials, saying: "We believe this is a misunderstanding." Further details remain hazy.

Figure 1: Anthropic's Claude Fable 5 is "currently unavailable"



Source: Anthropic's Claude desktop app, accessed June 14, 2026

Anthropic is in a race for funding in what could be a historic year for technology IPOs. After it announced on June 1 that it had confidentially filed for an IPO, OpenAI followed suit last week. Both are expected to take place later this year at valuations of more than \$1 trillion apiece.

What could this latest twist in the tale mean for a company growing at breakneck pace; its now smaller rival, whose ChatGPT chatbot is still the byword for AI outside Silicon Valley; a US government that sees an existential threat from China; and European governments already painfully aware of the continent's lagging position in the AI race?

Authors

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1. Short-term pain, but potentially long-term gain, for Anthropic

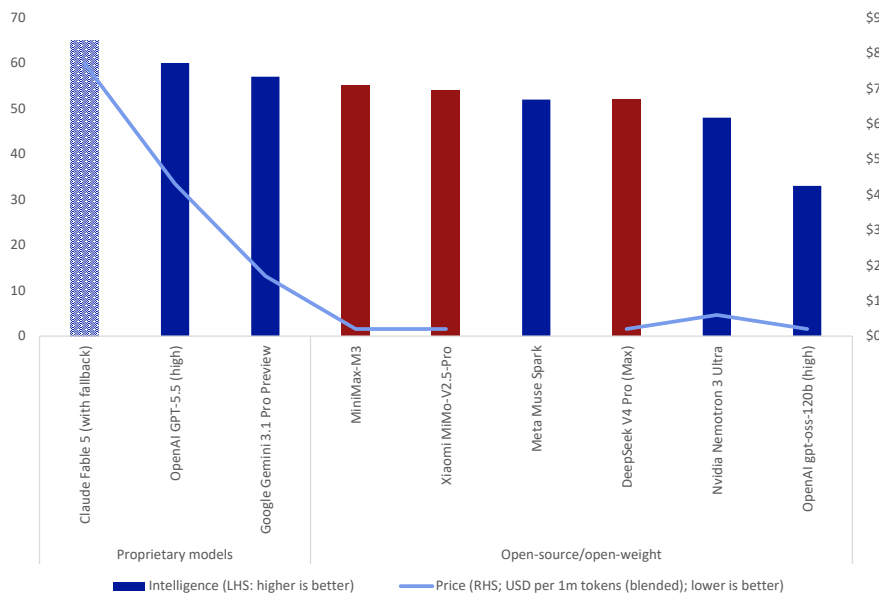
- What greater compliment than to be officially labelled too powerful to release? Nuanced questions of guard rails are likely to be overlooked in the face of marketing that money can't buy. It's like the Adidas running shoes that are so good they're banned in elite competitions. Even if Anthropic has to keep its most advanced models in a cage in the short term, in the longer term its reputation will likely have received a significant boost.
- The timing may be ideal, with a series of mega tech IPOs making huge valuations seem attainable, but also potentially forcing investors to pick a favourite or two amid concerns about how much the market can absorb.
- This is not the first time Anthropic has had a run-in with the US administration. In February, the Department of Defense designated the startup as a supply-chain risk, effectively blacklisting it from working with the US military and contractors, after Anthropic asked that its technology not be used for mass surveillance of Americans and lethal autonomous warfare. President Donald Trump posted that every federal agency should stop working with Anthropic. The company has taken the Pentagon to federal court. But if there were to be a change in the political winds, that could put Anthropic back on the government's books with its reputation stronger than ever.

2. Rising pressure on OpenAI's technology and business model

- OpenAI is under pressure as a potential IPO approaches. Anthropic has recently overtaken it in revenues (\$47bn annual recurring revenue at the end of May versus OpenAI's \$24bn at the end of March) and valuation (\$965bn post-money at the end of May in versus OpenAI's \$852bn in March). While models produced by the two companies have now been neck-and-neck for many months, this has been the first time that a model has been banned for being too smart.
- While OpenAI is still the leading consumer app by a country mile, reportedly having crossed the billion monthly users mark in May, only 50 million were paying for it at the end of February. Consequently, the company has sought to refocus around businesses, an area that Anthropic already dominates. OpenAI had nine million enterprise subscribers in February.
- The company is considering drastic price cuts ahead of similar cuts it expects at Anthropic, the Wall Street Journal reported last week, amid stiffer competition from open-source Chinese models that may be good enough for most applications at a fraction of the cost. When it was available, Claude Fable 5 cost \$7.70 per million tokens and GPT-5.5 cost \$4.30, according to a blended score from Artificial Analysis. Compare that with around 20 cents for a number of open-source models.



Figure 2: Slightly more intelligent proprietary models are significantly more expensive than open-source alternatives (chart includes selected models)



Source : Artificial Analysis Intelligence Index. "Fallback" mechanism routes safety-flagged messages to Opus 4.8 model. Meta Muse Spark price not available. US models in blue; Chinese models in red

3. Double-edged sword for US aim to dominate AI

- The move is the first time the US government has used export controls to stop access to a commercial AI model that is already widely used by the public. There is a risk that it backfires not only for a country that considers itself to be in an existential race for "global AI dominance" but also for US companies that rely on a global market to justify AI investments.
- The move echoes the now well-established policy of putting export controls on hardware, notably Nvidia's most advanced AI chips. Indeed, an unattributed report from Semafor suggested that the decision to ban foreign use of Mythos had been driven partly by suspicions that a China-linked group had accessed it.
- Five years ago, Chinese companies relied on US chips and developers built their software on its industry-standard CUDA platform. However, since the imposition of stiff export controls, starting under the Biden administration, Chinese companies led by Huawei have invested heavily in creating their own chips and AI ecosystem, eroding US leadership. While they are not yet at the cutting edge, they are gaining fast and may get a foothold in the rest of the world with unaligned countries that prefer cheaper alternatives to US technology.

4. Intensified calls for sovereign AI amid rising trade tensions

- The move casts an unforgiving light on more profound regulatory and strategic issues about sovereign AI: who controls the AI models, data and infrastructure. The US has half of the world's data centres and last year clocked up more than 20 times as much private investment in AI companies as China, the No. 2 country, with European companies far in the distance.
- The speed of the ban is likely to reinforce concern that access to all US technology could be suddenly switched off without warning. It may further propel efforts to diversify by companies such as financial institutions that depend on US technology for systemically critical infrastructure.



- The ban “further underlines Europe’s need for technological sovereignty,” the European Commission said over the weekend. The problem is that although Europe is home to a number of key AI-related engineering companies, it only has one notable AI model maker, France’s Mistral, and relies on non-EU countries for over 80 percent of key digital products, services, infrastructure and intellectual property. The Tech Sovereignty package announced on June 3 is an attempt to accelerate initiatives to build domestic capacity and autonomy across the stack.
- The move also plays into a wider theme of technological independence that has gained momentum with this year’s conflict in the Middle East and attacks on data centres highlighting the vulnerabilities of supply chains and dependence on other countries for data and compute.

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Capital allocation without Big Tech

June 15, 2026

The sharp sell-offs that have punctuated the AI-related stock market appreciation over the last few years have all come at times when investors have contemplated whether the AI business model is at a tipping point. This year's related IPOs have certainly entered into that realm.

The recent wobbles have also come as investors consider the financing methods of some of the largest hyperscalers which have signalled they are ready to turn to equity issuance to invest in AI capacity.

Apart from testing the pecking order theory of financing, these developments also have implications for aggregate capital allocation of the S&P 500 given the size of tech firms. So, in this short note we disaggregate buybacks, debt issuance and capex of the S&P 500 to quantify the impact of tech firms.

Amongst our key takeaways is that while tech dominance in the stock market is growing, other S&P 500 corporates offer solid fundamentals which means the bulk of companies in the index do not look overpriced. That should keep the potential fallout from any further AI disenchantment more contained than many currently fear.

Authors

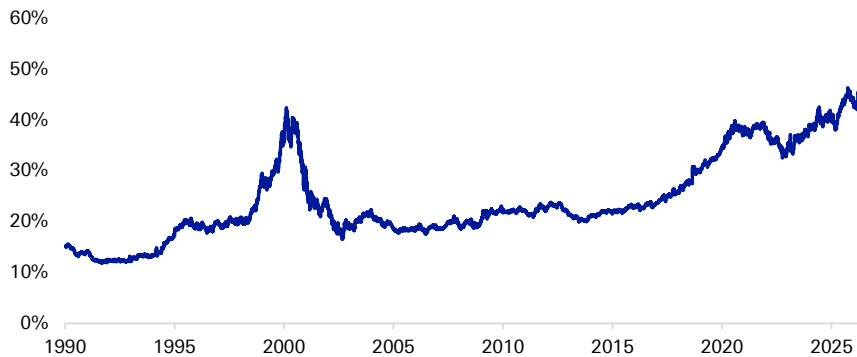
[Luke Templeman](#)
Research Analyst

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Research Analyst



Tech dominance is back but its ranks are changing

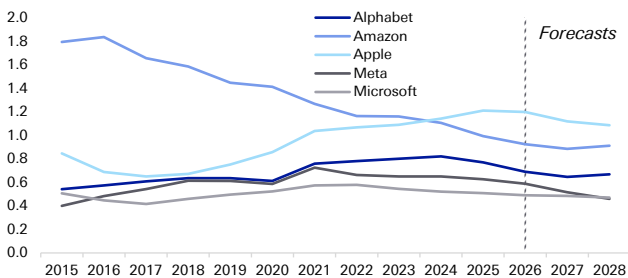
Figure 1: TMT companies market cap as % of S&P 500 total



Source : Bloomberg Finance LP, Deutsche Bank.

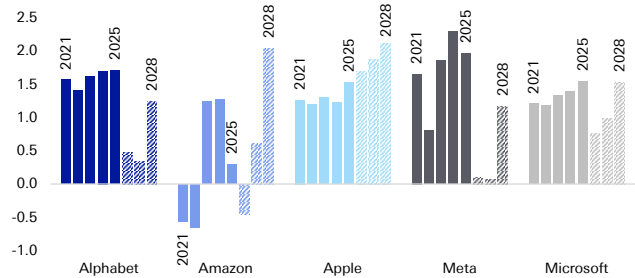
The run-up in the share of TMT companies' market cap relative to the S&P 500 total comes just as the biggest tech players that have propelled index fundamentals in recent years are expected to experience declines in free cash flow. That comes on top of slowing sales growth for some which will weigh on their asset efficiency, particularly given their capex plans.

Figure 2: Asset turnover of selected tech companies



Source : Bloomberg Finance LP, Deutsche Bank.

Figure 3: Free cash flow of selected tech companies, indexed to 2020 levels *

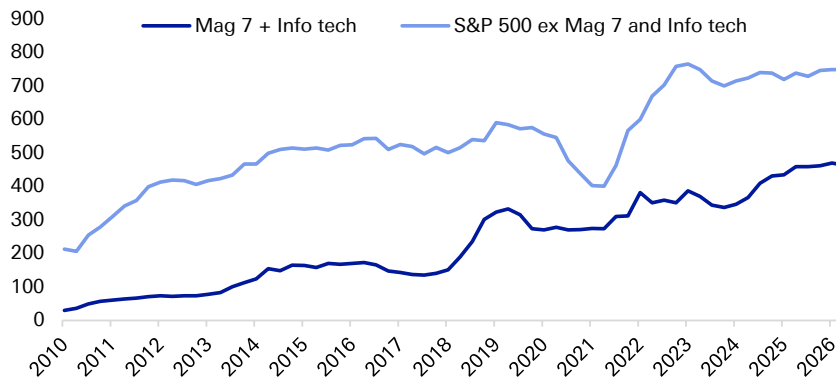


Source : Bloomberg Finance LP, Deutsche Bank. * Note: shaded areas are forecasts.

The cash drain from AI capex is a concern for investors in part because tech companies have offered strong payouts in recent years amidst record cash generation, including during covid when the aggregate for rest of the index declined.



Figure 4: Total payouts (\$bn, LTM) *



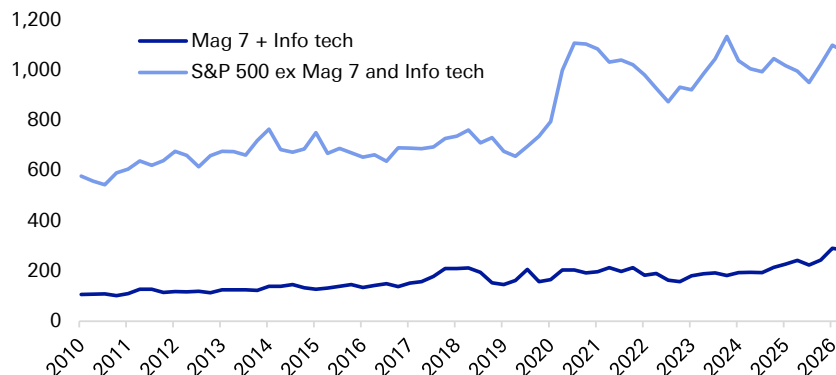
Source : Bloomberg Finance LP, Deutsche Bank. * Note: excluding financials, real estate and utilities.

When we now look at large non-tech American corporates, there are strong signs of resilience. This leads us to expect them to increase their investing and payout activities in the coming few years. That would be a pleasant contrast to the first half of this decade which was beset by various shocks and uncertainty amidst a changing geopolitical and economic backdrop.

Capital allocation trends of non-tech corporates look solid

Compared with the trend in payouts, the relative size of aggregate tech companies' actual cash pile is much smaller. While this reflects the differences in cash needs of business models and profitability, aggregate cash coffers are up by c 8% p.a. since 2019, outpacing inflation.

Figure 5: Cash & cash equivalents balances (\$bn) *

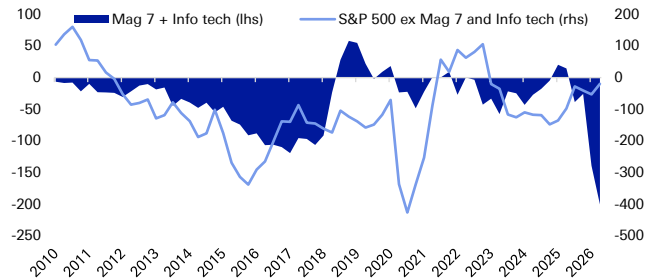


Source : Bloomberg Finance LP, Deutsche Bank. * Note: excluding financials, real estate and utilities.

Part of the reason for the increase in cash in non-tech companies is a turn towards relatively conservative capital allocation. Unlike tech companies, the rest of the S&P 500 corporates have been in cash-preservation mode, which is reflected in almost no new net borrowing in the last 12 months as well as a relatively slower increase in capex (+876% for tech and +62% for non-tech since 2019).

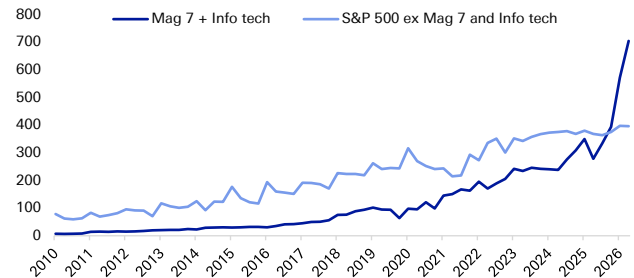


Figure 6: Cash spent on net debt (negative - more borrowing) *



Source : Bloomberg Finance LP, Deutsche Bank. * Note: excluding financials, real estate and utilities.

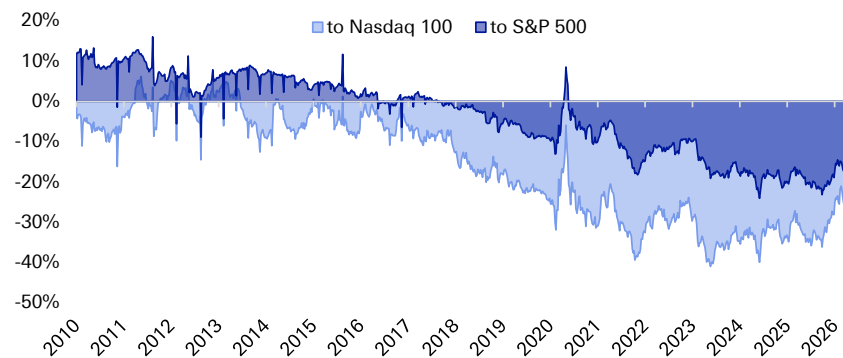
Figure 7: Reported capex, LTM \$bn *



Source : Bloomberg Finance LP, Deutsche Bank. * Note: excluding financials, real estate and utilities.

We believe this cash preservation is likely to change. Big M&A deals are already one of the signs that companies are looking to decrease their cash piles. This may help make non-tech corporates more attractive to investors and acquirors and, as the chart below shows, the equal-weighted S&P 500 continues to trade at c.15% discount to the market-cap index, and at 27% to the Nasdaq 100 (on a forward P/E basis). And while the non-tech firms are expected to have only 13.2% EPS growth in 2026 (23.7% for tech), that is ahead of Europe (11.2%) and Japan (10.5%).

Figure 8: Equal-weighted S&P 500, premium (+) / discount (-) to...



Source : Bloomberg Finance LP, Deutsche Bank.



Appendix 1

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Research

The House View: Snapshot 1999 meets 1990

Deutsche Bank

Macro views

World

- Global growth at 3.0% in 2026 and 3.2% in 2027.
- Rationale:** A US-Iran deal may be possible in June, which would reopen the Strait of Hormuz and see Brent crude fall to \$86/bbl in Q4 and \$80/bbl in 2027. Current AI boom is also buffering the US and North Asia from the energy shock.
- Key risks:** War continues/worsens, oil market impact from prolonged Strait of Hormuz closure; AI sell-off sparks major market rotation. Upside risk is that there is further momentum from AI adoption/investment.

Europe

- 2026 growth cut to 0.5%. Technical recession cannot be ruled out; recovery expected from Q4, with 2027 growth of 1.1%.
- Rationale:** The energy shock has significantly impacted European activity. Fading energy shock effects and Germany's fiscal spending (defence and infrastructure) will start to dominate the growth outlook in the latter part of the year.
- Key risks:** If the Strait of Hormuz remains closed beyond June, it will weigh even more on growth conditions. Non-linear effects could become stronger.

China

- 2026 GDP growth forecast at 4.7% (down from 4.9%) due to softer domestic activity and policy fine-tuning.
- Rationale:** Strong trade activities are expected to offset weaker domestic demand. Reflation is accelerating, driven by crude prices pass-through to downstream products. Fiscal spending slowed since April, but a rebound is likely.
- Key risks:** High fuel prices reduce domestic and external demand, AI could boost productivity while negatively impacting the labour market.

United States

- 2026 growth is projected at 2.2%, supported by easy financial conditions, fiscal support and AI-driven capex.
- Rationale:** Labor market has stabilized and we expect this to persist with unemployment near recent levels into year-end ~4.3%. PCE inflation forecast has risen to 3.3% YE given recent outperformance and lift from energy prices.
- Key risks:** A more substantial oil price shock could weaken real disposable income; financial conditions might tighten, perhaps driven by concerns around AI disruption or (private) credit risks.

Germany

- 0.5% GDP in 2026 & 1.3% 2027. Fiscal deficit to widen to 4.1% of nominal GDP upon weaker growth.
- Rationale:** Cost-related headwinds and inflation induced weakening of private consumption to persist into H2 2026. The expected normalization of the global environment, combined with the domestic fiscal stimulus, should boost growth in 2027.
- Key risks:** Germany remains highly exposed to the energy cost shock and consequent deterioration of its current account.

India

- 6.7% y/y GDP for FY27 (prev. 7.5%). CPI to rise to 4.6% y/y in 2026; 4.9% y/y in FY27 given higher oil prices and risks of a sub-normal monsoon.
- Rationale:** Energy shock and sub-normal monsoon challenge growth forecast.
- Key risks:** If monsoon disappoints, real GDP growth could fall further to 6.3% y/y.

Key downside risks

- H** **Iran war.** Prolonged Hormuz closure could push Brent towards \$150/bbl, significantly hitting global growth and tipping Europe into recession.
- M** **Higher inflation / hawkish central bank pivot.** Persistent inflationary pressures have also led to a shift in the monetary policy cycle, with the Euro Area and much of Asia expected to hike rates this year, whilst the Fed also faces hawkish risks.
- M** **AI.** AI's job-loss and disinflation questions are overshadowed by the Iran War but will resurface post-conflict. Risks are concentrated in software/private credit, with defaults more likely a 2027–2028 story.

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June 2026

Deutsche Bank AG/London

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Market views

	Market Sentiment	❑ Risk assets are outperforming amidst a return of AI excitement, in addition to recent moderation in oil prices. ❑ Equity and credit markets have been remarkably resilient against geopolitical shocks, yet sovereign bonds like 10yr Treasuries are still trading in a tight correlation with oil prices.
	Equities	❑ S&P 500 YE-2026 target: 8000; STOXX 600 YE-2026 target: 640. ❑ Strong Q1 earnings reinforce constructive outlook on equities, driven by Tech on the back of AI-related demand. ❑ Overweight Financials, Industrials & Healthcare; prefer US & Europe over Japan.
%	Rates	❑ 10yr UST at 4.70% YE-2026; 10yr Bund at 3.20% YE-2026. ❑ Fed is done cutting and on-hold at nominal neutral. Bund forecast revised based on view the ECB will hike 25bps in Jun & Sep, taking the deposit rate to 2.50%.
	Credit	❑ YE 2026 targets: \$IG 82bps; \$HY 305bps; €IG 95bps; €HY 345bps. ❑ Global credit spreads show resilience, particularly in the US, which displays low recession risks. ❑ Turn underweight €-credit over \$-credit due to Europe's economic sensitivity and ECB tightening.
	FX	❑ YE-2026 targets: EUR/USD to 1.20. ❑ Turn neutral on the dollar as Strait of Hormuz stalemate persists. An escalation in Iran and more hawkish Fed pricing may push EUR/USD below 1.15; conversely a quick reversal in oil prices plus surprise Ukraine-Russia peace deal could support a move above 1.25. Risks are finely balanced.
	Oil & Gold	❑ Brent \$99/bbl in Q3-26; \$86/bbl in Q4-26; WTI at \$94/bbl in Q3-26, \$81/bbl in Q4-26. Based on expectation that the Strait of Hormuz will normalize in June. ❑ Gold: YE-2026 target: \$5,800/oz, average \$6,000/oz through.
	Monetary Policy	❑ Fed: On hold at long-run neutral Fed Funds rate of 3.63%, but hawkish risks increasing. ❑ ECB: 25bp hikes in June and September, taking deposit rate upper end of neutral to 2.50% in 2026. ❑ BoJ: Quarterly 25bps hikes starting in June, bringing policy rate to 1.75% by April 2027. ❑ BoE: Bank rate unchanged at 3.75% in 2026. ❑ PBoC: Unchanged at 1.4% through 2026.

Key macro & markets forecasts

GDP growth (%)			Central bank policy rate (%)				Key market forecasts			
	2026F	2027F		Current	Sep 26F	Dec 26F		Current	Sep 26F	Dec 26F
Global	3.0	3.2	US: Federal Funds Rate	3.625	3.63	3.63	US 10Y yield (%)	4.54	4.60	4.70
US	2.2	2.2	Eurozone: Deposit Facility Rate	2.00	2.50	2.50	EUR 10Y yield (%)	3.04	3.10	3.20
Eurozone	0.5	1.1	Japan: Policy Balance Rate	0.75	1.00	1.25	S&P 500	7,387	7,500	8,000
Germany	0.5	1.3	UK: Bank Rate	3.75	3.75	3.75	Gold	4,168	5,500	5,800
Japan	0.7	0.7	China: 7d OMO Rate	1.40	1.40	1.40	EUR/USD	1.15	-	1.20
UK	1.0	1.2					USD/JPY	160	-	150
China	4.7	4.5								

2026 Macro events calendar

June 2026			July 2026			August 2026		
10	CA	Bank of Canada Decision	8	US	FOMC Meeting Minutes	6	EZ	ECB Economic Bulletin
11	EZ	ECB Decision	15	CA	Bank of Canada Decision	19	US	FOMC Meeting Minutes
15-16	JN	Bank of Japan Decision	23	EZ	ECB Decision			
17	US	Federal Reserve Decision	29	US	Federal Reserve Decision			
18	UK	Bank of England Decision	30	UK	Bank of England Decision			
25	EZ	ECB Economic Bulletin	31	JN	Bank of Japan Decision			



Recent editions

- ❑ [1999 meets 1990](#), 9 June 2026
- ❑ [The Impact of the Middle East](#), 10 March 2026
- ❑ [Anything but dull](#), 1 December 2025

Analyst Certification

The views expressed in this report accurately reflect the personal views of the undersigned lead analyst(s). In addition, the undersigned lead analyst(s) has not and will not receive any compensation for providing a specific recommendation or view in this report. Marion Laboure / Camilla Siazon / Jim Reid



Appendix 1

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Deutsche Bank
Research

Geopolitics beyond Iran

Events, trends and themes markets should be paying attention to

June 2026

Dr. Helen Belopolsky | Miha Hribernik

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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Beyond Iran: What should markets be paying attention to?

The Iran crisis has rightly dominated market attention since 28 February, given its far-reaching implications for the Middle East region and the global economy. But the world hasn't stood still during the conflict. This report takes us beyond the immediate headlines to the emerging geopolitical dynamics that could redefine market trajectories and create the next wave of strategic opportunities.

Even as US attention shifted toward the Middle East, the last several months saw signs of movement toward a potential ceasefire in **the war in Ukraine**, and a deal remains possible amid mounting war fatigue. Over the next year, Europe will face a number of key elections that will likely serve as major watchpoints for the popularity of **far right movements and candidates**, with potential EU-wide implications. **China's efforts to position** itself as a reliable geopolitical partner amid rising global uncertainty appear to be bearing some fruit, including in the form of improved global perceptions and signs of success with RMB internationalization.

Meanwhile, **global defence spending** has continued to soar, and Asia is fast emerging as one of the centres of a global defence spending 'supercycle', both as a buyer and seller of arms. **Drones** are increasingly at the centre of this structural trend, as countries across the globe race to absorb the lessons of the wars in Ukraine and Iran – which should add further impetus to defence spending.

But this list is hardly exhaustive as global geopolitical fragmentation continues to accelerate. We've also asked our proprietary model **dbLumina** for its take on the geopolitical themes beyond Iran that markets need to pay attention to.

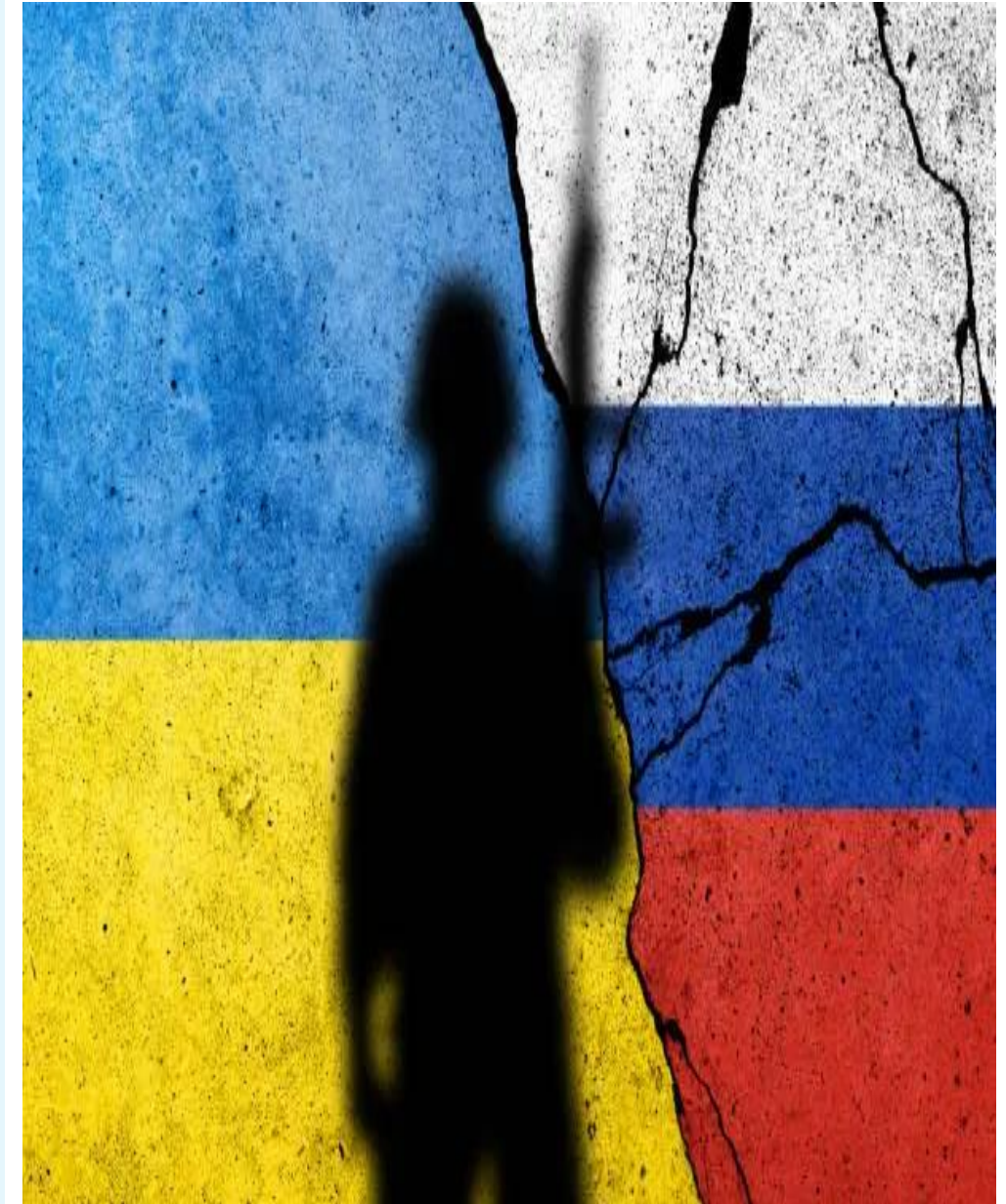
War in Ukraine

Is there an end in sight?



Though trilateral negotiations (US/Russia/Ukraine) are stalled as a result of the war in Iran, a number of recent developments suggest we **may be nearer a ceasefire than many believe**, and this could potentially move quickly. While military barrages continue, as war fatigue mounts, the rising costs of protracted conflict may eclipse its diminishing returns, shifting the strategic calculus. Certainly, polling is demonstrating a marked shift. Russian support for peace talks reached a record high of 67% in March while 72% of Ukrainians say they could support a peace plan that freezes the conflict along the current front line. In addition, casualties in the war have reached the highest rates in modern warfare. Some signs that we may be seeing progress towards a ceasefire:

- On 4 June, Ukrainian President Zelenskyy wrote an open letter to Russian President Putin urging a meeting to agree on an end to the war. Putin has so far rejected the proposal but agreed to another round of negotiations after 22 June.
- Russian President Putin publicly said the war may be “coming to an end” and his recent language contrasts with earlier maximalist messaging. The Kremlin is increasingly speaking more about security arrangements and mediation possibilities than outright military victory.
- US, Russian and Ukrainian officials reportedly have continued negotiations.
- A senior Ukrainian delegation recently travelled to Washington for renewed negotiations with US officials on possible frameworks for freezing the war.
- **Washington may well increase pressure on Moscow and Kyiv** if negotiations remain stalled into the summer. US pressure could be the decisive catalyst, especially if we see shifts to actively conditioning Ukraine support on negotiations or signalling a capped long-term commitment. For Russia, the US could leverage both “carrots” and “sticks” by, for example, offering a pathway to sanctions relief, access to frozen assets, or security architecture talks and/or through increased sanctions or the provision of long-range weaponry to Ukraine.
- If we see increased US pressure, **a deal could emerge relatively quickly**. While a ceasefire over the coming months is far from guaranteed, even credible steps toward a deal would have effects for markets. Any ceasefire would have significant implications for the post-conflict landscape. This could include a possible easing of some sanctions on Russia, the potential for increased Ukrainian agriculture exports, a pathway for Ukrainian accession to the EU and for enabling reconstruction – with implications for investments in minerals, gas infrastructure, technology, city reconstruction, defence, data centres and artificial intelligence.



Source: iStock: Tomas Ragina

Far right politics in Europe

What may upcoming elections mean for the future of Europe?



Far right parties have made significant gains in the 2024 EU elections and recent national polls, notably in France, Germany, and Austria. The **"normalization" of far-right ideas in Europe carries significant implications** for the continent's future, potentially reshaping its political, social, and environmental landscape. Mainstream parties are increasingly adopting certain themes from far-right narratives, potentially as a strategy to appeal to a broader electorate. Looking ahead, key far-right parties' priorities that we should watch in upcoming elections:

Migration and Border Control: This is perhaps the most unified policy area for far-right parties. They consistently advocate for hardening borders, stricter immigration laws, and processing asylum claims outside the EU. Even mainstream parties have adopted more restrictive migration policies in response to the far-right's influence. Successes for far-right parties in upcoming elections could translate into tightening of EU migration policies with possible implications for the EU labour market.

Climate Policy: Many far-right parties express scepticism of ambitious climate policies, including the European Green Deal. There is a risk of a "green backlash," where far-right influence could slow or dilute the implementation of climate-related directives.

National Sovereignty and EU Integration: Far-right parties generally prioritize national sovereignty over further EU integration. While some have moderated their calls for leaving the EU entirely, many advocate for reforming it to reduce its mandate. This focus on national interests could lead to increased challenges in reaching consensus on critical EU issues, including the development of a unified foreign and security agenda, the allocation of the EU budget, and efforts to enhance the EU's overall economic competitiveness..

Economic Policy: While there are internal divisions within the far-right on specific economic policies, a common thread is a push for greater national control over economic decisions. Some advocate for strong fiscal discipline, while others oppose EU-induced fiscal rules. Protectionist economic policies, such as tariffs and non-tariff trade barriers, could also gain traction.

Key European Elections in 2026/2027 with Far-Right Touchpoints	
Germany	State elections in Baden-Württemberg, Rhineland-Palatinate, Saxony-Anhalt, Berlin, and Mecklenburg-Vorpommern – MAR to SEP 2026
Denmark	General election is set for OCT 2026
Latvia	Parliamentary election is scheduled for OCT 2026
Sweden	General election is scheduled for SEP 2026
France	Presidential election is scheduled for APR 2027
Italy	General election will be held no later than DEC 2027
Poland	Parliamentary election will be held by NOV 2027
Spain	General election will be held no later than AUG 2027
Finland	Parliamentary election will be held on APR 2027
Greece	Parliamentary election is scheduled to be held before JUL 2027
Slovakia	Parliamentary election is scheduled to be held by SEP 2027

Uncertainty as Opportunity for China's Global Influence

Beijing is benefitting from perceptions of stability amid global geopolitical uncertainty

Rising geopolitical uncertainty should continue to generate upsides for China. Over the past year, Beijing has sought to position itself as a more predictable global power in contrast with US foreign, trade and security policy volatility. While there are probably limits to upsides from this approach – amid deep-seated structural differences and continued derisking in major markets such as the EU – Beijing's positioning appears to have had some success. This is reflected in [evidence](#) of improving global perceptions of China and a flurry of senior foreign visits to Beijing during 2026.

These visits included the leaders of Canada, Ireland, Finland, the UK, Vietnam, and Germany, and have – among other deliverables – resulted in a US\$15bn UK [investment pledge](#) into the pharmaceutical industry in China; progress on a planned [China-Vietnam railway link](#); a deal for [120 Airbus jets](#); and cuts to Canadian [tariffs on Chinese EVs](#).

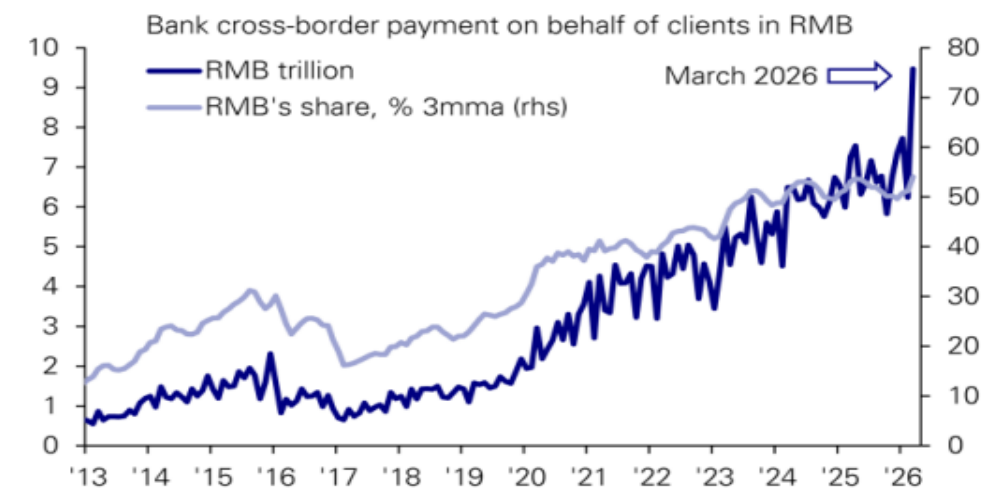
The global energy shock driven by the Middle East crisis could lead to further upsides for Chinese renewables, EV exports, the RMB, and Hong Kong. While not immune to the war's growing economic cost, China's economy has also shown resilience, which included a 14% y/y rise in exports in April (and an [estimated](#) 15% y/y rise in May), in part attributed to global stockpiling amid the Iran war. Combined with perceptions of geopolitical stability and structural advantages owing to investment in strategic industries, this could leave China with a strengthened geopolitical position after the war.

Chinese renewables companies and EV manufacturers have seen an export boom in recent months as consumers in key markets, including Europe, shift away from fossil fuels amid high prices (this has continued, although slowed following the cancellation of China's solar panel export tax refund programme on 1 APR). A likely acceleration in state-led decarbonization efforts after the war across many parts of the world could generate sustained demand for these companies over the medium term.

As our colleagues have argued, there are [tentative signs](#) of an acceleration in RMB internationalization – one of Beijing's key strategic objectives. This was reflected in a 30% jump in cross-border RMB payments during the first month of the war, and in China successfully securing [yuan pricing](#) for some Australian iron ore shipments. The Iran conflict could be a catalyst for the erosion of the petrodollar, and could result in the beginning of a '[petroyuan](#)'. Hong Kong could see further upsides from the conflict, including in the form of inflows of capital, skilled workers and family offices from the Gulf.

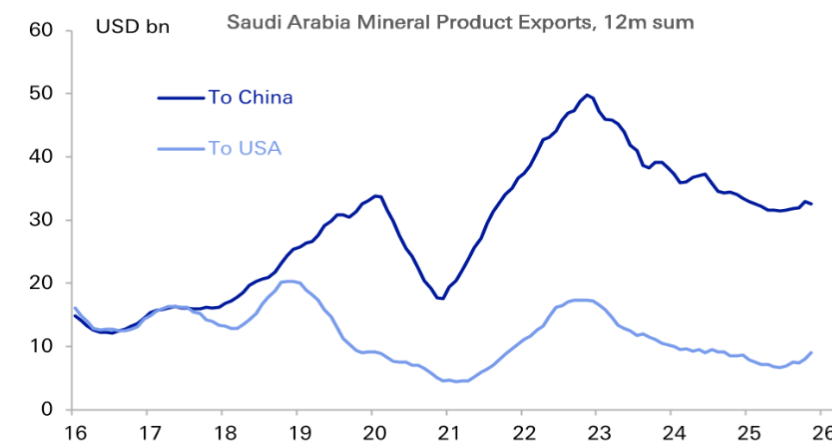


Cross-border RMB payment surged by 30% in March



Source: Deutsche Bank Research. See: [The Big FX View: War and Peace](#)

Saudi Arabia now sells four times as much oil to China as to the US; a trend that could undermine petrodollar foundations



Source: Deutsche Bank. See: [FX Special Report: What Iran means for the dollar: a perfect storm for the petrodollar](#)

Spotlight on Defence: A New Era of Growth in Asia

Asia is emerging as one of the global centres of a new defence spending supercycle, both as a buyer and seller of arms



The fracturing of the post-1945 international order is fuelling a structural defence spending supercycle. Global defence outlays reached [USD2.6 trillion](#) in 2025. In Asia, spending grew by 5.7% y/y last year, reaching [USD573bn](#). We expect geopolitical factors – such as European concerns over Russia, US requests for allies to boost spending and capabilities, China's military modernization, and [intra-NATO friction](#) – to sustain further growth.

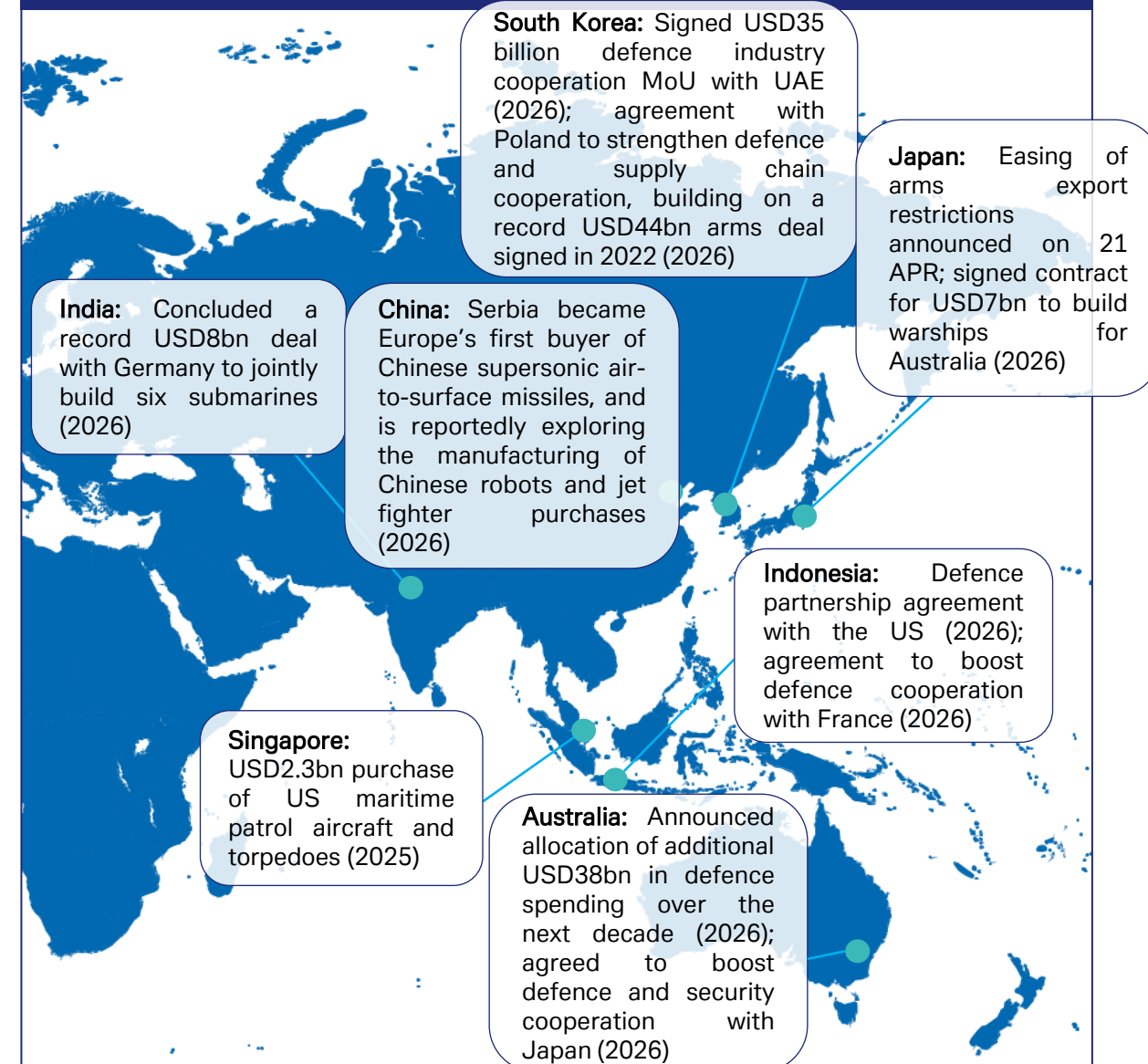
We assess that this supercycle is unique due to Asia's emerging role as a major arms exporter, not just a buyer. The past year has seen significant activity, including record arms deals and major policy shifts (see map). Asia is unlikely to displace leading suppliers but competitive pricing, access to advanced technology, and increasingly integrated supply chains could boost Asian market share in countries traditionally dominated by non-Asian suppliers.

China and South Korea are already two of the world's largest arms exporters. South Korean exports surged by a [reported](#) 83% between 2024 and 2025, and the country is now the [second largest](#) arms supplier to NATO, behind the US. Poland and the UAE are among its biggest buyers, and both could increase purchases amid rising security risks in their regions. The Iran war could boost Gulf defence spending, as the region [reportedly](#) focuses on South Korean anti-missile systems given potential delays in the replenishment of US-made munitions.

Asian arms exports are set to accelerate. Japan's eased restrictions on 21 APR could lead to the biggest increase in Japanese arms exports since WWII, with Poland and the Philippines [reportedly](#) among potential buyers. Meanwhile Taiwan's status as a producer of 90% of the world's advanced chips means its technology is seeing widespread use by the US military, including in F-35 fighter jets and for cutting edge AI development, including for military applications.

Rising Asian defence spending is also creating opportunities for domestic and overseas suppliers. Regional defence spending appears set to climb further, fuelling purchases of defence systems, from submarines to fighter jets and drones. This includes recently proposed or approved spending increases in South Korea, Japan, Taiwan, India, Australia, and the Philippines. Underlying geopolitical imperatives suggest the trend is unlikely to reverse over the coming years, with global implications, including in the form of a potentially weaker dollar (see [FX Special Report: How defence spending will shrink global imbalances and weaken USD](#)).

Selected policy developments and arms deals in Asia, 2025-26



Source: Deutsche Bank

Spotlight on Defence: The Unmanned Revolution

Competition for leadership in the drone revolution is set to intensify



The proliferation of inexpensive defensive and offensive drones is reshaping military doctrine, defence spending, and procurement. The wars in Ukraine and Iran have demonstrated the limitations of costly military assets in the face of numerous cheap and disposable drones. The range and capabilities of these systems continue to rapidly evolve, with the integration of ground robots fast emerging as the next frontier. Ukraine has announced plans to use robots for 100% of frontline logistical needs, and plans to more than double the number of unmanned ground vehicles by mid-2026. However, once there is no active combat and after a certain amount of stockpiling, production will likely slow significantly given how quickly capabilities change.

We expect significant global increases in investment as governments absorb the lessons of ongoing wars. Ukraine has emerged as a global leader in unmanned operations and production, manufacturing around 4 million drones annually (more than the rest of NATO combined), with a [planned](#) increase to 7 million in 2026. Ukrainian production is expanding to facilities in countries including the UK, Denmark, Norway, and Germany, fostering technological diffusion and the emergence of a Europe-wide supply chain. Ukraine is also pursuing joint production with countries in the Middle East, Europe and the Caucasus, and has struck 10-year cooperation agreements with Saudia Arabia, the UAE and Qatar. But recent developments also illustrate the rapid pace of development beyond Ukraine, where policy incentives and spending hikes are set to fuel a massive expansion of drone arsenals over the coming years (see table).

National security concerns will likely drive efforts to reduce critical dependencies, insulate supply chains, and secure an edge in AI. China continues to dominate drone supply chains, including for critical inputs such as motors, batteries, lenses, processed critical minerals, and rare earth magnets. We anticipate increasing efforts by governments, including the US and EU, to reduce these dependencies over the coming years. This will likely take a three-pronged approach: a) curtailing market access to Chinese technology; b) accelerating investments to reduce critical dependencies (e.g., in critical minerals extraction/processing, manufacturing, and AI capabilities) and c) countering vulnerabilities through advancing technological solutions that include rare-earth-free e-engines and more sustainable navigation and power systems for drones. The EU's cooperation with Taiwan, for example has led to a [749% y/y growth](#) in Taiwan's drone export sector.

Precedent suggests that China could seek to slow such derisking efforts. On 24 APR, Beijing banned dual-use item exports to seven European companies due to their links to Taiwan, [purportedly](#) to counter EU-Ukraine-Taiwan security cooperation and drone supply chains.

Selected Recent Developments, including Planned or Proposed Spending Increases

United States	The US Department of War has requested a 24,000% increase in FY2027 funding for its Defense Autonomous Weapons Group, which oversees mass-produced drones with AI capabilities. The Pentagon has placed an order for 30,000 one-way attack drones (the largest such order to date), and expects to receive more than 300,000 by early 2028
France	Plans to expand its stock of explosive drones by up to 400% by 2030
Australia	Intends to more than double funding for anti-drone capabilities over the next decade
Germany	German defense minister stated that Germany is planning to spend EUR10 billion (US\$11.5 billion) on drones over the coming years
Japan	Allocated US\$800 million in FY2026 for a coastal defence system that includes unmanned systems; may allocate US\$1.2 billion for new drones through FY2027
South Korea	Intends to train 500,000 military drone operators and procure 11,000 commercial drones for military use in 2026
European Union	"Drone Wall" or "Drone Shield" – multi billion EURO commitment

What does AI say?



While attention is firmly focused on the crisis in the Middle East and the state of the Strait of Hormuz, we asked our proprietary AI model, dbLumina, to review our report and identify any additional geopolitical themes beyond the Iran crisis. The below assessment is entirely generated by the AI model.

European Geopolitical Alignments: Fallout from US Actions

The current US foreign policy posture could trigger an unprecedented crisis for NATO and broader European unity. The watchpoint is how European states respond to such events: whether they choose to pursue economic measures against the US, and how traditional alliances evolve when faced with US unilateralism. This dynamic will redefine transatlantic relations and European strategic autonomy.

Greenland: Escalating US Control & Resource Scramble

President Trump explicitly indicated the US would act on Greenland "whether they like it or not". This signals an intense push for strategic control and access to Greenland's vast resources. The watchpoint is the potential for increased US military presence, long-term lease negotiations, or even more assertive actions, despite unified Danish and Greenlander opposition.

Taiwan Strait: Persistent Tensions & Cross-Strait Stability

The ongoing strategic rivalry between the US and China, particularly concerning Taiwan, remains a critical flashpoint. The broader context of "digital sovereignty and tech decoupling," "rewiring of global trade," and intensified competition for "critical minerals" directly implicates the Taiwan Strait due to its centrality in global semiconductor supply chains and major maritime trade routes. Any escalation here could have immediate and cascading global geopolitical and economic consequences.

Re-evaluation of Multilateralism & Global Governance

US unilateral actions are actively challenging established international law and norms. This trend creates systemic uncertainty for global governance mechanisms, impacting trade agreements, environmental accords, and the enforceability of international contracts. Watch for key international forums (e.g., UN, G7/G20 summits) to see if major powers continue to bypass or actively undermine multilateral institutions, leading to further fragmentation and a less predictable global operating environment.



Source: Growtika on [Unsplash](#)

Selected key geopolitical events to watch in 2026

For a detailed look at key monthly geopolitical events, please also see our [This Month in Geopolitics](#) publication series.

Date	Market(s)	Comment
15-17 June	France/G7/China	France will host the G7 Leaders’ Summit in Évian. Talks will likely include efforts to coordinate a response to major geopolitical issues, including the global impacts of the Iran war, the Russian invasion of Ukraine, competition with China , and supply chain resilience.
1 July	US/Canada/Mexico /China	The start of the mandatory six-year joint review of the US-Mexico-Canada (USMCA) trade agreement . Efforts to reduce reliance on China , and to curb the role of Chinese companies in North American supply chains, will be a watchpoint.
7-8 July	NATO Summit	The NATO summit in Türkiye is expected to be one of the most important alliance meetings in years. Major issues include the future of support for Ukraine; European defence spending; NATO’s posture toward Russia; burden-sharing tensions with the US; and security coordination in the Black Sea and the Middle East.
24 July	US/Global	The US global 10% Section 122 tariffs are set to expire. These will likely be replaced by new tariffs introduced as a result of two Section 301 trade investigations. Dozens of countries could face higher US tariffs as a result.
Early August	China	China’s leaders will gather for their annual informal summer retreat in Beidaihe near Beijing. The venue provides an opportunity for Party leaders to exchange views on major policy issues, including relations with the US.
8 September	UN General Assembly in New York	The meeting in New York will bring leaders together amid increasing geopolitical volatility and the process to choose the next UN Secretary-General before Antonio Guterres leaves office at the end of 2026.
13 September	Germany	German state and local elections may impact Chancellor Friedrich Merz’s governing coalition amidst economic pressures, immigration debates and the continued rise of the far-right Alternative for Germany (AfD).
September	India/BRICS	India will host the BRICS Summit in New Delhi. Watchpoints include any announcements that could suggest efforts to move further away from the US dollar. This includes BRICS Pay (an initiative meant to facilitate cross-border payments in BRICS currencies) and the BRICS ‘Unit’, a prototype gold-backed digital currency unveiled in 2025.
24 September	China/US	China’s President Xi is expected to visit the United States following the China-US Summit in Beijing in May.
3 November	US Midterm elections	These midterms will see voters choose 435 seats in the House of Representatives, 35 Senate seats and governors in 39 states and territories. The US midterms will be vital in determining control of the Congress and the Senate, which will significantly influence President Trump’s ability to pass laws and shape national policy.
18-19 November	APEC Summit	China will host the APEC Economic Leaders’ meeting in Shenzhen . The summit could see a Xi-Trump meeting on the sidelines, and may shape Indo-Pacific economic architecture for years to come.
14-15 December	G20 Summit	The 2026 G20 Summit in Miami may become contentious due to Ukraine-related divisions, trade and tariff disputes, climate financing disagreements and questions around global governance reforms. The summit could also see a fourth meeting between presidents Xi and Trump in 2026.



Source: [Nils Huenefruerst](#) on [Unsplash](#)

Appendix 1



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Assessing AI in finance with Evident Co-CEO Alexandra Mousavizadeh

June 9, 2026

Banks are racing to invest heavily in AI to avoid being left behind, even though the impact is yet to show up significantly in their bottom line, cost-to-income ratios or share prices, Alexandra Mousavizadeh, Co-Founder and Co-CEO of Evident, said in a [Deutsche Bank Research Institute](#) dbInSights webinar last week.

The industry has spent the past three years testing AI and is now entering a new phase of production at scale, said Mousavizadeh of [Evident](#), whose AI Index is the global standard benchmark of AI maturity for financial services companies.

“We are still in the foothills of the journey,” she told Adrian Cox, Thematic Strategist for the Institute, adding that investment in building in-house research teams, creating an engineering culture and thinking through workflows and architectures over the past decade is paying off for early movers.

“The longer you wait, the harder it is going to be,” she said.

Click [here](#) to watch the half-hour webinar. (You will need to download this report as a pdf to access the link.)



It will probably take another two years to see the results at the leading banks and another eight to ten years for the technology to be fully diffused throughout the industry, she said.

The most impactful cases are not always the most sophisticated, Mousavizadeh said. Streamlining Know Your Client (KYC) activities can provide an instant revenue lift, often from simply re-engineering old workflows, with AI layered on top to automate and accelerate them. In the meantime, more complex use cases are emerging, such as AI agents to stitch together different processes and departments, she said.

The webinar also includes debates on why Mousavizadeh sees no jobs apocalypse in finance, what differentiates successful adopters of AI, how different regions are diverging, and what advice she would give a young professional entering the workforce.

Authors

[Adrian Cox](#)
Research Analyst



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Deutsche Bank
Research

dbDataInsights Geopolitics Survey

UK Politics: Perspectives on leadership transition and key priorities

June 2026

Dr. Helen Belopolsky | Sanjay Raja | Debbie Jones

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

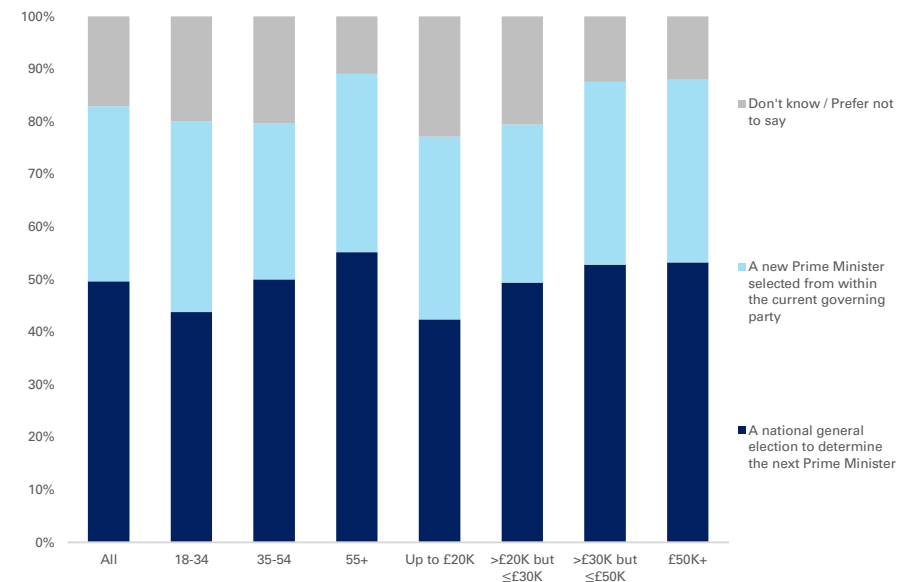
Deutsche Bank
Research Institute

Population would prefer election to determine the next Prime Minister



- Prime Minister Starmer's administration is currently navigating a complex political landscape, contending with [economic pressures](#), the implementation of key policy initiatives, and recent instances of political controversy. In addition, Labour saw [poor results](#) in the latest local elections. By mid-May, ~100 Labour MPs had called on PM Starmer to resign or set a departure timetable – though 111 backbenchers signed a letter opposing a leadership challenge. Speculation around the Prime Minister's position remains, with rival Andy Burnham running to return to Westminster as part of the Makerfield by-election.
- Our dbDataInsights survey suggests that voters are considering the potential for significant political change and its implications. **In a scenario where the Labour Party were to elect a new successor, a notable 50% of respondents expressed a preference for a national general election** to determine the next Prime Minister, rather than an internal selection process.
- This sentiment is particularly salient as PM Starmer is currently only one of the last five PMs to have entered Downing Street as a result of a general election victory. The preference for broader electoral participation is observed to be strongest among the oldest age cohort, indicating a desire for a more direct public mandate in future leadership changes.

UK participants were asked: if the UK Prime Minister did change soon, which of the following would you most prefer: a new PM selected from within the governing party or a general election?



Methodology

These results are derived from a monthly online nationally representative survey. We survey ~550 respondents in European countries and the UK, and ~650 respondents in the US. Our protocol ensures proportional representation across key demographics such as age, income, location, and gender to accurately reflect population diversity. Participants are restricted from taking the survey more than once every six months. The geopolitics questions are integrated into a larger monthly survey conducted by the dbDataInsights team.

Source: dbDataInsights

Responses collected in May 2026 from an Online Nationally Representative sample of ~650 US respondents.

Cost of living an overwhelming concern for Britains



- Cost-of-living concerns are seen as the key priority to be tackled by any new Prime Minister. This is particularly salient given the volatile global environment, where escalating trade tensions, persistent tariff pressures, and broader geopolitical uncertainty fuel inflationary pressures on essential goods and services (namely, energy and food). Such external factors inevitably translate into significant impacts on household purchasing power and overall consumer sentiment domestically.
- This is reflected in the survey data, with an overwhelming **75% of respondents identifying the “cost-of-living crisis”** as the most critical immediate issue for a new Labour Prime Minister. The consistently high salience of this figure across all demographic groups underscores its importance. Addressing it would be a foundational priority for any incoming government seeking broad public support.
- Beyond economic pressures, the **National Health Service** also stands out as a significant immediate focus, identified by 49% of respondents – a reflection of an ageing society. This highlights a widespread public expectation that the health service requires urgent attention and resources beyond the multi-billion-pound investment delivered thus far.
- Furthermore, demographic-specific priorities are evident within the survey results. A notable **62% of the oldest age cohort surveyed identified immigration** as a key issue requiring the attention of any incoming Prime Minister.

Source: dbDataInsights

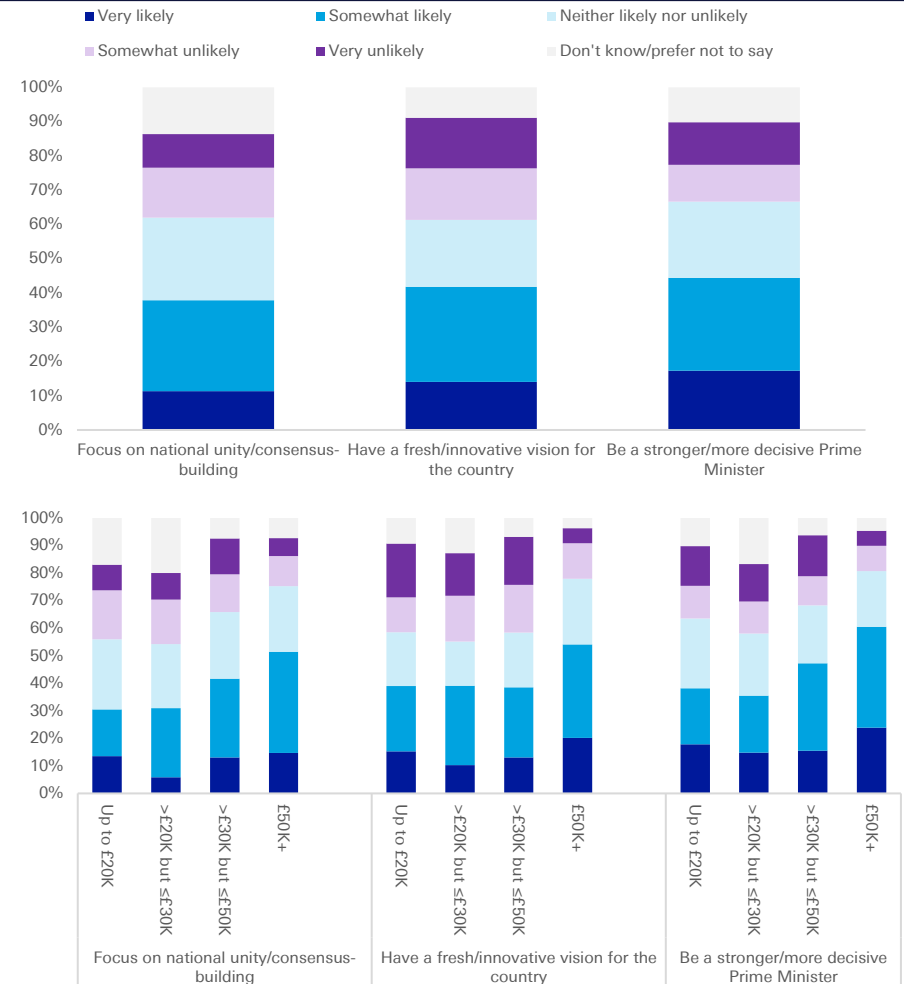
Responses collected in May 2026 from an online nationally representative sample of ~550 respondents in European countries and the UK, and ~650 in the US. See slide 1 for further information.

Focus on unity through decisiveness



- 40% of respondents believe a new Labour Prime Minister should primarily concentrate on **fostering national unity**. This figure rises slightly to 44% among those who anticipate a stronger and more decisive leadership approach, suggesting a segment of the electorate desires clear and firm direction in navigating national challenges.
- Approximately 42% of respondents anticipate that a new Labour Prime Minister would introduce **a fresh or innovative vision for the country**. This sentiment is particularly pronounced among higher-income demographics, with 61% of those earning over £50,000 annually expressing confidence in such a prospect.
- This suggests that while there is a general appetite for new ideas, specific segments of the electorate, particularly those with higher incomes, may hold stronger hopes for transformative policy directions under a new Labour leader.
- “Faith” in a new leader appears to diminish with age, suggesting younger cohorts are more optimistic about a fresh start.

UK participants were asked: How likely or unlikely do you think it is that a new Labour Prime Minister would focus on national unity/consensus-building, have a fresh/innovative vision for the country, and/or be a stronger/more decisive Prime Minister?



Source: dbDataInsights

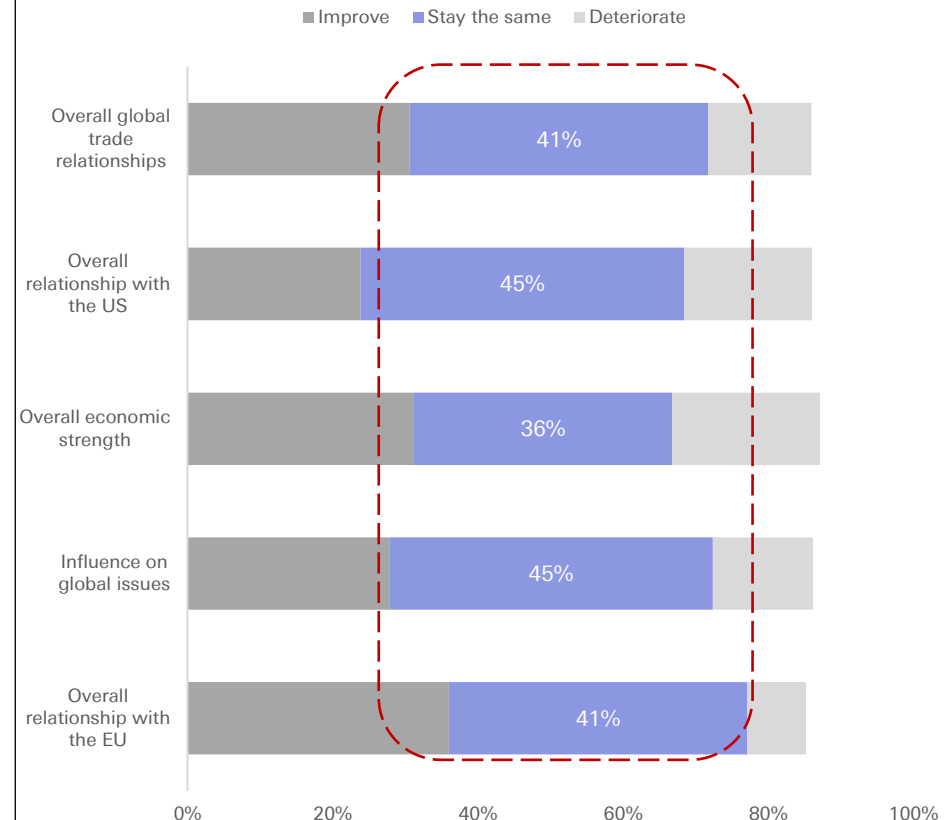
Responses collected in May 2026 from an online nationally representative sample of ~550 respondents in European countries and the UK, and ~650 in the US. See slide 1 for further information.

Expectations for UK's international standing and economic future



- The data points to a public mood that is guarded, indicating that any new Prime Minister would face the challenge of not only delivering tangible improvements but also shifting perceptions about the nation's global trajectory.
- 45% of respondents **expect the UK's influence on global issues and the UK's relationship with the US to remain unchanged** under a new Prime Minister. Only 28% and 24%, respectively, foresee improvement. This could indicate a perception that these areas are particularly entrenched or less amenable to rapid positive change, regardless of a new Prime Minister's efforts.
- In every category, a plurality of respondents anticipates that current conditions will "stay the same", which suggests a prevailing sentiment of cautious realism, or perhaps a lack of strong optimism for immediate positive shifts across critical areas such as international relations, global influence, and economic performance.
- While expectations that "economic strength" and "global trade relationships" would "stay the same" are also prominent (36% and 41%, respectively), a somewhat higher proportion still hopes for improvement (31% in both cases), suggesting slightly more fluidity or a greater desire for change in these economically relevant domains.

UK participants were asked: Under the new Labour Prime Minister, do you think the following would improve, stay the same, deteriorate or don't know: overall relationship with the EU, influence on global issues, overall economic strength, overall relationship with the US, and overall global trade relationships?



Source: dbDataInsights

Responses collected in May 2026 from an online nationally representative sample of ~550 respondents in European countries and the UK, and ~650 in the US. See slide 1 for further information.

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Deutsche Bank
Research

Private capital monitor

Media narratives v the hard reality

3 June 2026

Luke Templeman | Galina Pozdnyakova

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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Executive summary

- Private capital managers' confident Q1 performance is a stark contrast to the media and investor narratives that have focussed on concerns about credit quality, AI disruption and redemptions. We take a closer look at what this divide means for PC in 2026.
- AI disruption: Our risk scoring and indexes show that PC managers were more worried about the 2022-23 rate rises, but today, the media is far more worried about AI.
- Overall in Q1, PC managers became more attuned to various external risks including AI disruption. Still, the large managers remain confident about their portfolios and fundraising.
- Managers emphasised that retail fundraising woes have been more concentrated and less severe than the media presents.
- We show our proprietary survey of individuals in the UK and the US that highlights the virality of negative headlines in the financial press.
- We track decompression in BDCs' non-accrual rates to show that the deterioration comes after quarters of strong growth. Thus, further defaults will require significant stress in the equity layer.
- We continue to expect PC dealmaking to recover as the most intense adverse headlines subside.
- Last, we take a look at the impact of market volatility amidst geopolitical tensions on realisations and dealmaking. We expect both to be resilient, including IPOs, as both corporates and dealmakers have become accustomed to market turbulence this decade. We also note that softness in sponsored deals is less pronounced outside tech and managers are eager to invest in key themes such as infrastructure, energy and defence.

The background of the slide is a photograph of two hands holding a large orange and a large red apple. The hands are positioned at the bottom, with fingers visible. The orange is on the left and the apple is on the right. A semi-transparent dark blue rectangle covers the middle portion of the image, serving as a backdrop for the text.

01. Q1 2026 results

What do earnings calls from large, listed managers say about the private capital environment?

Key themes

Q1 was characterised by a complex macro environment, with managers navigating geopolitical tensions, fears of AI-driven disruption, and a bifurcation in private credit markets. Despite this, management teams are confident amidst strong fundraising, accelerating demand for AI-related infrastructure, and strategic advantage of diversified, scaled platforms. Firms are differentiating themselves through operational expertise and a disciplined investment processes, while being ready for opportunities from market dislocations.

1. AI as a generational investment catalyst, especially in infrastructure

- The AI revolution is the most dominant theme, viewed as a massive, multi-decade tailwind. The focus is less on direct software investment and more on providing the essential physical infrastructure required to power AI. Management consistently highlighted the enormous capital required for data centres, power generation, transmission, cooling systems, and the broader energy transition.
 - Blackstone noted its portfolio of data centres exceeds \$150 billion, with a \$160 billion prospective pipeline, and stated AI infrastructure is the "single most important thing for the performance of our clients". Ares sized the third-party data centre market opportunity at around \$900 billion.
- Firms are positioning themselves as beneficiaries of the AI build-out, regardless of which software models win.
- Internally, firms are adopting AI to enhance productivity. TPG noted that nearly 80% of its staff uses its customized AI tools daily

2. A bifurcation in private credit markets with strong institutional backing

- Despite negative headlines and "noise" surrounding retail-focused BDCs, managers reported robust institutional demand for private credit, emphasizing a "flight to quality." Several firms noted a clear divergence. While retail channels experienced redemption requests, institutional demand remained strong.
 - KKR noted "meaningful inbound interest from institutions around our direct lending business".
 - Blackstone stated its institutional and insurance clients, representing 75% of its credit AUM, "continued to commit large-scale capital".
- Managers stressed their disciplined underwriting, low LTVs, and focus on senior secured debt as key risk mitigants. Diversification away from sponsor-backed direct lending into areas like asset-based finance (ABF) and investment-grade private credit is a major trend.

3. Fundraising strength driven by scale and client consolidation

- Despite the environment, fundraising was a significant bright spot. Across the board, large, diversified managers are benefiting from a trend of investors consolidating their relationships with fewer, strategic partners who can offer a wide range of products – clients are increasingly seeking multi-asset, customized solutions rather than single-fund products. The GP-led secondaries market is seen as a major growth area, driven by the need for liquidity solutions for fund sponsors.

4. Cautious optimism on monetisations amid geopolitical uncertainty

- The M&A and IPO environment is viewed with caution due to geopolitical instability, but firms with high-quality, well-performing assets see a path to monetization. Firms are actively exiting assets at strong multiples.

Source: Bloomberg Finance LP, Deutsche Bank.



What do earnings calls from large, listed managers say about the private capital environment?

Key risks

The primary risks centre on the disruption of software and other "white-collar" industries by AI, the potential for a broader credit cycle turn, and redemption pressures in the retail wealth channel for private credit.



1. AI disruption to software and "white-collar" business models

- While AI is seen as an opportunity, its potential to disrupt existing business models, particularly in software, is a frequently cited risk. While there is a general acknowledgment that the full impact of AI is unknown and will play out over several years, managers are actively reviewing their software portfolios for AI risk:
 - Ares had a review that concluded that 86% of its portfolio had low AI disruption risk, 13% had medium risk, and only 1% had high risk.
 - TPG noted one of its most exposed funds has 7% exposure to companies in a "mitigate category" where material AI risk is perceived.
 - Blackstone mentioned that software "has come into focus as an at-risk area".
- The consensus is to be defensively positioned, focusing on software with proprietary data and high barriers to entry, while avoiding areas like content generation or simple productivity tools.

2. Redemption pressures in retail private credit vehicles

- The slowdown and net outflows in non-traded BDCs and other retail credit funds are a headwind, though managers frame it as manageable and cyclical.
- While the overall impact on AUM and fees is presented as minimal to firm-wide earnings, it is causing a near-term deceleration in fee growth from these specific products.
- Managers also defended the redemption gates (typically 5% of NAV per quarter) as an essential feature to protect investors from forced asset sales and align the fund's liquidity with its illiquid underlying assets.

3. Geopolitical Instability and Macro Headwinds

- The conflict in Iran and broader geopolitical tensions were cited as creating uncertainty, volatility, and a more challenging operating environment – the conflict has caused a near-term "pause" in transaction activity as sponsors and corporates re-evaluate the landscape.
- Despite the risks, managers believe periods of uncertainty favour scaled, patient investors.
 - Ares noted it experienced its two fastest growth periods during the GFC and COVID crises.

Source: Bloomberg Finance LP, Deutsche Bank.

What do earnings calls from large, listed managers say about the private capital environment?

Scoring key concerns using earnings transcripts and our AI tool, dbLumina – Q4 vs Q1

Managers became on average more concerned about risks last quarter. They are also relatively more concerned about AI disruption and it is now the top 3 risk in our list, though it still trails macroeconomic environment risk by half. Importantly, managers continue to stress they're comfortable with own credit portfolios and fundraising.

AI & digital infrastructure

- Earlier this year, AI was discussed more as an opportunity. Software exposure was noted, but the risk of disruption was less of a focus. Now, the risk of AI-driven disruption to software business models has become a central topic of questioning and a key area of risk management. Firms are proactively quantifying and managing this risk.

Private credit:

- The narrative has become more nuanced. While underlying credit quality remains strong, the sector is experiencing a turbulence in the US retail wealth channel, while institutional demand and opportunities in non-corporate or non-US credit are accelerating.

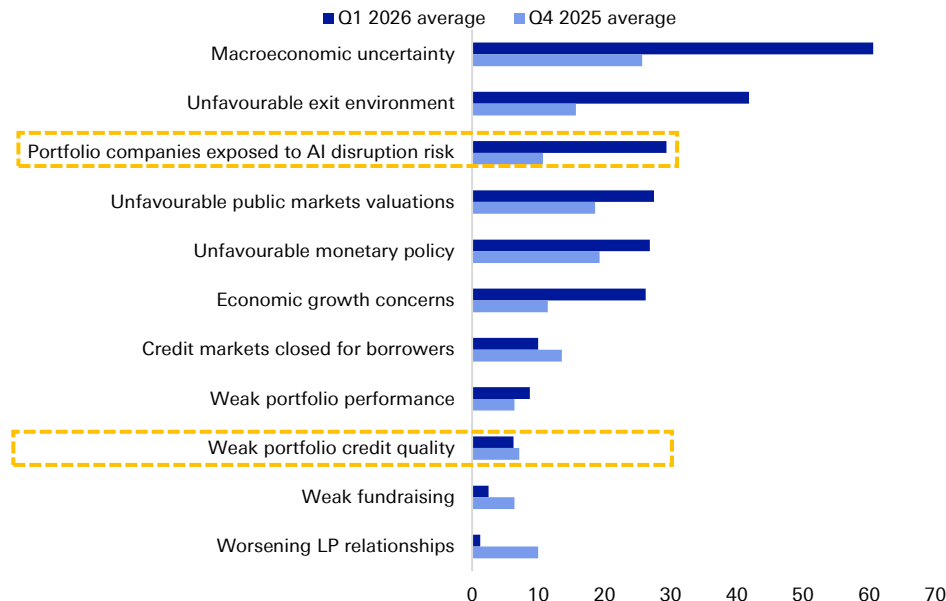
Realisations & IPOs

- The strong monetisation momentum from late 2025 was tempered by early 2026 volatility. However, the underlying pipelines remain robust, suggesting that a stabilisation in markets could lead to an acceleration in H2.

Retail channel volatility

- The wealth channel was a key growth engine, with record inflows reported by many firms for 2025. But the "social media and press reporting" on private credit led to a tangible slowdown and increased redemptions in US-focused retail credit funds – the risk of pro-cyclical flows in the retail channel, previously a more theoretical concern, materialized in Q1 2026.
- Managers also noted that redemption requests are often concentrated among a small number of larger, often institutional-like, investors within these funds, rather than a broad-based retail exodus.
- Firms are emphasizing the structural features of these semi-liquid vehicles and the fact that they have met all redemption requests, which they believe will ultimately build long-term confidence.

Average concern score across 8 large listed managers (from 0, least concerned, to 100, most concerned)



Source: Bloomberg Finance LP, Deutsche Bank.

Q1 earnings: reported performance by asset class – performance has slowed outside infrastructure



KKR	Q1	LTM
Private equity		
Traditional PE portfolio	1%	10%
Real assets		
Opportunistic real estate	-1%	2%
Infrastructure	2%	10%
Credit		
Leveraged credit composite	-1%	5%
Alternative credit composite	-1%	4%
Apollo	Q1	LTM
Direct origination	0.5%	8.5%
Opportunistic credit	2.5%	11.4%
Multi-credit	1.0%	7.9%
Asset-backed finance	-1.0%	8.2%
Flagship PE	-0.3%	4.7%
Carlyle	Q1	LTM
Global private equity		
Corporate PE	-2%	3%
Real estate	1%	3%
Infrastructure & natural resources	9%	25%
Global credit	4%	15%

Ares	Q1	LTM
Credit		
Alternative credit	3.9%	15.3%
Opportunistic credit	-0.2%	11.7%
US senior direct lending	2.1%	11.8%
US junior direct lending	1.5%	14.8%
European direct lending	2.6%	9.2%
APAC credit	2.5%	20.3%
Corporate private equity	-1.1%	3.5%
Real assets		
Americas real estate equity	0.6%	5.9%
European RE equity	0.1%	3.1%
Infrastructure debt	1.9%	9.0%
Secondaries		
Private equity secondaries	0.9%	7.2%
Real estate secondaries	2.5%	7.9%
Blue Owl	Q1	LTM
Direct lending	-0.4%	8.5%
Alternative credit	1.7%	11.0%
Net lease	2.6%	14.7%
Real estate credit	1.9%	8.5%
Digital infrastructure	3.9%	2.6%

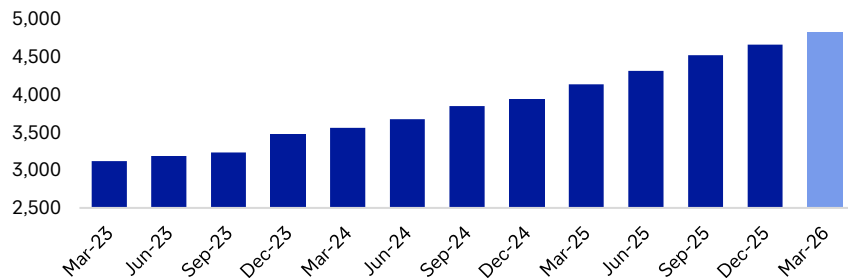
Blackstone	Q1	LTM
Real estate		
Opportunistic	-0.9%	-1.6%
Core+	0.8%	2.6%
Private equity		
Corporate private equity	3.2%	15.7%
Tactical opportunities	3.6%	10.9%
Secondaries	0.7%	12.7%
Infrastructure	7.8%	24.8%
Credit & insurance		
Private credit	0.6%	8.9%
Liquid credit	-1.3%	4.1%
TPG	Q1	LTM
Capital	-2.3%	6.7%
Growth	-1.5%	2.0%
Impact	0.3%	12.5%
Credit	2.2%	10.8%
Real Estate	1.9%	8.3%

Source: Company reports, Bloomberg Finance LP, Deutsche Bank. Note: Performance reflects gross returns or appreciation depending on the manager, strategy and time period.

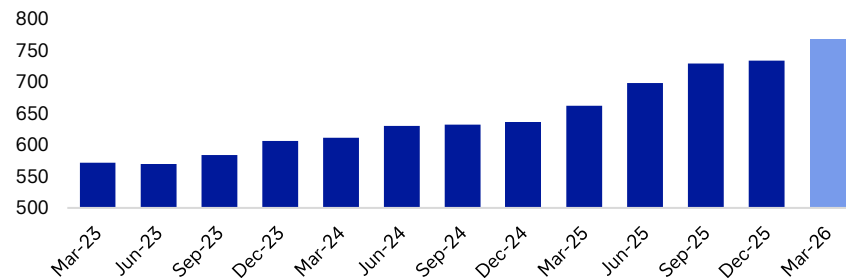
Growth in key metrics for selected large private capital managers – strong AUM growth but softer earnings QoQ



Total AUM in \$bn



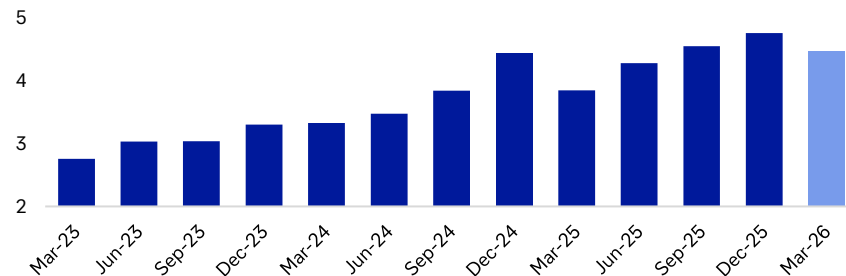
Dry powder in \$bn



Average fee-related earnings margin (%)



Fee-related earnings in \$bn



Source: Bloomberg Finance LP, Deutsche Bank.

Management comments – focus on resilience amidst a more volatile backdrop



- **Stephen Schwarzman of Blackstone:** “Blackstone delivered outstanding first-quarter results despite the turbulent environment, highlighted by almost \$70 billion of inflows and positive appreciation across nearly all of our flagship strategies. Our all-weather model protects us in these times of disruption while also allowing us to invest where we see the greatest opportunity”
- **Joseph Bae and Scott Nuttall of KKR:** “Against a volatile backdrop, monetization activity accelerated, and over the past 12 months we’ve invested more capital on behalf of our clients than at any point in our history. We are highly confident in the strength of our platform and our long-term positioning.”
- **Marc Rowan of Apollo:** “Our first quarter results set a strong tone for the year, with record fee-related earnings and assets under management surpassing \$1 trillion – a testament to the trust our clients place in us and a reminder of the value we create at scale. Whether we’re originating investment grade credit, delivering retirement solutions, or developing new products and capabilities, innovation and discipline continue to drive us forward.”
- **Harvey Schwartz of Carlyle:** “Our first quarter results reflect continued momentum executing against our strategic plan. Carlyle AlInvest delivered another quarter of exceptional growth, fundraising for both institutional and wealth clients had a strong start to the year, and we had a record quarter for U.S. Buyout realizations – each reinforcing our confidence in the path to achieving the 2028 targets...”
- **Michael Arougheti of Ares:** “We reported strong first quarter results highlighted by continued growth across our key financial metrics, including record first quarter fundraising of \$30 billion, up more than 45% year over year. We are on track for another record year of fundraising as we continue to see broad-based investor demand across our platform. We also continue to see strong fundamental performance across our investment portfolios despite the volatile market environment”.”
- **Jon Winkelried of TPG:** “TPG’s strong first quarter performance reflects the significant momentum across our global platform, as the scale of our franchise and consistent execution continue to translate into powerful results. Over the past year, despite an uncertain macro environment, we delivered step function growth across capital formation, deployment, and realizations. Our resilient business model is intentionally built to navigate complexity and capitalize on opportunity. As our clients deepen their engagement with TPG, we are confident in our positioning and ability to deliver long-term growth and differentiated value for our stakeholders..”
- **Doug Ostrover and Marc Lipschultz of Blue Owl:** “Blue Owl’s results for the first quarter of 2026 demonstrate the power of our three differentiated and scaled platforms, each of which has contributed to our continued expansion to \$315 billion of AUM. Our financial results reflect stability, stemming from our durable capital base, and growth, driven by fundraising and ongoing capital deployment.”

Source: Company reports, Deutsche Bank.



02.

Spotlight section: the key narratives weighing on private capital



1. Private capital outflows fears

Private capital managers argue that the "headline-driven" anxiety, particularly in the press and social media, is disconnected from the fundamental reality they are observing. While acknowledging a cyclical slowdown and elevated redemptions in some US retail vehicles, managers assert that the impact is contained and far less severe than portrayed. They point to strong portfolio performance, structural resilience of perpetual funds, and robust demand from institutional clients as counterweights. They highlight investor education, product diversification, and frame the current dislocation as a market share opportunity for scaled, trusted platforms.

Redemptions are concentrated and not systemic:

- Ares: for its non-traded BDC, the firm noted that the "majority of repurchase requests... came from a limited number of family offices and smaller institutions in select regions... Over 95% of our investors did not request redemptions"
- Blue Owl: for its OCIC fund, redemption requests were "concentrated with 1% of investors representing a majority of tenders" and approximately 90% of the investor base did not tender at all.
- Blackstone: on redemptions: "contrary to this popular idea that it's small investors that are leading the charge here, it is actually a smaller number of large investors who are double the size basically on average of the typical investor in these vehicles."

In sharp contrast to the retail wealth channel, managers report strong demand from institutional and insurance clients, who view the dislocation as an opportunity.

- Blackstone: mentioned the "striking the difference in terms of what we've seen from the institutional and insurance clients relative to the wealth channel," and noted that Q1 was one of their best quarters for credit fundraising from these clients on record.
- Ares: stated that "the institutional investor is not anxious. They're not allocating away from private credit. And in fact, I think they're looking at this as a huge opportunity to take advantage of a bizarre dislocation and bring liquidity into the market to capture excess return."
- KKR: received "meaningful inbound interest from institutions around our direct lending business" in the weeks leading up to their call, with clients viewing the dislocation as an "interesting entry point".

While acknowledging the reality of the outflows, managers assess their financial and strategic impact as minimal and manageable, saying that the growth in other areas far outweighs the softness in specific US wealth products.

- Ares: said its two private credit wealth products account for only 4.5% of its overall fee-paying AUM and that in a "very unlikely scenario" where if these funds experienced 5% quarterly redemptions for a full year with no new inflows, it would affect total firm FPAUM by only approximately 1% annually.
- Blue Owl: Net outflows from its two BDCs were less than 6 basis points of the firm's beginning-of-period AUM, with the funds collectively making up less than 17% of total AUM.
- KKR: Its private BDC footprint is "even smaller, around \$3 billion of AUM or 0.4% of our AUM in total".

There is a consensus that the softness is cyclical and temporary. Managers are not altering their long-term plans for the wealth channel. Blackstone said it will continue to expand its wealth team and product suite at a "fairly rapid pace". Ares confirmed it has not changed its fundraising guidance.

- Blackstone, drawing a parallel to the earlier noise around his firm's real estate vehicle, stated, "We're going to get through this like we've always gotten through these moments, and the products will continue to grow".
- Blue Owl expects to "continue to be in a softer environment in wealth" but can maintain its margin targets regardless.

Managers defend the redemption gates (typically 5% of NAV per quarter) as a fundamental and necessary feature of the products, designed to protect investors from forced asset sales and match the illiquid nature of the underlying loans.

- Blackstone noted, "These caps on redemptions are not a bug. They're a feature of these products... The key question is, are you offering a premium in exchange for giving up this liquidity?"
- Blue Owl noted that for its OCIC fund, regular-way paydowns from underlying loans were almost \$3 billion in the quarter, providing 3x coverage for the \$1 billion in gross redemptions, before considering new inflows or credit lines.

Product and channel diversification: Capital that might have gone to a US BDC can be seamlessly deployed elsewhere.

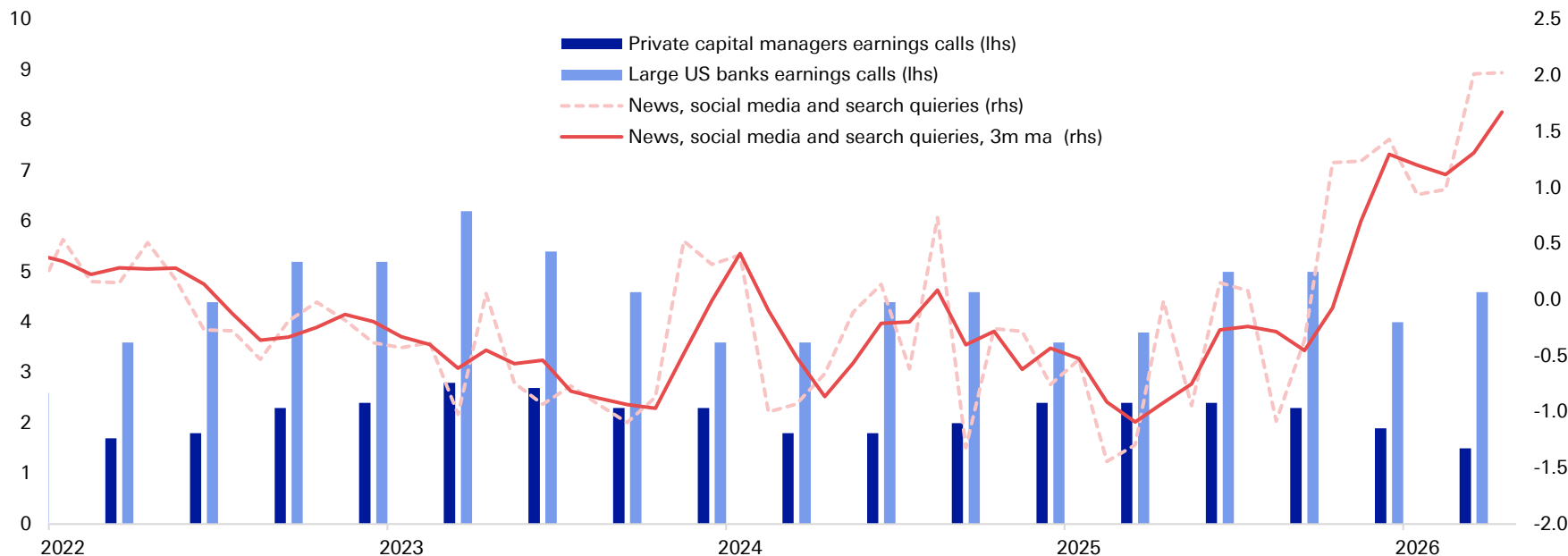
- Ares explicitly stated that any capital not deployed in its non-traded vehicles will be "taken up by other traded and institutional funds and SMAs with limited to no impact to our current year profitability". They also highlighted very strong inflows into their European direct lending and SME funds.

Source: Bloomberg Finance LP, Deutsche Bank.

When we asked our AI tools to contrast sentiment around private credit risks, it is clear that there is a big disconnect between sentiment in the media, which is the most negative it's been for a while, and private capital managers, who have been more concerned about this topic in 2023 when impact of rate hikes and recession risks were at the forefront. This is true for large US banks as well.



AI analysis of sentiment around private credit risk
(higher = more concerns; maximum of 10 for private capital and banks sentiment)

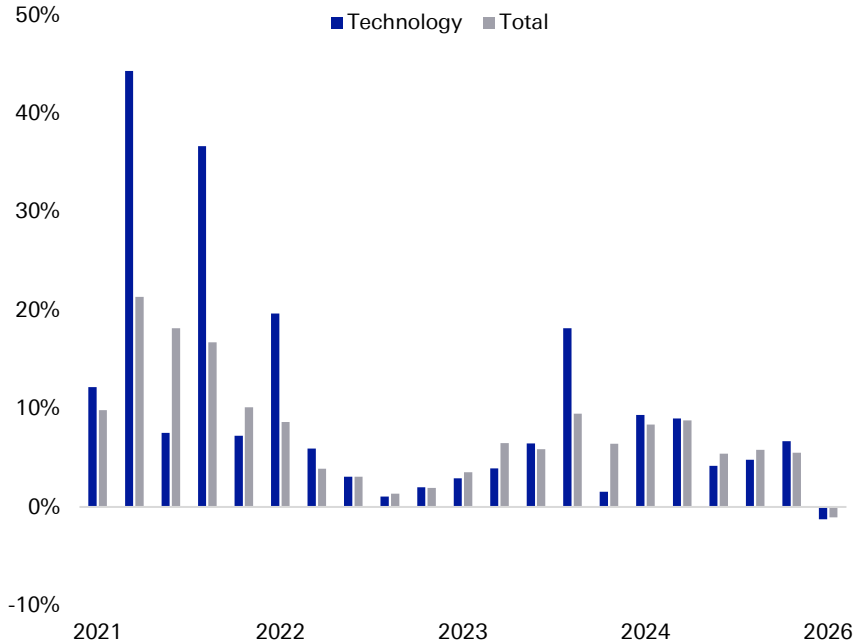


Source: Bloomberg Finance LP, Google Trends, Deutsche Bank.

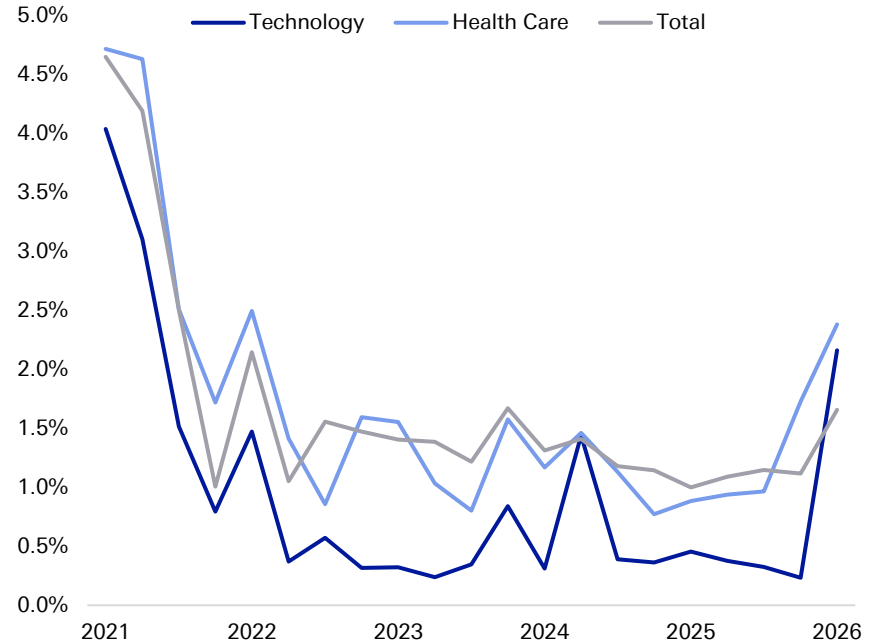
While there has been a lot of focus on net outflows and rising default rates in BDCs, these metrics are worsening after quarters of strong growth and low credit stress economy-wide. Some decompression is reasonable as sentiment gives a pretext for some overdue write-downs, especially in tech. Nevertheless, the equity layer will need to deteriorate more to see big increases in overall default rates. Commentary from PE-driven managers suggests this is not an immediate concern



Growth of fair value of total assets of all selected BDCs *



Non accruals as % of assets at cost, all selected BDCs *



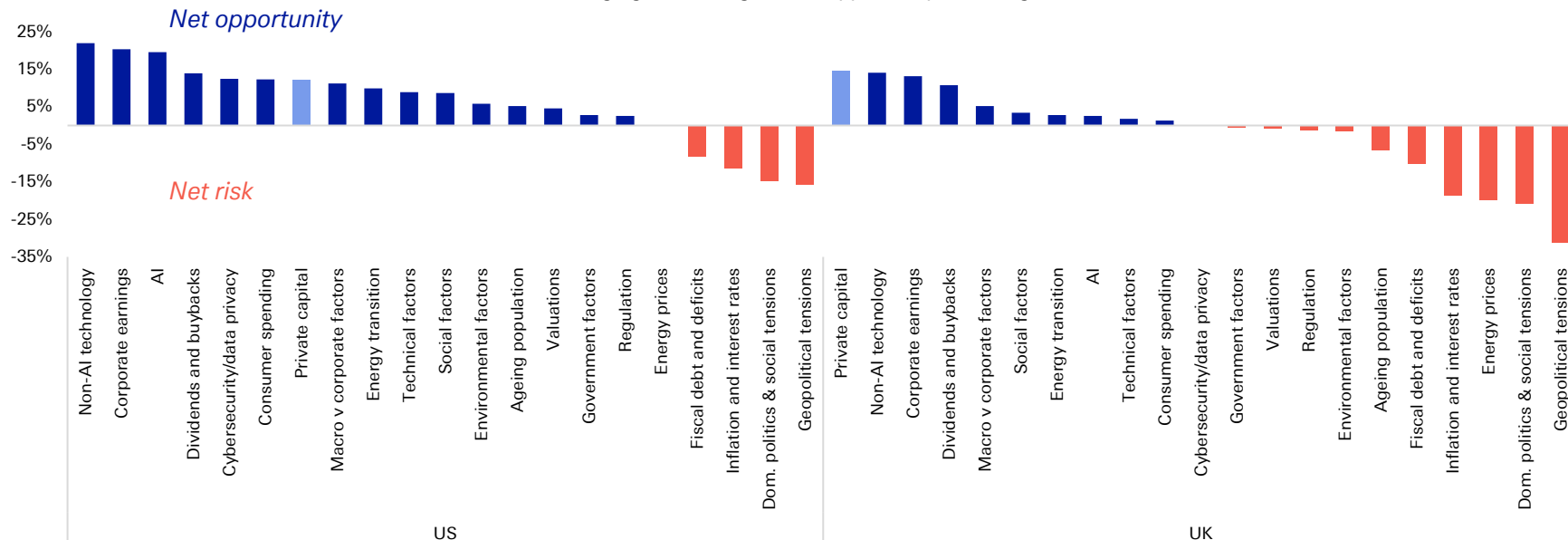
Source: Bloomberg Finance LP, Deutsche Bank. * Note: as tracked by Bloomberg, total assets of c \$420bn.

Our survey of around 500 people in both the US and UK (a representative sample of the population) shows that the sentiment amongst the broader public towards private capital is sanguine. This is in contrast to market participants, which placed private capital risk third in our survey at the end of 2025 when asked about the biggest risks to market stability in 2026, which shows the impact of financial media sentiment.



For each investment theme listed below, please indicate whether you perceive it as a risk or an opportunity for investors in 2026

Mean score ranging from 1 (significant opportunity) to -1 (significant risk)



Source: dbDataInsights, Deutsche Bank.

2. How serious is AI disruption risk for portfolios?

Still an equity question first. While services and tech make up c50% of BDC assets, we will need to see bigger markdowns by sponsors first to signal a lack of support from the equity cushion. Big managers assessment of their equity exposure shows this is not likely soon.



Managers are taking AI risk seriously, particularly within software, but generally view their direct exposure as limited and well-underwritten. They have quantified their exposure and stress-tested portfolios, concluding that the immediate risk to their credit-focused strategies is manageable:

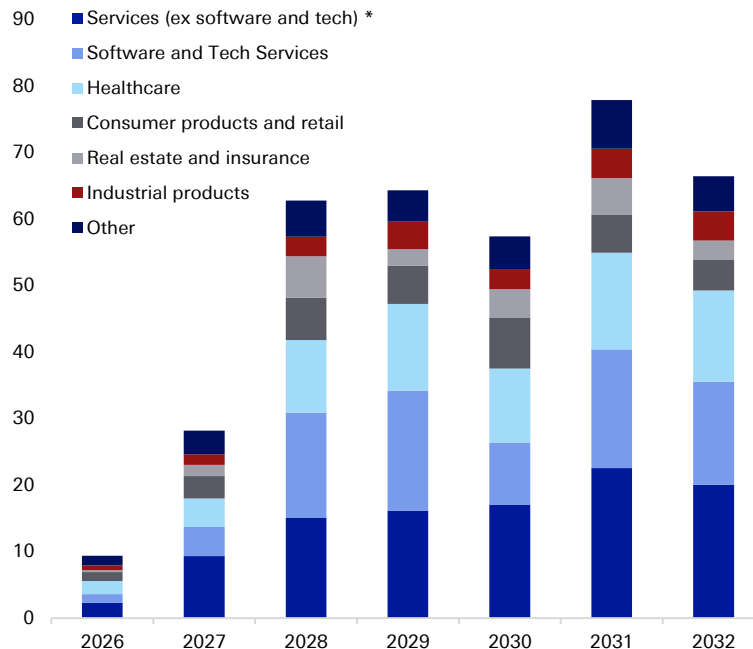
- **KKR:** Software represents **7%** of total AUM, **15%** in private equity, **5%** in credit, and **2.5%** in Global Atlantic. While public market weakness led to markdowns, the underlying software companies are seeing healthy single-digit revenue and EBITDA growth.
- **Ares:** Conducted a nine-week independent review of its software portfolio (6% of total AUM, <8% of private credit AUM). The study concluded the portfolio is "very well positioned," with **86%** classified as low-risk, **13%** as medium-risk, and only **1%** as high-risk of AI disruption. This high-risk portion represents less than 1% of the firm's total AUM.
- **TPG:** Has undertaken a "systematic review of the risks in and outside of software." In its TPG 8 fund, work showed that only **7%** of the remaining value fell into a "mitigate category" where material risk is perceived.
- **Blue Owl:** Acknowledges a "software maturity wall" coming in 2028-29 but stresses that its tech portfolio consists of very large companies (average EBITDA of \$320 million) with significant equity cushions from sponsors. They view the current situation as an "equity question, not a debt question".

How are managers mitigating the risk?

Mitigation strategies include proactive portfolio re-underwriting, strategic divestment, internal AI adoption for productivity, and a thematic pivot toward AI-beneficiary assets.

- **Blackstone** is assessing companies based on their incumbency models, proprietary data, and workflow integration. **Ares** is focusing its senior lending on companies providing core operational software for regulated industries, avoiding more vulnerable areas like content generation.
- A cross-firm strategy is to invest heavily in the infrastructure enabling the AI revolution, mostly data centres. The need for data centres directly feeds into the need for more power. **Blackstone** is the most active private investor in the US utility sector.
- Leveraging AI in portfolio companies: **KKR** has deployed AI across over 150 portfolio companies to automate workflows and enhance products, using its operational team, Capstone, to ensure lessons are shared across the firm. **TPG** is investing to help its software portfolio companies leverage AI opportunities.

Selected BDCs, maturities by industry (\$bn)

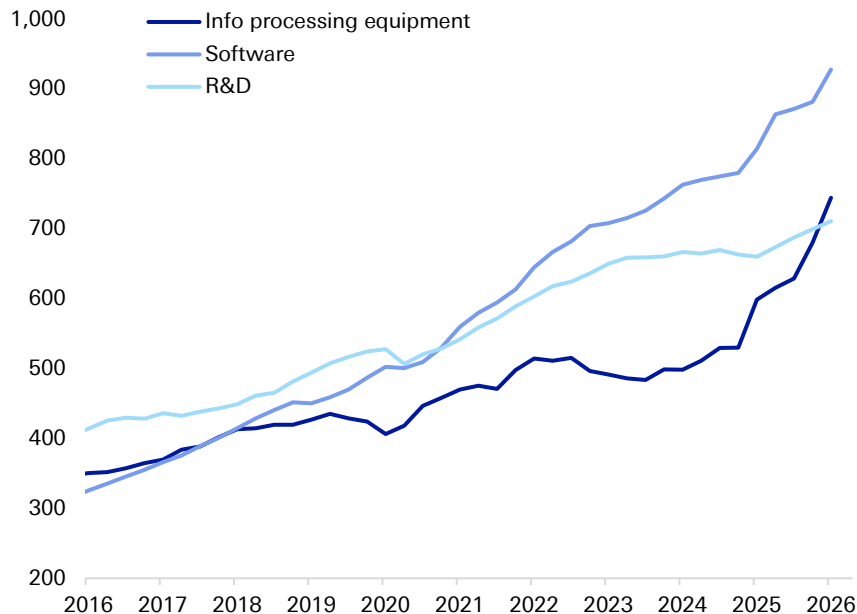


Source: Bloomberg Finance LP, Deutsche Bank. * Note: services include industries such as industrial, consumer and financial services; BDCs as tracked by Bloomberg, total assets of c \$420bn.

In the meanwhile, managers are doubling down on infrastructure investments amidst investor appetite for the AI theme, and this is likely to further fuel AI build out. Power is amongst the bottleneck private capital firms are likely to tackle further.

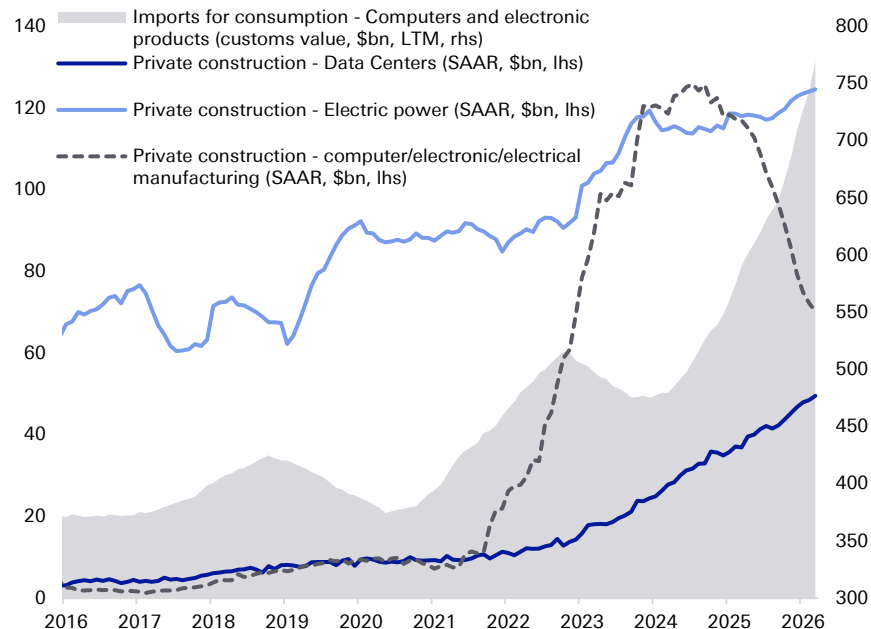


Real private non-residential fixed investment (SAAR, bn 2017 chained dollars)



Source: Bloomberg Finance LP, Haver Analytics, Deutsche Bank.

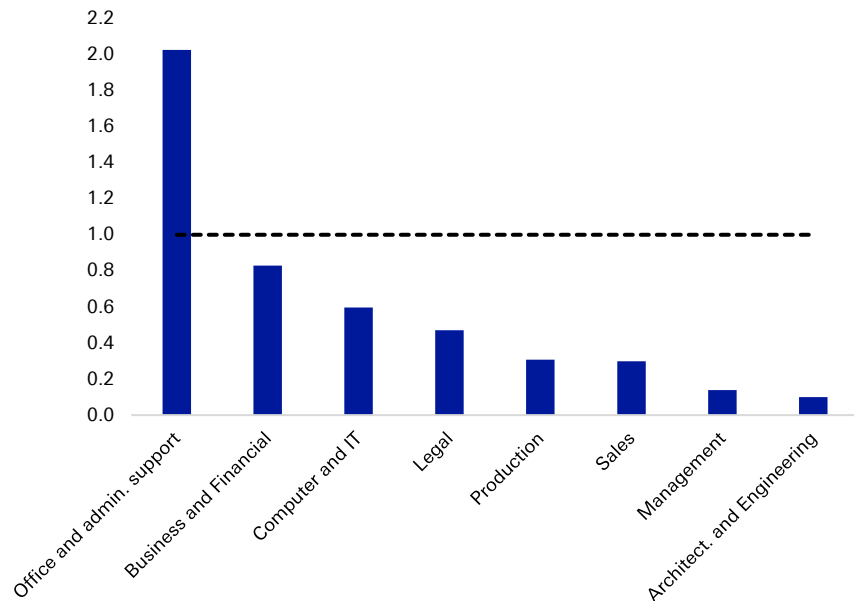
Data centre construction has run ahead of new (electric) power investments



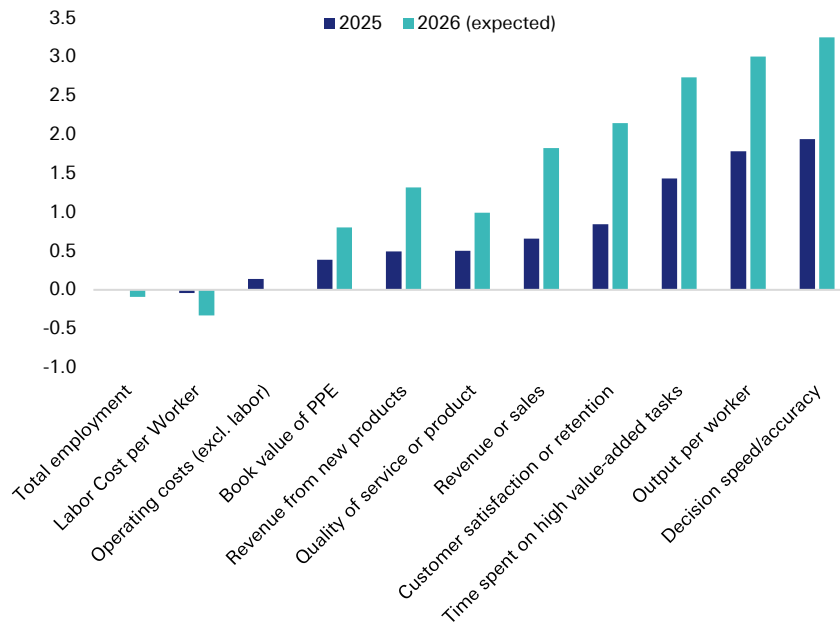
For software, the wide range of potential outcomes of AI's impact means a more benign path than the media suggests may be plausible. Indeed, US CFOs expect the fastest growth in AI's impact in 2026 to be in, first, generating more sales and then some productivity gains and do not describe most corporate roles as outright replaceable or see a significantly negative impact on headcount.



CFO survey: AI negative exposure index by occupation group
ratio of replacement mentions to enhancement mentions, >1 indicates that AI is more frequently described as replacing rather than enhancing tasks



The impact of AI usage on firm outcomes
(%, averages across bucketed options)

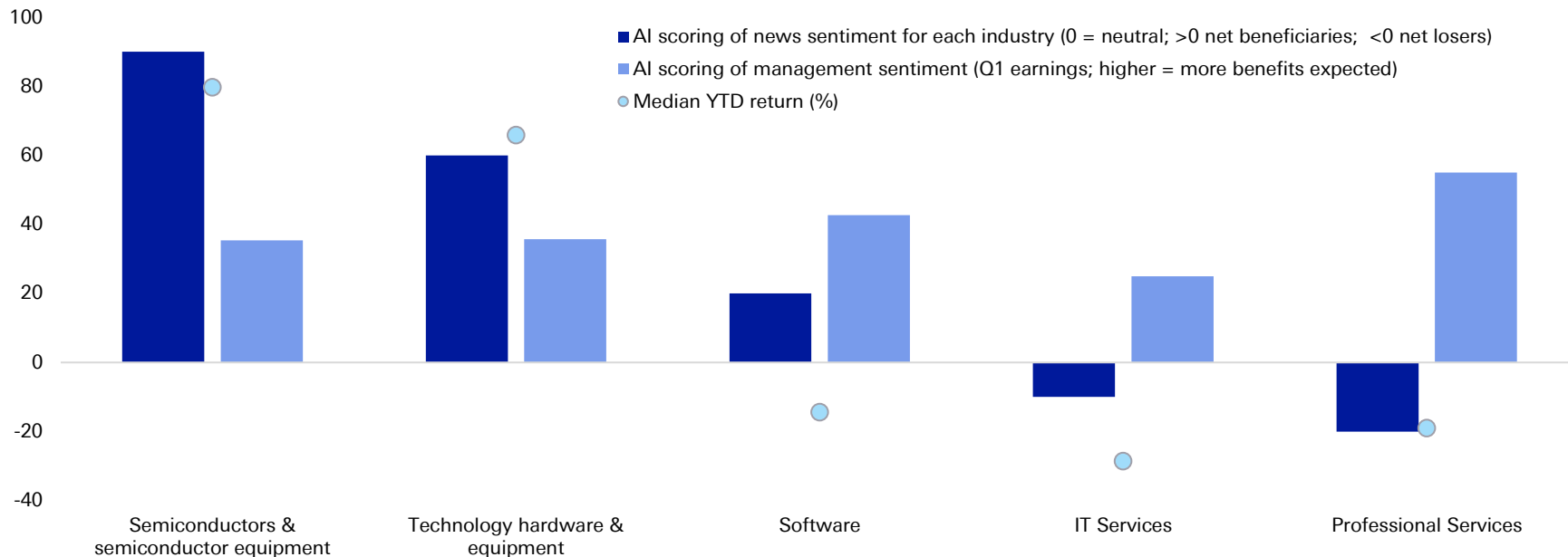


Source: Bloomberg Finance LP, Richmond Fed CFO survey, Federal Reserve Bank of Atlanta "Artificial Intelligence, Productivity, and the Workforce: Evidence from Corporate Executives", Deutsche Bank.

Moreover, our AI scoring of software risk sentiment again shows a big disconnect between what companies and media are saying about AI disruption risk, and, as the result, the potential mispricing in tech and services shares.

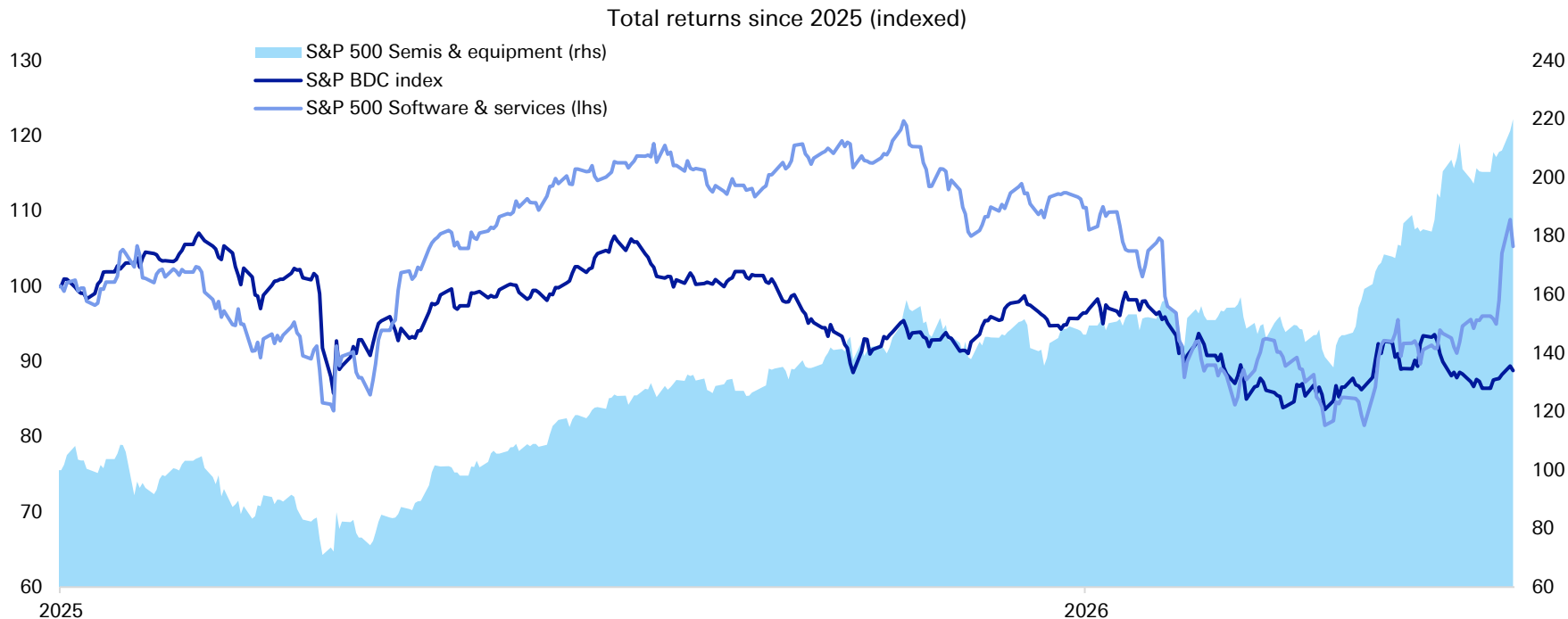


AI scoring of media and management sentiment around AI disruption risk for selected industries and their YTD performance (S&P 500 stocks)



Source: Bloomberg Finance LP, Google Trends, Deutsche Bank.

So there is still room for fears of AI doom in software to allay. This should return some confidence in the BDC market as well.



Source: Bloomberg Finance LP, Deutsche Bank.

3. Geopolitical volatility

Managers view the recent conflict in Iran as the latest in a series of "market-moving events" since 2020 that are challenging periods to navigate, but which do not derail long-term strategy. While sponsored dealmaking has been soft in recent months, it held up better ex. tech. Private capital is likely to further channel deployment into areas with thematic tailwinds like energy and security for now.



1. The message is that geopolitical shocks create near-term volatility but ultimately dealmaking recovers.
2. Managers view the conflict and broader geopolitical splintering as an accelerant for pre-existing investment themes such as defence and energy security.
3. Geopolitical uncertainty is a key factor impacting the monetization environment and the views are mixed.

How serious is the impact on portfolios and markets?

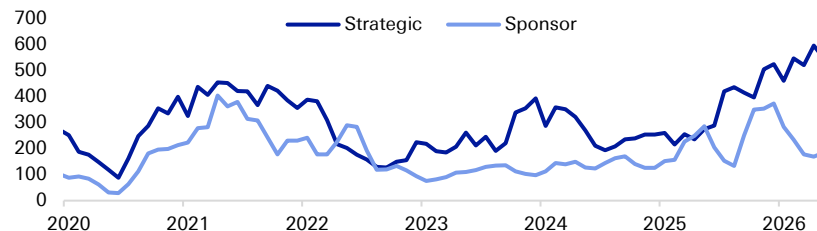
- Managers assess the direct impact on portfolio fundamentals as limited, but acknowledge the significant indirect impact on market sentiment, deployment pacing, and monetization timelines. The primary effect is a slowdown in M&A and capital markets activity as buyers and sellers pause to evaluate the new landscape.
- The conflict in Iran is cited as a reason for a pause in transaction activity. **Ares** noted that "when you see things like the conflict in Iran, you get a little bit of a pause as everybody just evaluates." However, they also reported that their aggregate pipeline is at a record level and that the direct lending pipeline is re-engaging, drawing a parallel to a similar pause after tariff announcements in April of last year, which was followed by a record deployment year.
- The increase in oil prices reinforces the strategic value of portfolios tilted towards energy and infrastructure. **TPG's** climate platform is benefiting from the increased focus on energy security, with the firm noting "The Strait of Hormuz may be bad for many things, but it's good for our business here," as it drives demand for renewables.

How are managers mitigating risk and finding opportunity?

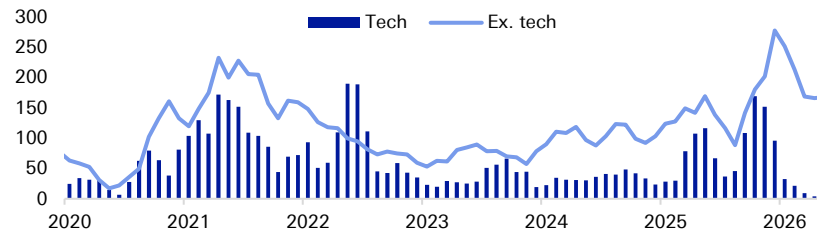
- Managers repeatedly stress that their global and strategic diversification is the primary defence against localized geopolitical shocks.
- Managers are increasing exposure to physical assets and sectors benefiting from the focus on security and re-industrialization.
- In general, managers see the volatility as a chance to invest

Source: Bloomberg Finance LP, Dealogic, Deutsche Bank.

US M&A activity has bifurcated between strength in corporate and weakness in sponsored deals...
(\$bn 3m sum)



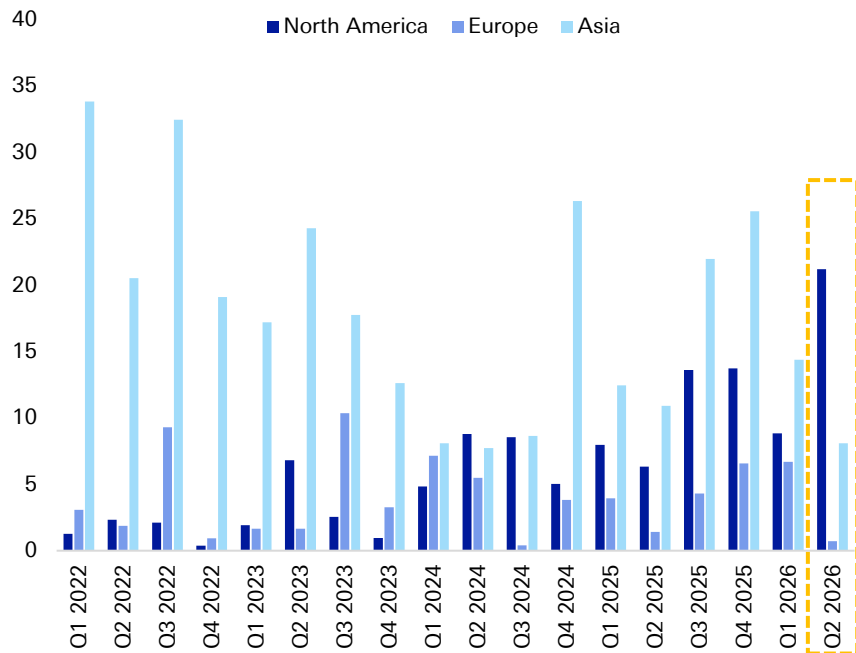
... but much of that PE weakness is tech-related. Other thematic sectors are in focus amidst all the dry powder
US sponsored M&A activity (\$bn 3m sum)





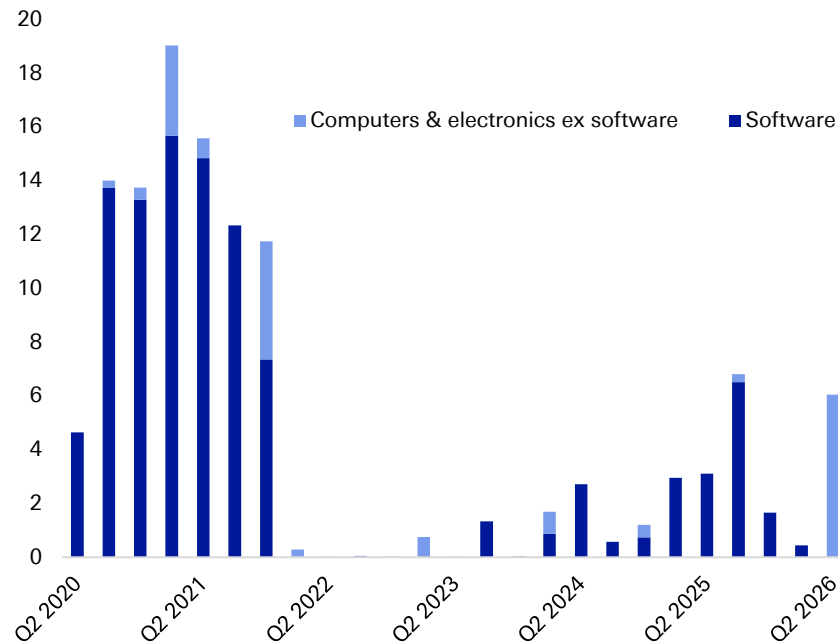
For realisations, we expect IPO recovery momentum to continue through the rest of the year, with M&A activity also resilient. Although there are some near-term cooling risks, corporates have navigated a vast array of shocks and now are well-adapted to expect risks and volatility, supported by cash buffers and liquidity. This should keep a few options for sponsors to exit portfolio companies even if some market volatility persists. See more [here](#).

IPO deal values by region (\$bn)*



Source: Dealogic, Deutsche Bank. * Note: excluding financials, Q2 2026 is QTD as of May 18th.

North America IPOs, computer & electronics (\$bn) *





03.

Market conditions

Shares of US-listed private capital managers are down -20% YTD amidst persistent negative sentiment and despite the recovery in the broader markets in Q2



Private capital managers total return performance relative to broader markets (USD)

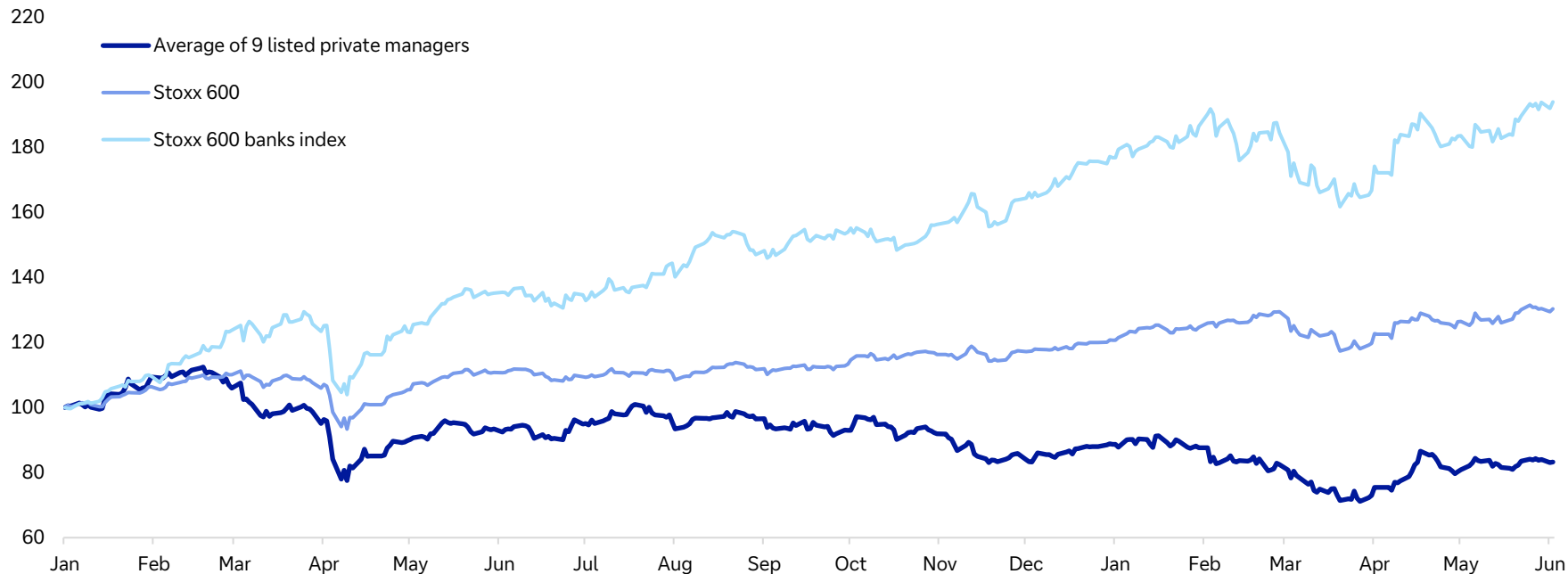


Source: Bloomberg Finance LP, Deutsche Bank.

The underperformance of private capital firms in Europe has been less stark this year (-5%)



Private capital managers total return v broader markets in 2025 (EUR)

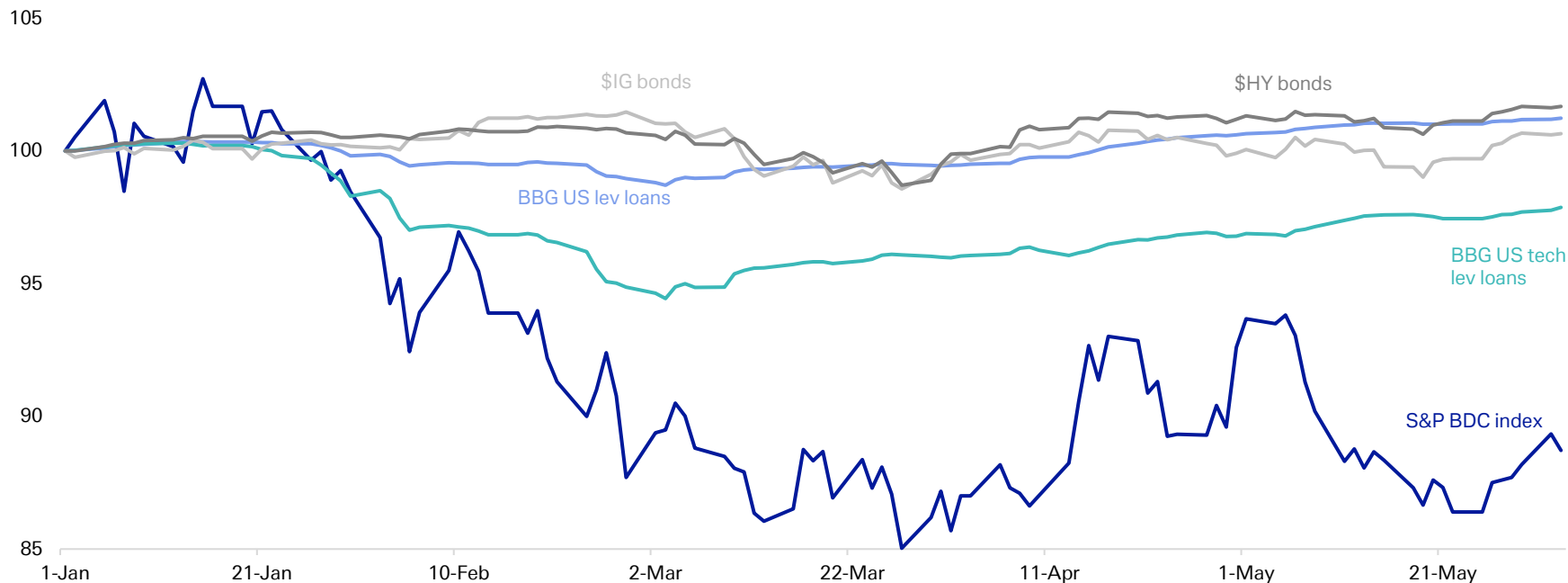


Source: Bloomberg Finance LP, Deutsche Bank.

It's a similar story for BDC returns vs the broader credit markets

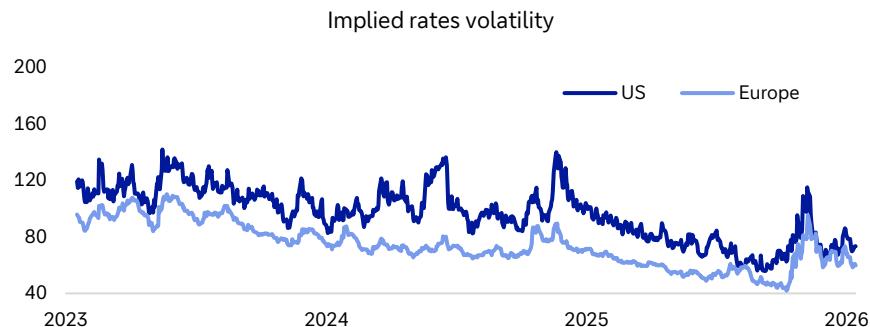
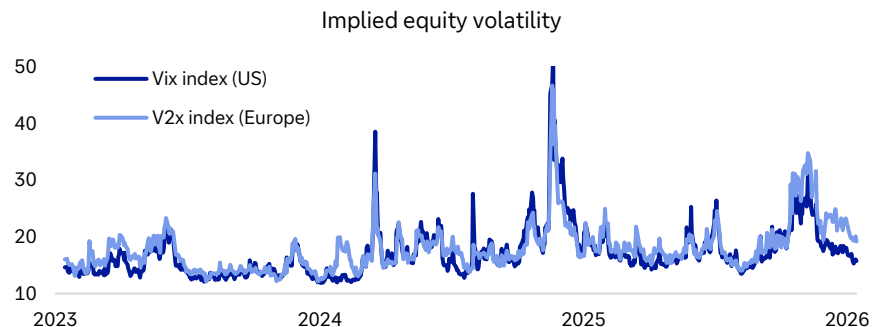
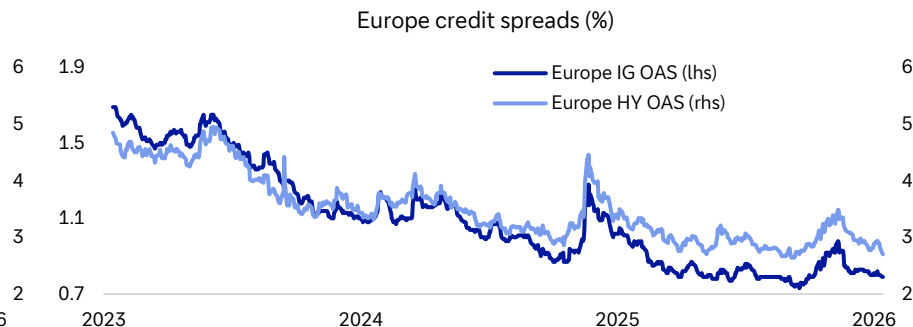
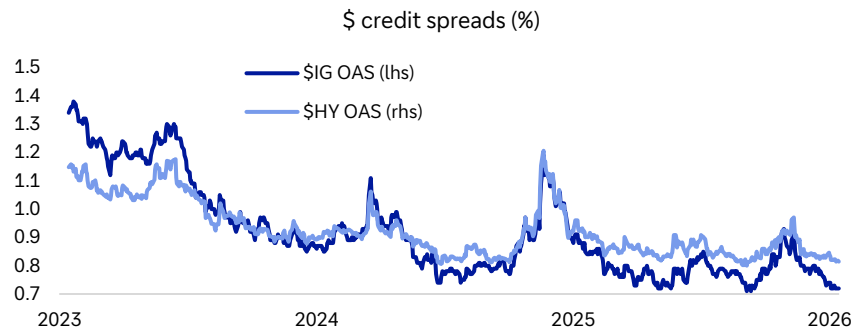


Credit assets YTD performance



Source: Bloomberg Finance LP, Deutsche Bank.

Volatility is back to somewhat subdued in the US, with Europe's disadvantage amidst the Iran conflict more pronounced

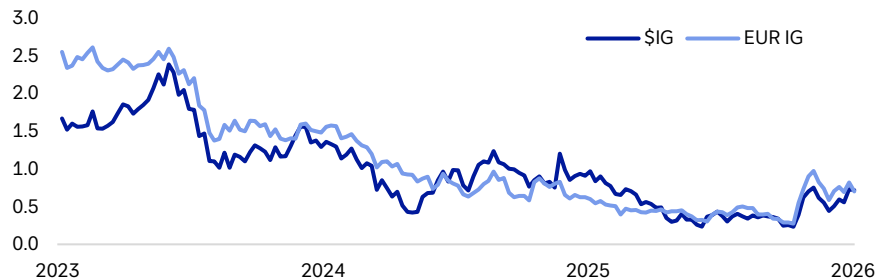


Source: Bloomberg Finance LP, Deutsche Bank.

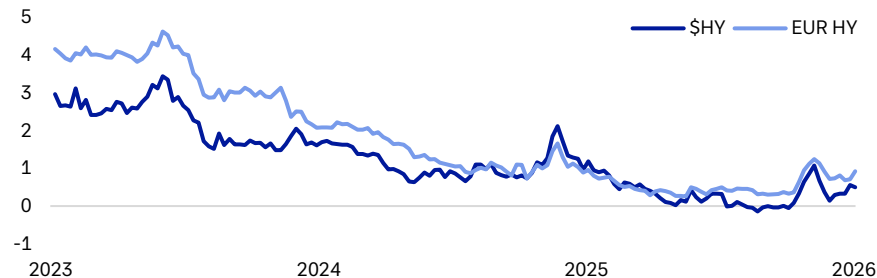
Rising refinancing costs may deter some debt issuance but overall credit conditions remain favourable



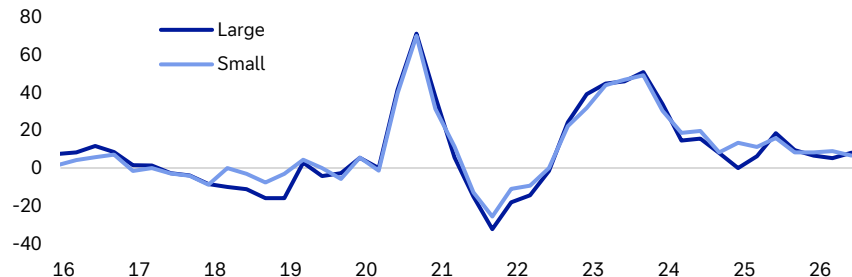
IG bonds - Refinancing proxy in %



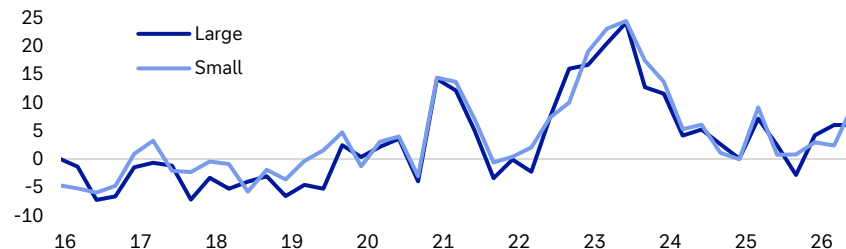
HY bonds - Refinancing proxy in %



Fed SLOOS: banks tightening standards for C&I loans to firms (%)

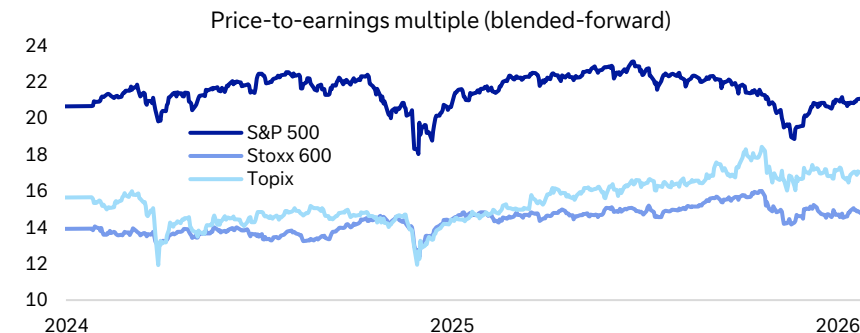
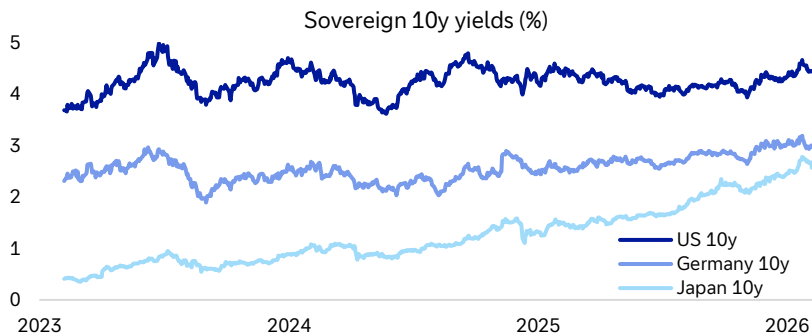
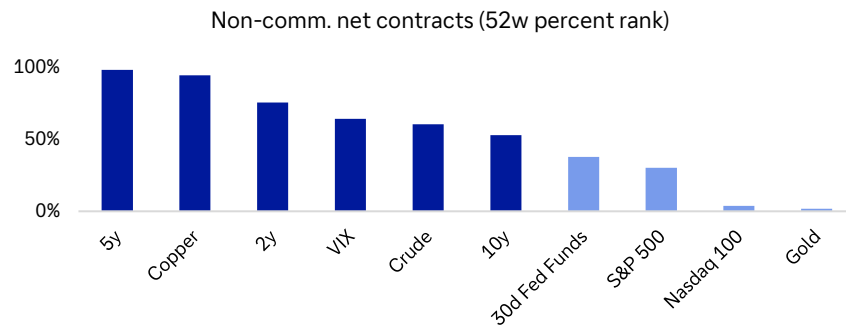
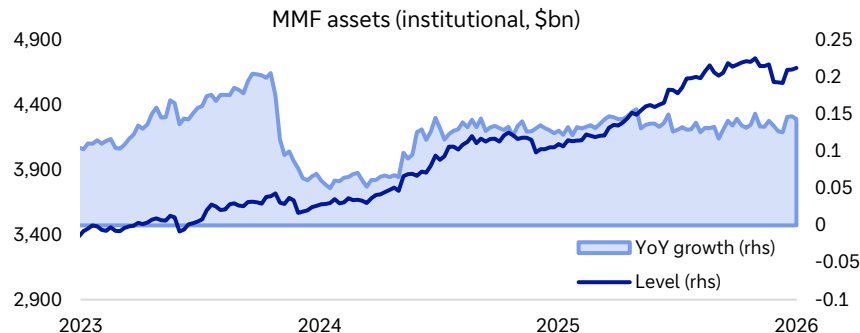


Euro Area bank lending survey: change in credit standards for business loans



Source: Bloomberg Finance LP, Haver Analytics, Deutsche Bank.

The extent of upward pressure on sovereign yields is understandably a key topic for markets. While the US economy and markets have been accustomed to rates in the current range in the past couple of years, the picture is more negative in Europe and Asia.

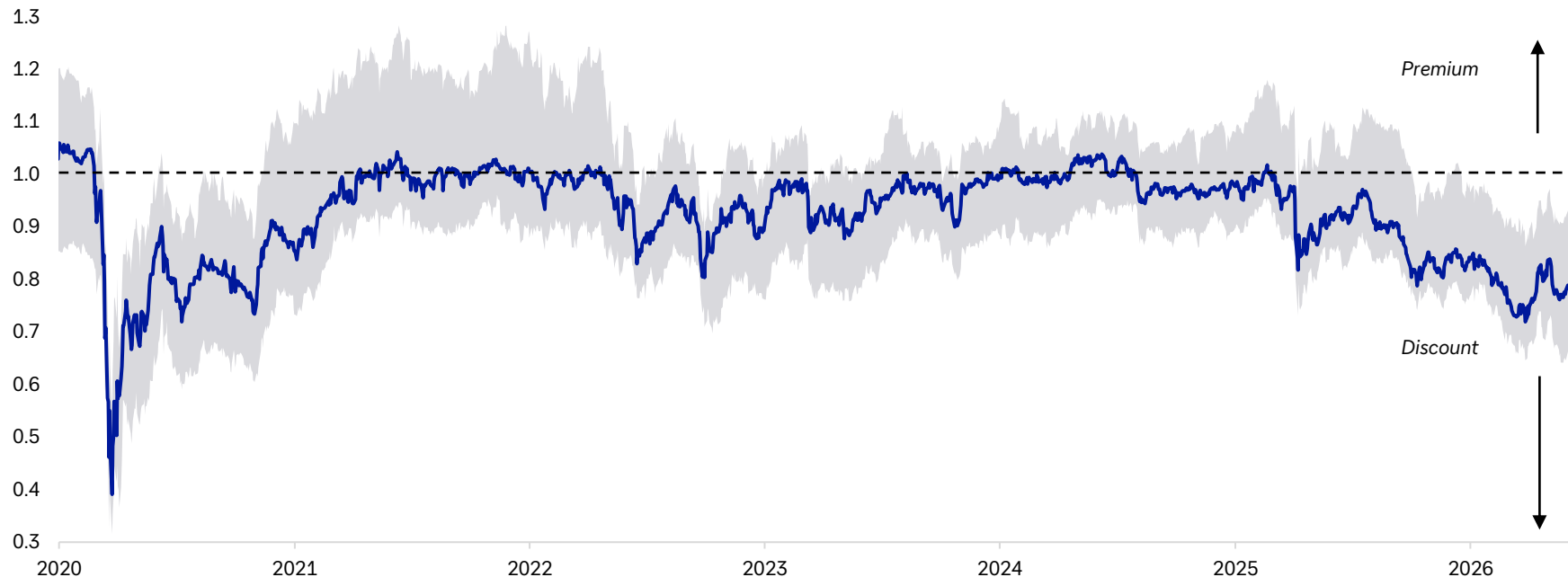


Source: Bloomberg Finance LP, Haver Analytics, Deutsche Bank.

The median BDCs discount to NAV is still wide and now stands at c 23%. Around 25% of BDC investments tend to be in tech, suggesting severe concerns beyond just AI disruption risks.

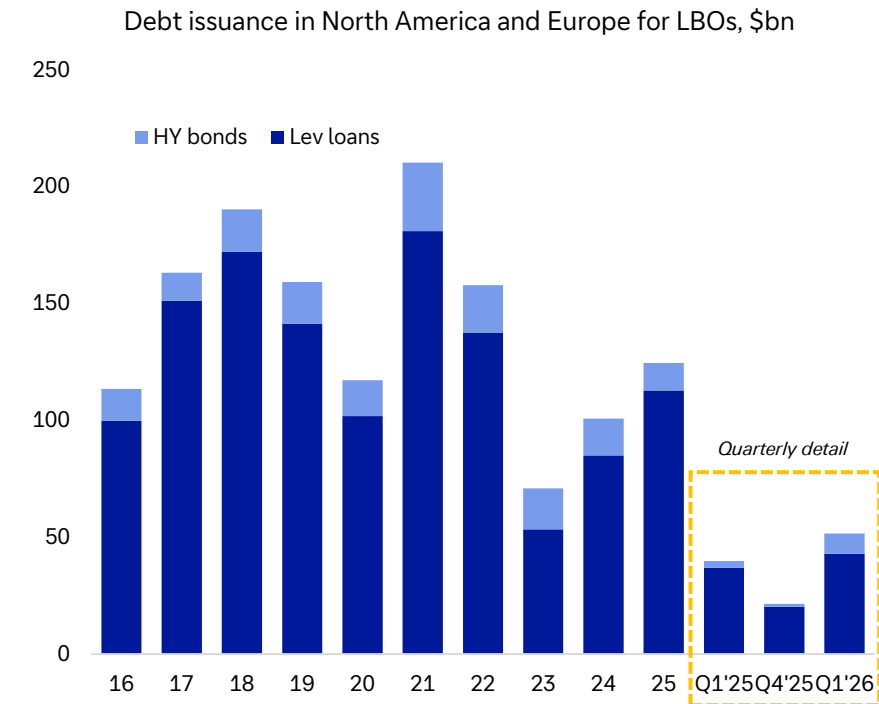


Median price-to-NAV ratio of funds in the S&P BDC index

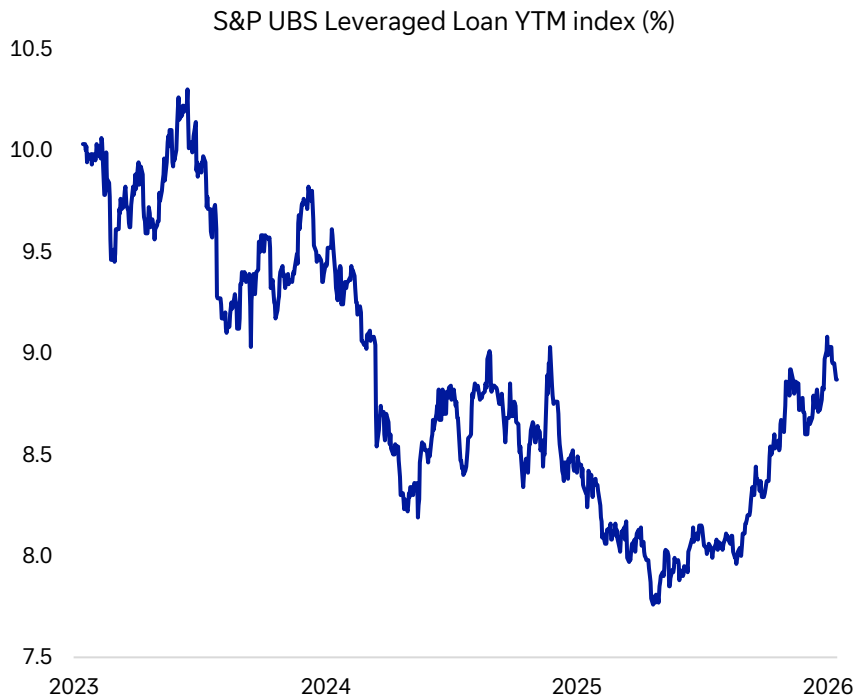


Source: Bloomberg Finance LP, Deutsche Bank.

Q1 debt issuance for LBOs is already at 40% of the 2025 total. Higher yields and spreads decomposition will help attract investor demand for more deals.



Source: White & Case, Bloomberg Finance LP, Deutsche Bank.





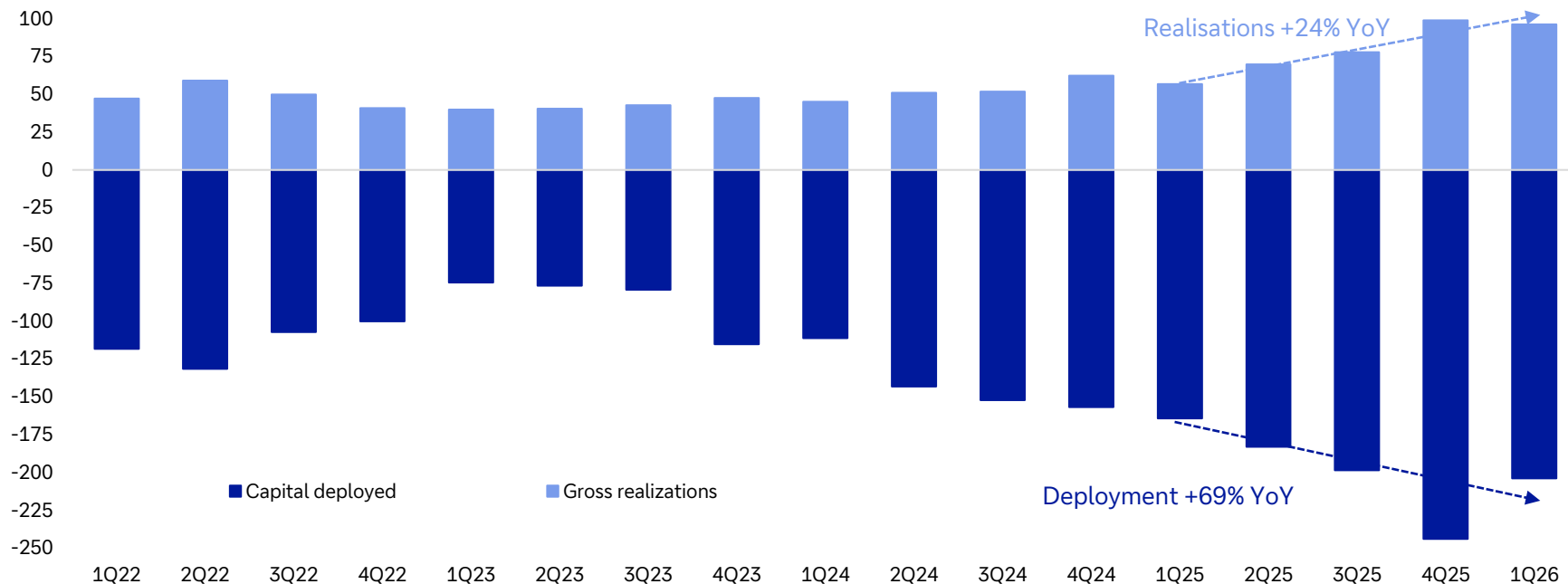
04.

Fundraising environment

Despite a QoQ drop, both deployment and realisations have been solid YoY



Selected private managers, deployment v realisations (\$bn)

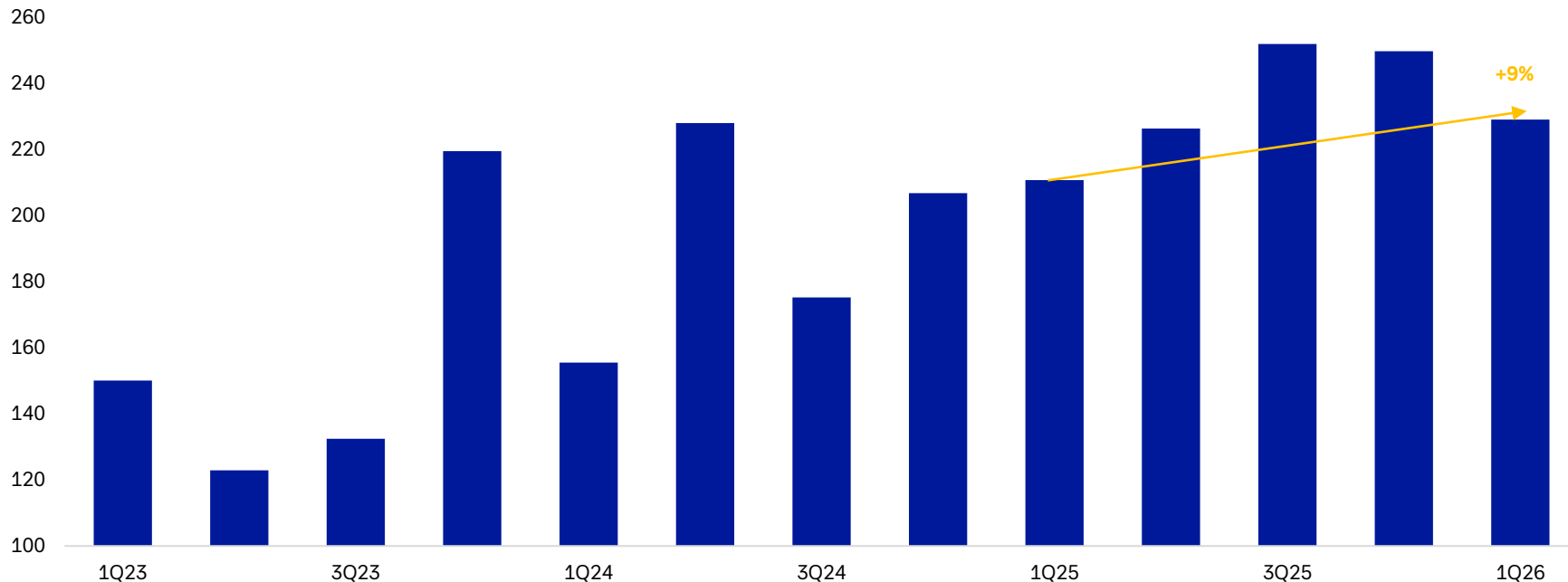


Source: Deutsche Bank company research – [US Brokers, Asset Managers & Exchanges - Deutsche Bank Research](#), Deutsche Bank. * Note: Deutsche Bank company research forecasts.

Aggregate fundraising growth continued in the negative territory. New fund launches will be the ultimate bellwether of LP sentiment and good progress on exits will help.



Fundraising by selected large, listed managers (\$bn)

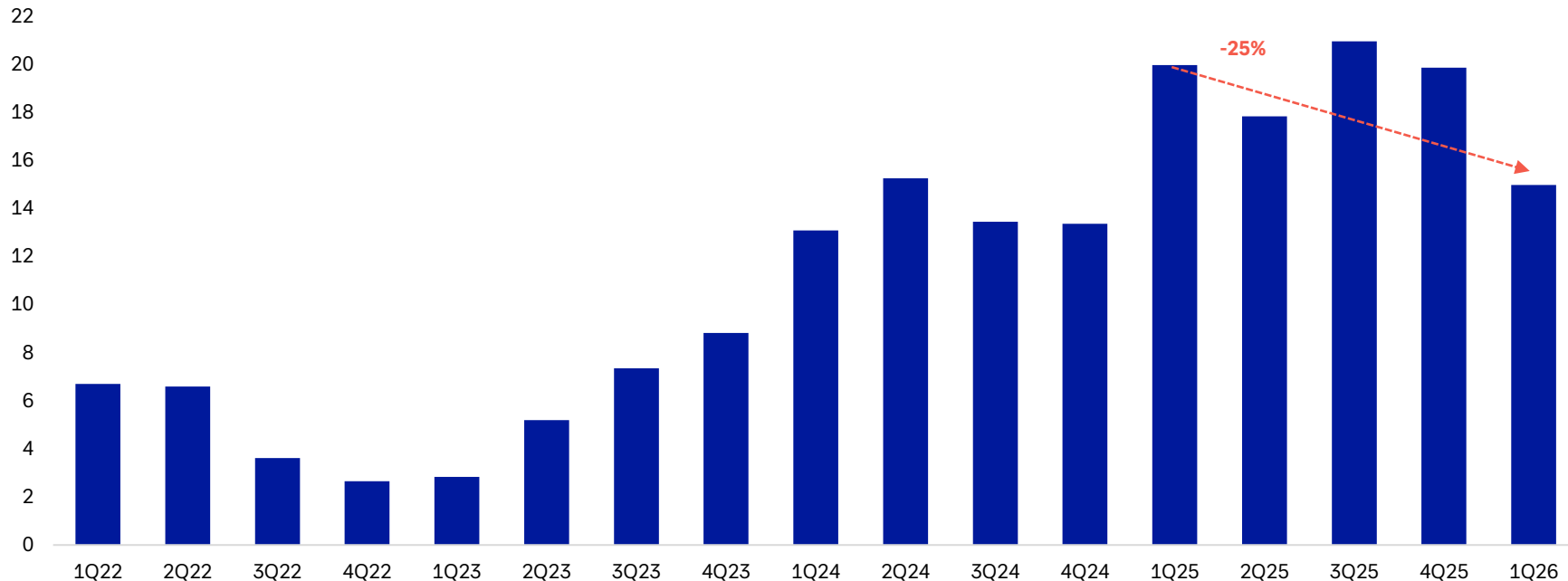


Source: Deutsche Bank company research – [US Brokers, Asset Managers & Exchanges - Deutsche Bank Research](#), Deutsche Bank.

Gross retail fundraising (not taking into account redemptions) is down by a quarter. Managers emphasise that the issue is headline driven and concentrated, but the sentiment needs to improve significantly to restore strong growth.



Retail fundraising by largest listed private capital managers *

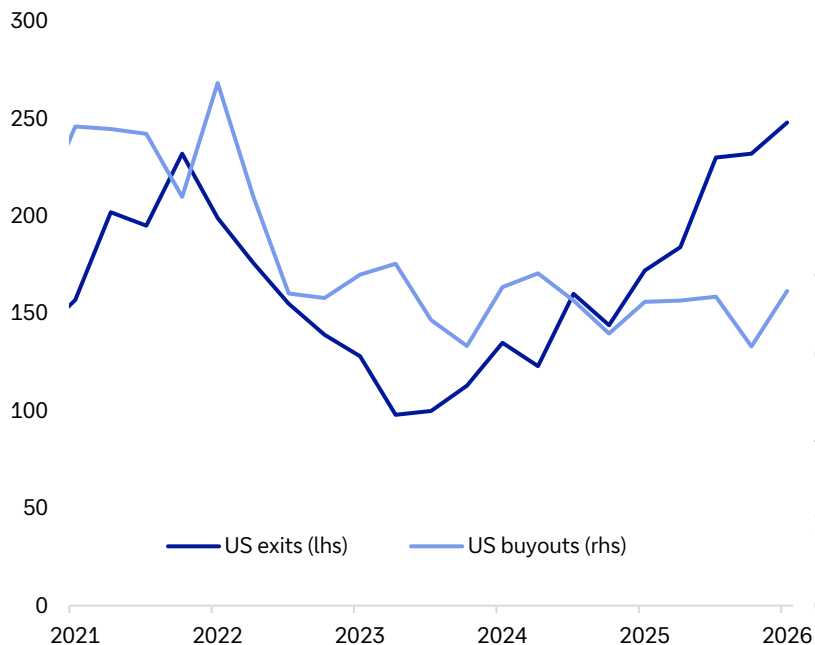


Source: Deutsche Bank company research – [US Brokers, Asset Managers & Exchanges - Deutsche Bank Research](#), Bloomberg Finance LP, Deutsche Bank. * Note: gross, only funds with available Q1 data.

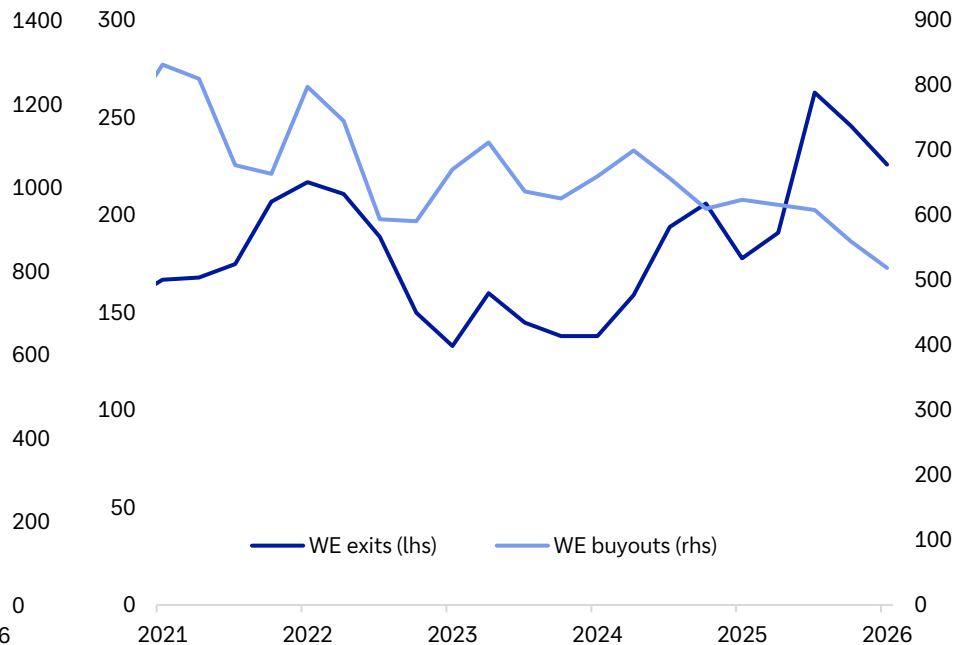
Strengthening exit activity should assuage investor concerns about private equity and free up more capital to do deals



US PE exits have accelerated past 2021 highs



European exits are high but somewhat slowing

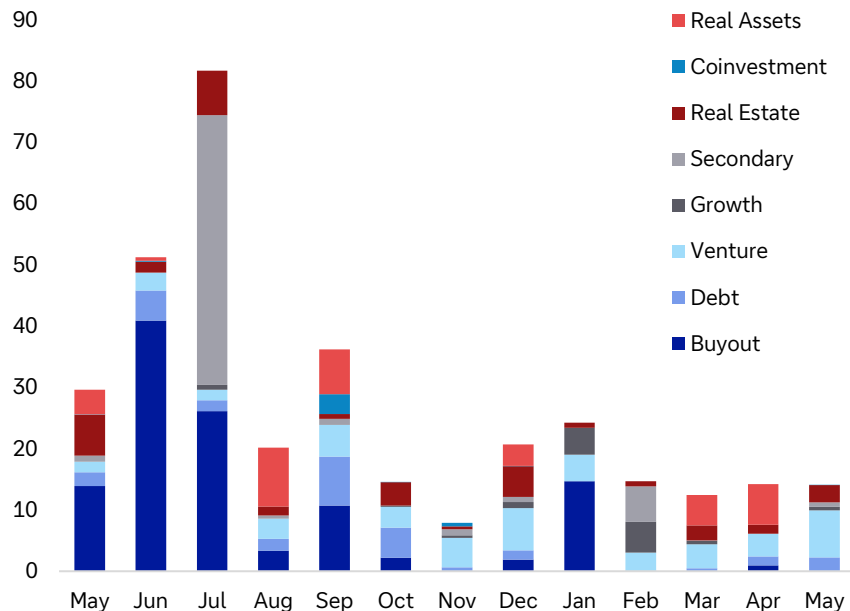


Source: White & Case, Deutsche Bank.

New fund launches – private equity gathering pace in recent months

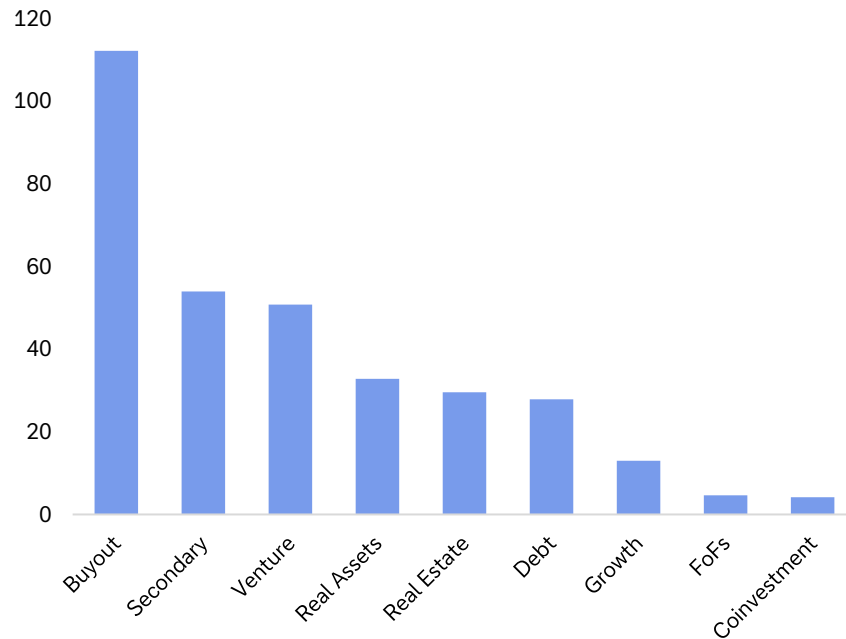


Fundraising funds - target in \$bn by strategy type and launch month, US and Europe



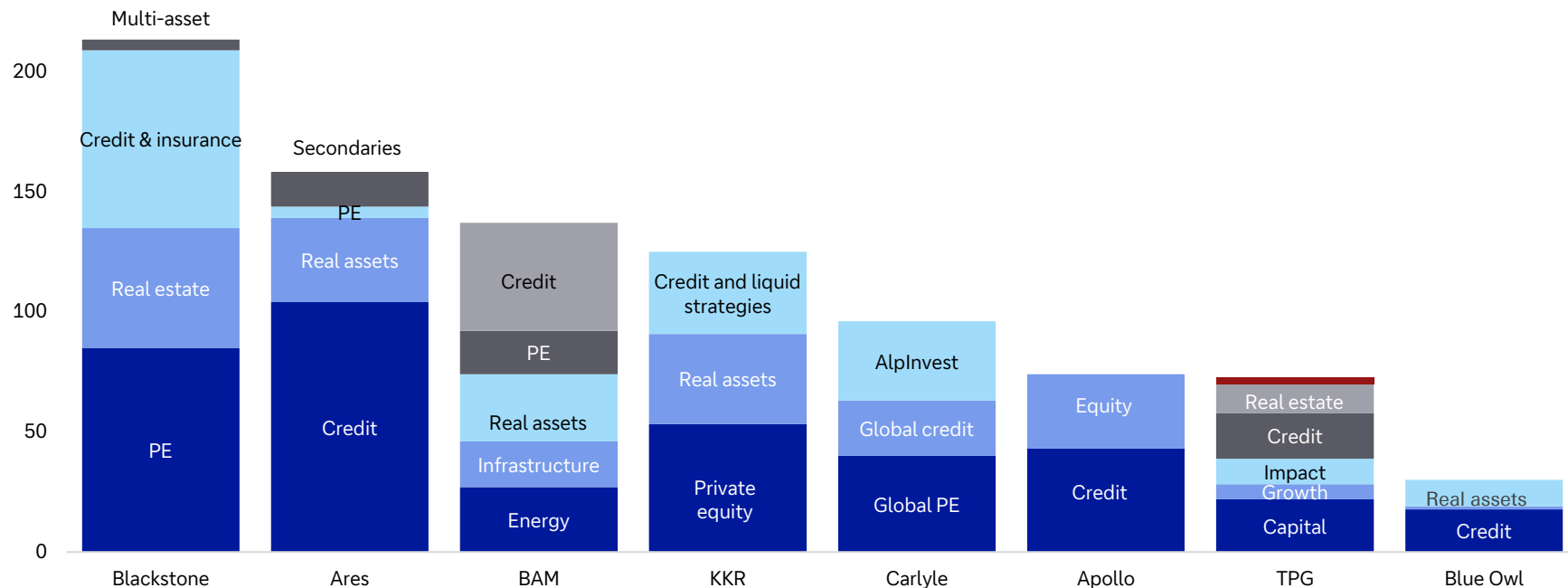
Source: Bloomberg Finance LP, Deutsche Bank.

Fundraising funds in US and Europe, last 12m in \$bn





Dry powder by manager and reported segment, Q1 2026 (\$bn)

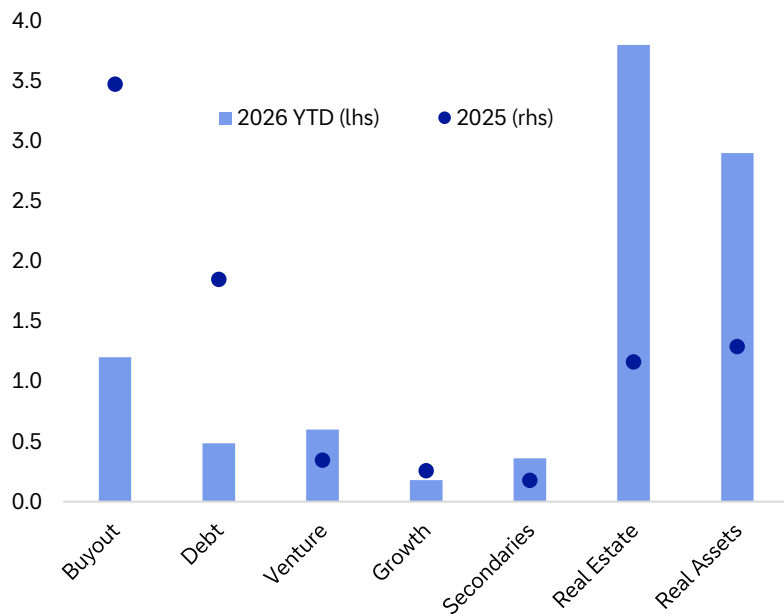


Source: Company reports, Deutsche Bank.

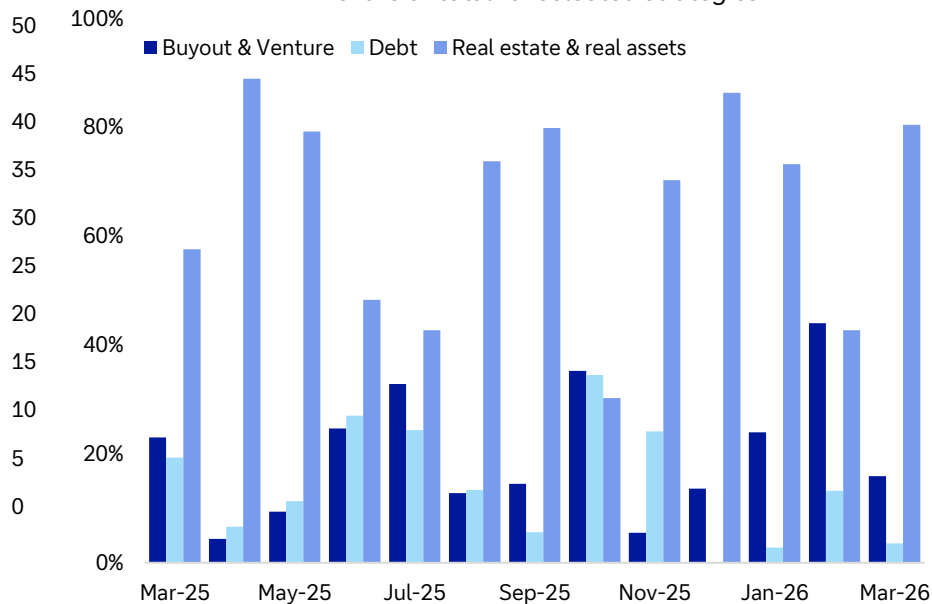
Recent disclosed LP commitments



Disclosed LP commitments by strategy in \$bn



Disclosed LP commitment targets by month and strategy, share of total for selected strategies



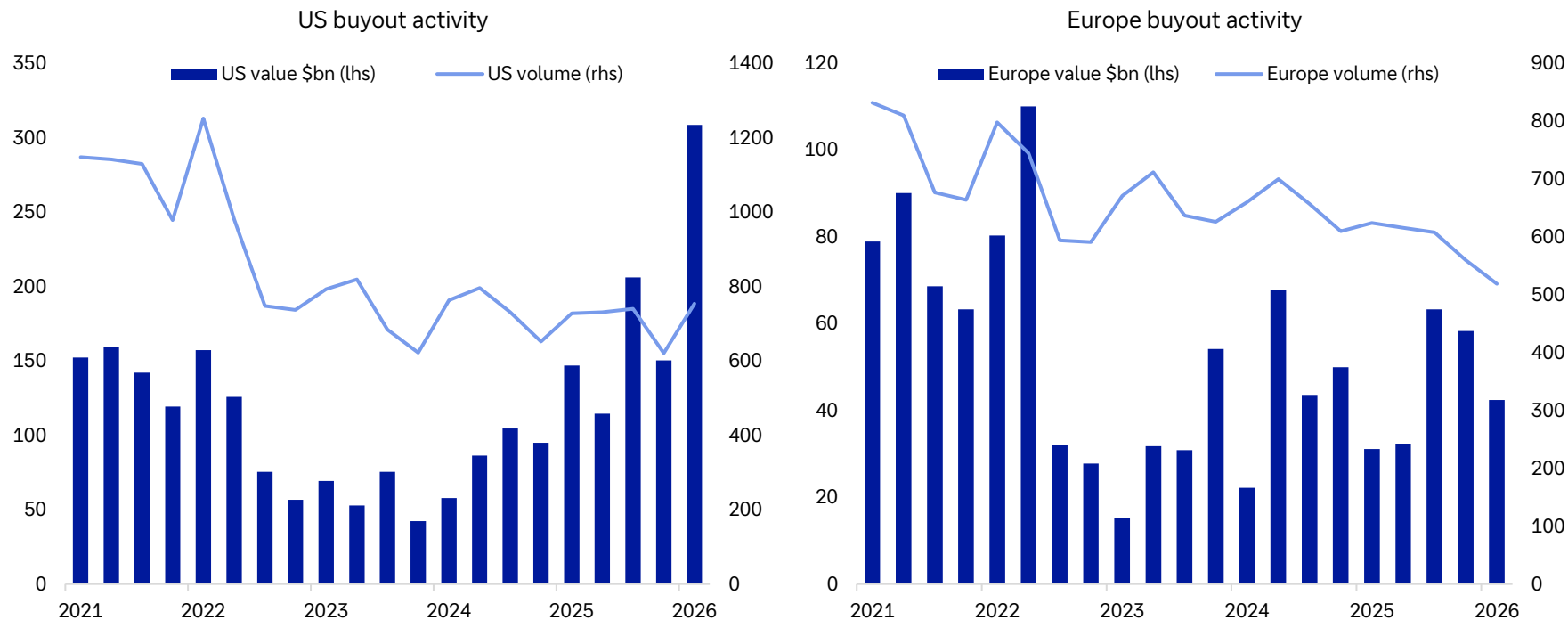
Source: Bloomberg Finance LP, Deutsche Bank.



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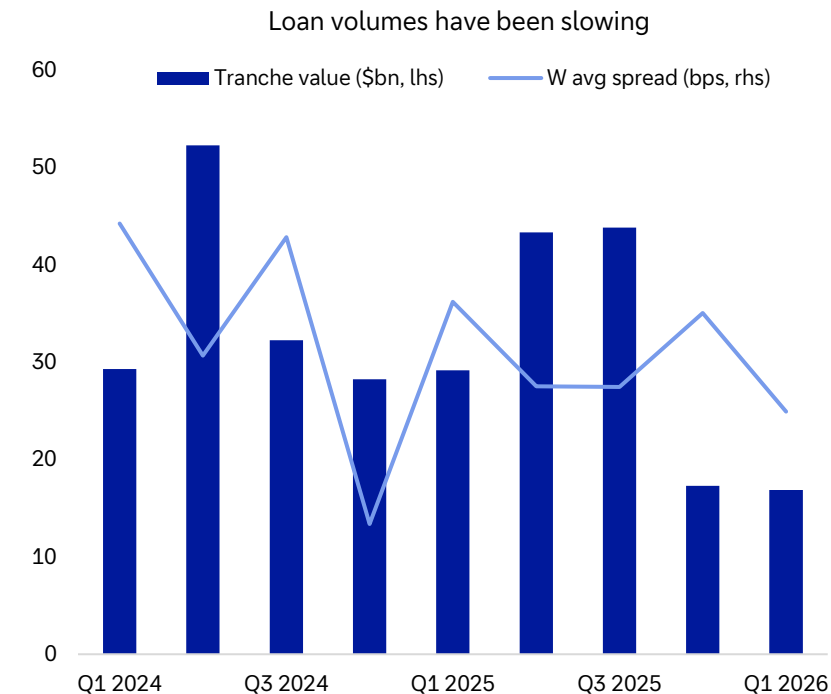
Dealmaking activity

Dealmaking activity in buyouts

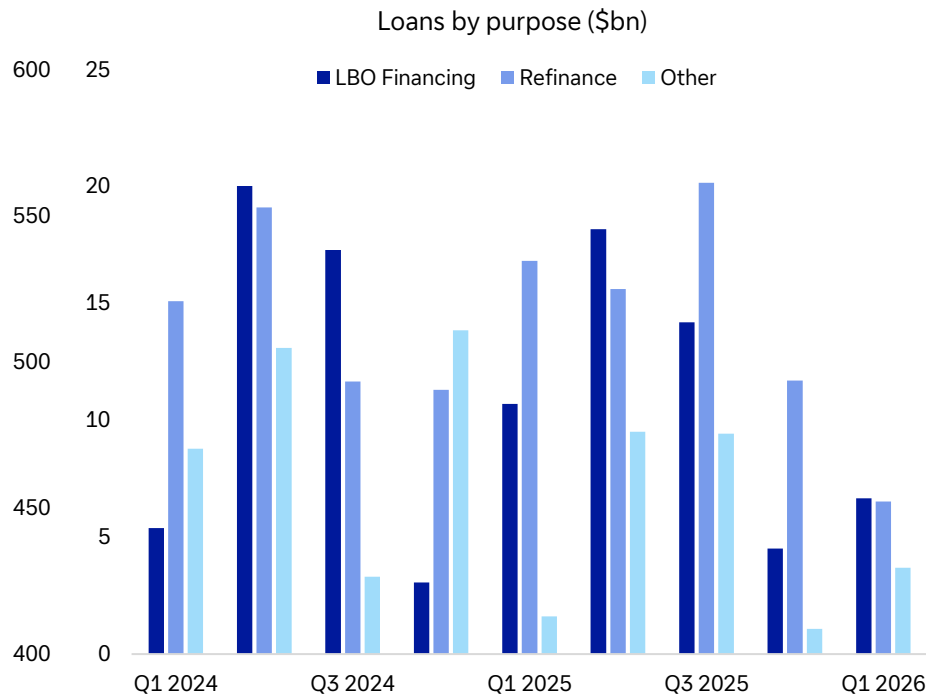


Source: White & Case, Deutsche Bank.

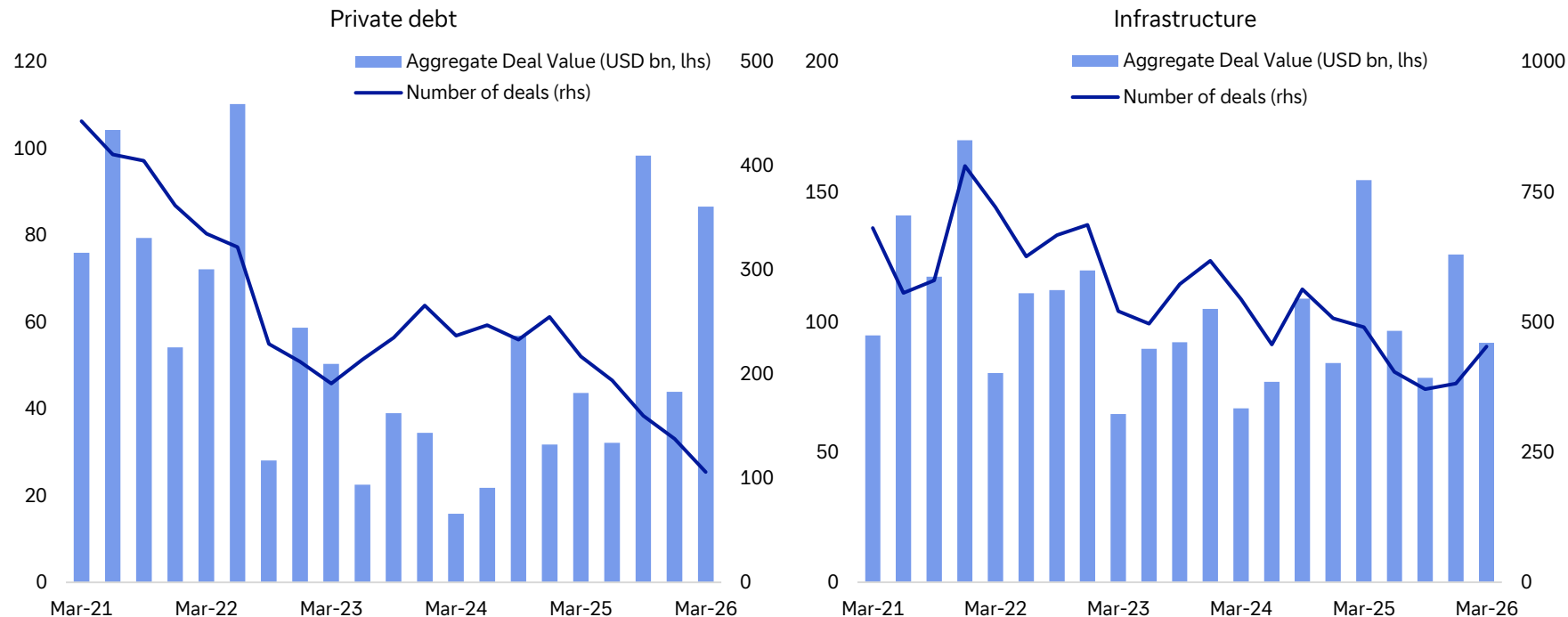
Origination in large private credit loans – loan volumes steady QoQ but down YoY as both LBOs and refinancing demand slowed



Source: Bloomberg Finance LP, Deutsche Bank.



Global dealmaking activity in private debt and infrastructure



Source: Preqin, Deutsche Bank.

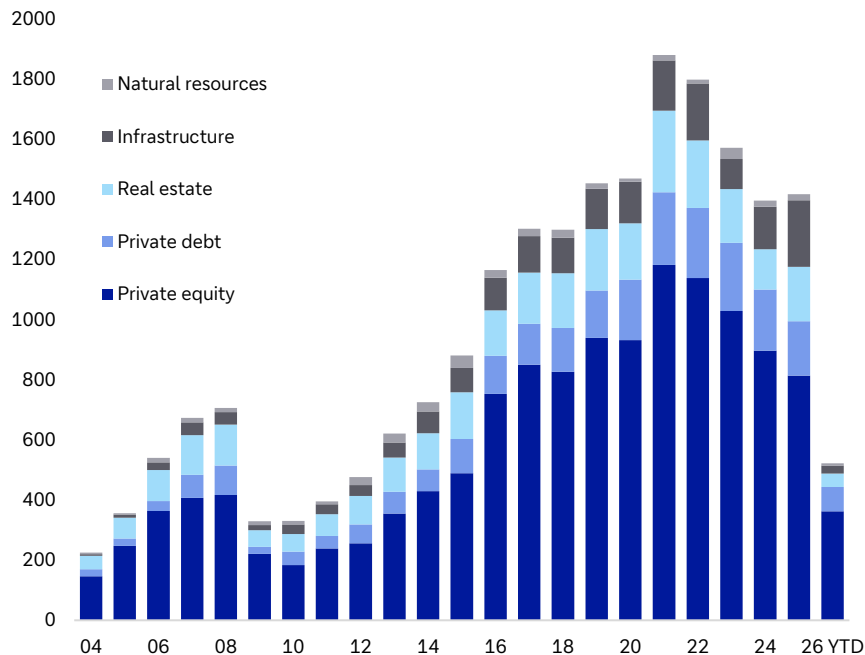


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Investor allocations

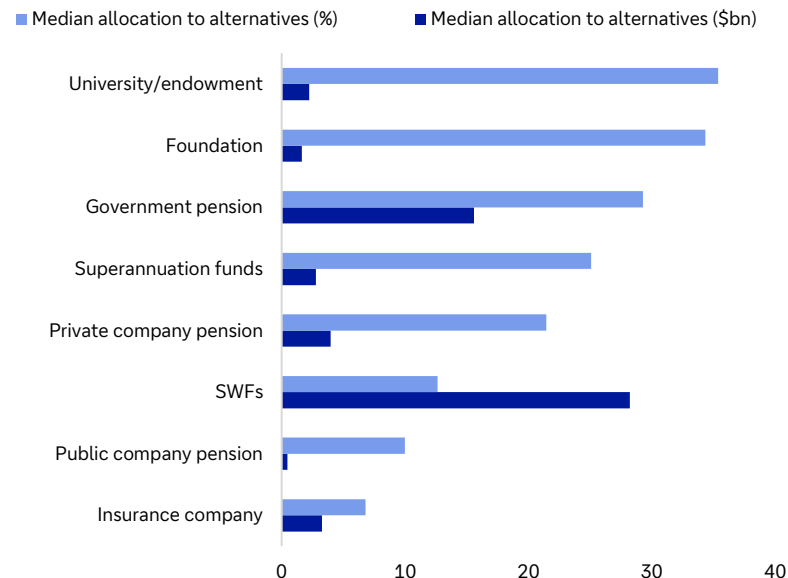


Private capital sum of funds raised (\$bn)



Source: Preqin, Bloomberg Finance LP, Deutsche Bank.

Top 100 LPs in each category, allocations to alternatives



Appendix 1



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